Particle Physics 3 Supersymmetry And Grand Unification

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Contents

1 Supersymmetry & Grand Unification: Lecture 1 **1**

1.1 Renormalization as Coarse-Graining . . . 1

1.2 Molecules as the Prototype Example . . . 2

1.3 From Effective Potentials to Field Theory . . . 3

1.4 Scalar Fields, Propagators, and Mass Renormalization . . . 4

1.4.1 Question & Answer . . . 6

1.5 From Scalar Renormalization to the Higgs Hierarchy Problem . . . 6

1.5.1 Question & Answer . . . 7

1.6 Why Fermions Behave Better . . . 7

1.6.1 Question & Answer . . . 9

1.7 Vacuum Energy and the Supersymmetry Outlook . . . 9

1.7.1 Question & Answer . . . 10

1.8 Summary . . . 10

2 Supersymmetry and Grand Unification: Lecture 2 **11**

2.1 Rotations, 2π, and the Topological Prehistory of Supersymmetry . . . 11

2.2 Spin States Under Rotation and the Interference Test of a Sign Change . . . 12

2.2.1 Question & Answer . . . 12

2.3 Bosons, Fermions, and Exchange Symmetry . . . 13

2.4 Why a Closed Fermion Loop Carries a Minus Sign . . . 14

2.4.1 Question & Answer . . . 14

2.5 Vacuum Energy, Boson Self-Energy, and the Zeroth-Order Supersymmetry Idea . . . 15

2.6 Broken Supersymmetry, Naturalness, and Experimental Reach . . . 16

2.6.1 Question & Answer . . . 17

2.7 Exact Supersymmetry Would Destroy Ordinary Chemistry . . . 19

2.7.1 Question & Answer . . . 19

2.8 Summary . . . 21

3 Divergences, Propagators, and the Supersymmetric Motive 22

3.1 Why We Are Still Talking About Divergences . . . 22

3.2 Propagators in Position Space and Momentum Space . . . 23

3.2.1 Question & Answer . . . 23

3.3 Scalar Fields, Dimensional Analysis, and the Massless Propagator . . . 23

3.4 The Simplest Scalar Self-Energy Diagram and the Fine-Tuning Problem . . . 24

3.4.1 Question & Answer . . . 26

3.5 Fermions: Dirac Lagrangian, Field Dimension, and Short-Distance Propagator . . . 26

3.6 Closed Fermion Loops, Yukawa Couplings, and the Supersymmetry Motive . . . 27

3.6.1 Question & Answer . . . 27

3.7 Summary . . . 29

4 Symmetry, Supersymmetry, and Grassmann Numbers 30

4.1 Symmetries As Operations On States . . . 30

4.1.1 Question & Answer . . . 31

4.2 Infinitesimal Transformations And Generators . . . 31

4.3 Why Commutators Encode The Group . . . 32

4.4 Symmetry And The Hamiltonian . . . 33

4.4.1 Question & Answer . . . 36

4.5 Why Supersymmetry Is Different . . . 37

4.6 Supersymmetry Generators And Statistics . . . 37

4.6.1 Question & Answer . . . 37

4.7 Grassmann Numbers . . . 39

4.8 Summary . . . 40

5 Supersymmetry as Odd Symmetry and Superspace 41

5.1 Why Supersymmetry Begins with Grassmann Numbers . . . 41

5.2 Grassmann Numbers, Functions, and Calculus . . . 41

5.2.1 Question & Answer . . . 44

5.3 Ordinary Symmetry as the Comparison Case . . . 44

5.4 A Minimal Supersymmetry Algebra from Oscillators . . . 45

5.4.1 Question & Answer . . . 46

5.5 From the Hamiltonian to Superspace . . . 47

5.6 Summary 49

6 Vacuum Energy, Unification, and Grassmann Superspace 51

6.1 Vacuum Energy, Casimir Effect, and Honest Evidence 51

6.1.1 Question & Answer 52

6.2 What a Supersymmetry Generator Really Does 52

6.2.1 Question & Answer 53

6.3 Why Low-Energy Supersymmetry Is Worth Taking Seriously 53

6.4 Running Couplings and Higher-Precision Unification 54

6.5 Vacuum Energy in Cosmology and the Friedmann Balance 55

6.5.1 Question & Answer 55

6.5.2 Question & Answer 57

6.6 Grassmann Numbers, Superspace, and Superfields 58

6.7 Grassmann Integration as the Calculus of Supersymmetry 59

6.7.1 Question & Answer 59

6.8 Summary 61

7 Lecture 7: Supersymmetry, Superfields, and Chirality in the One-Mode Toy Model 62

7.1 Bosons, Fermions, and the Zero-Momentum Toy System 62

7.2 The Bosonic Oscillator in Lagrangian Form 63

7.3 Grassmann Variables and the Fermionic Lagrangian 64

7.3.1 Question & Answer 64

7.4 Supercharges and the Basic Supersymmetry Algebra 65

7.5 From Supercharge Algebra to Superspace Geometry 66

7.6 Superfields, Superactions, and the D-Term 67

7.7 Powers of Superfields, Derivative Subtleties, and Chirality 67

7.7.1 Question & Answer 68

7.8 Summary 70

8 Supersymmetry in Four Dimensions 71

8.1 From Time-Only Supersymmetry to Lorentz Invariance 71

8.2 Massless Chiral Fermions and the Hamiltonian $H = \sigma \cdot p$ 71

8.2.1 Question & Answer 72

8.3 The Four-Dimensional Superalgebra 73

8.4 Chiral Superfields as a Consistent Constraint 74

8.4.1 Question & Answer . . . 74
8.5 Solving $\bar{D}\Phi = 0$ and Expanding the Chiral Superfield . . . 75
8.6 Superspace Actions and the Free Example $\Phi^*\Phi$ . . . 76
8.7 Mass Terms, Interaction Terms, and the Role of the Auxiliary Field . . . 78
8.7.1 Question & Answer . . . 78
8.8 Feynman-Diagram Interpretation and the Supersymmetric Payoff . . . 80
8.9 Summary . . . 81

9 Supersymmetry Breaking and the Opening of Grand Unification 82
9.1 Symmetry Breaking as Level Splitting . . . 82
9.1.1 Question & Answer . . . 83
9.2 Vacuum Criteria and Order Parameters . . . 83
9.3 Recap of the Chiral Superfield . . . 84
9.4 From the Superpotential to the Scalar Potential . . . 85
9.4.1 Question & Answer . . . 87
9.5 Consequences of Supersymmetry Breaking . . . 88
9.6 The First Step Toward $SU(5)$ . . . 90
9.7 Summary . . . 92

10 SU(5), Multiplets, and Proton Decay 93
10.1 SU(5) as a Concrete Grand-Unification Example . . . 93
10.2 Representations of $SU(N)$: N, $\bar{N}$, and Tensor Products . . . 94
10.3 Symmetric and Antisymmetric Two-Index States . . . 94
10.3.1 Question & Answer . . . 96
10.4 From $N \otimes \bar{N}$ to the Adjoint, and Then to the Five-Plet . . . 96
10.4.1 Question & Answer . . . 97
10.5 The Antisymmetric Matrix and the Remaining Fermions . . . 98
10.6 Gauge Bosons, Leptoquark Transitions, and Proton Decay . . . 100
10.6.1 Question & Answer . . . 102
10.7 Unification Scale, Running Couplings, and the Supersymmetry Epilogue . . . 102
10.8 Summary . . . 103

Chapter 1

Supersymmetry & Grand Unification: Lecture 1

These notes follow Leonard Susskind's opening lecture in the supplementary course on supersymmetry and grand unification, curated by LazyingArt LLC. The lecture begins one step earlier than supersymmetry itself. It begins with renormalization: first as a way of throwing away microscopic detail that does not matter for the question at hand, and then as a way of estimating what the discarded short-distance physics must do to the parameters we measure. We will keep that order. The chapter therefore unfolds as the lecture unfolds: coarse-graining, then dimensional analysis, then scalar mass renormalization, then the Higgs hierarchy problem, then the much better behavior of fermions, and finally vacuum energy.

1.1 Renormalization as Coarse-Graining

Susskind opens the course by saying that many of the ideas to come grew out of puzzles in the renormalization of the Standard Model. But he does not begin with counterterms, regularization schemes, or formal perturbation theory. Instead he asks us to remember that physics repeatedly changes its effective language as the scale of interest changes.

Definition 1.1. In the sense emphasized throughout this lecture, renormalization has two ingredients:

1. we eliminate from the description degrees of freedom associated with distances too small, or motions too fast, to matter for the problem we are actually asking;
2. we replace their net effect by effective parameters in a coarser description.

The second half of the subject is dimensional analysis: once the short-distance modes are identified, dimensional analysis often tells us how large their effects must be.

The lecture then climbs, deliberately and patiently, through a ladder of effective descriptions. If we are doing nuclear physics, we may begin with quarks, because quarks are the microscopic constituents. But once we have extracted the masses, spins, and interactions of protons and neutrons, we are free to describe the nucleus as a nonrelativistic many-body system built from those slower composite objects. If we move up to atomic physics, we need even less microscopic information. The detailed

structure of the nucleus mostly drops out; its mass and charge are the main data that survive. Once we know those, we start over again and treat nuclei and electrons as the relevant elementary ingredients.

The same step is taken yet again when we go from atoms to molecules. At each stage we solve just enough of the finer theory to determine the parameters that matter at the next scale, and then we forget how those parameters were produced. The description becomes less microscopic, but more useful. That is the sense in which renormalization is already part of ordinary physics long before we apply the word to quantum field theory.

1.2 Molecules as the Prototype Example

The lecture's first real mathematical example is a pair of atoms, reimagined as two heavy nuclei together with their electrons. The aim is to replace the detailed electronic cloud by an effective force between two slow, heavy objects. Susskind's motivation is physical before it is formal: we would like to describe atoms as particles with forces between them, not as bowling balls permanently accompanied by a swarm of tiny fast flies.

Because relativity is irrelevant at molecular scales, he starts from nonrelativistic quantum mechanics. A cleaned reconstruction of the Hamiltonian written on the board is

$$H = \frac{P_1^2}{2M} + \frac{P_2^2}{2M} + \frac{e^2}{R_{12}} + \sum_i \frac{q_i^2}{2m} + \sum_{i<j} \frac{e^2}{r_{ij}} - \sum_i \left(\frac{e^2}{R_{1i}} + \frac{e^2}{R_{2i}} \right). \tag{1.1}$$

Here P_1, P_2 are the nuclear momenta, M is the nuclear mass, q_i are the electron momenta, m is the electron mass, R_{12} is the distance between the nuclei, r_{ij} is the separation of two electrons, and R_{1i}, R_{2i} are the distances from the i-th electron to the two nuclei.

Remark 1.2. The board image fixes the structure of the Hamiltonian more securely than it fixes every symbol. In particular, the electron momentum notation and the electron-nucleus attraction terms are partly obscured in the frame. The formula above is therefore a cautious transcript-backed reconstruction rather than a literal blackboard transcription.

What matters next is the separation of time scales. The nuclei are heavy. The electrons move much more rapidly. Susskind's picture is that the nuclei move so slowly that, in first approximation, we may nail them down and solve only for the electronic motion in that fixed nuclear background. We group all electron-dependent terms together and treat them as the Hamiltonian of the electrons alone, with R_1 and R_2 held fixed.

We then solve the electronic Schrödinger problem and keep the lowest eigenvalue,

$$E_{\text{el}} = E_{\text{el}}(R_1, R_2). \tag{1.2}$$

This is the ground-state energy of the electrons in the background of stationary nuclei. Since it depends only on the nuclear positions, it can be absorbed into an effective potential for the nuclei:

$$V_{\text{eff}}(R_1, R_2) = \frac{e^2}{R_{12}} + E_{\text{el}}(R_1, R_2). \tag{1.3}$$

The lecture emphasizes the qualitative shape. At very short separation the Coulomb repulsion of the nuclei dominates, so the potential is strongly repulsive. At very large separation the interaction

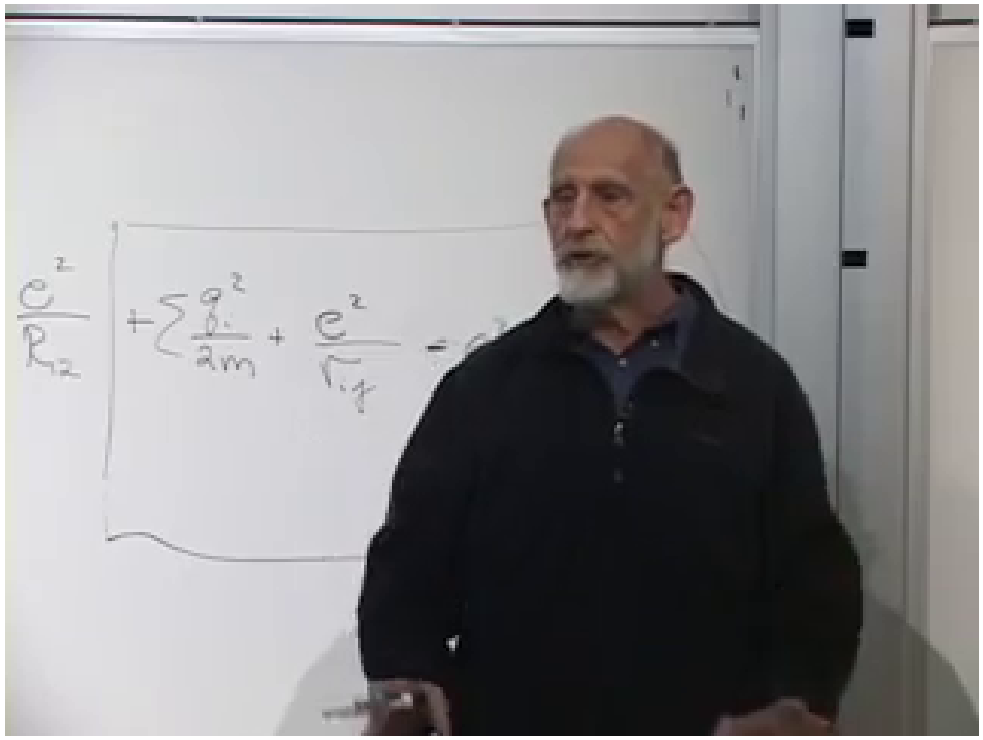

Figure 1.1: Molecular Hamiltonian with the electron sector visually grouped as a subproblem.

becomes weak. Between the two there is an attractive region due to the electrons. Once that effective potential has been found, we no longer track the electronic cloud explicitly. The fast microscopic degrees of freedom have been eliminated and replaced by a renormalized Hamiltonian for the slow ones.

That is the prototype. Susskind explicitly pauses here to say that this, too, is renormalization.

1.3 From Effective Potentials to Field Theory

Only after the molecular example has done its work does the lecture pivot to field theory. The transition is exact: in quantum field theory the modes we wish to eliminate are the very short-wavelength, very high-frequency, very high-energy ones. Small usually goes with fast. The ultraviolet modes are the field-theory analog of the fast electrons in the molecular problem.

Before launching into diagrams, however, Susskind deliberately interrupts the story and inserts a small toolkit. “Before we do that,” he says in effect, “let us do some dimensional analysis.” This is not a digression. It is what will allow the loop estimates to be read off rather than computed in detail.

Setting $c = \hbar = 1$, one independent dimension remains. We may take it to be length. Then

$$[M] = [E] = [p] = L^{-1}, \qquad [x] = [t] = L. \tag{1.4}$$

So mass, energy, and momentum all count as inverse length, while space and time count as length. Once this is fixed, the size of propagators and couplings becomes almost forced upon us.

1.4 Scalar Fields, Propagators, and Mass Renormalization

The lecture now chooses the cleanest possible setting: one scalar field ϕ. On the board the theory is introduced schematically by first writing the local Lagrangian density,

$$\mathcal{L} = (\partial_\mu \phi)^2 - V(\phi), \tag{1.5}$$

and then, on the next line, the action,

$$S = \int d^4x \, \mathcal{L}. \tag{1.6}$$

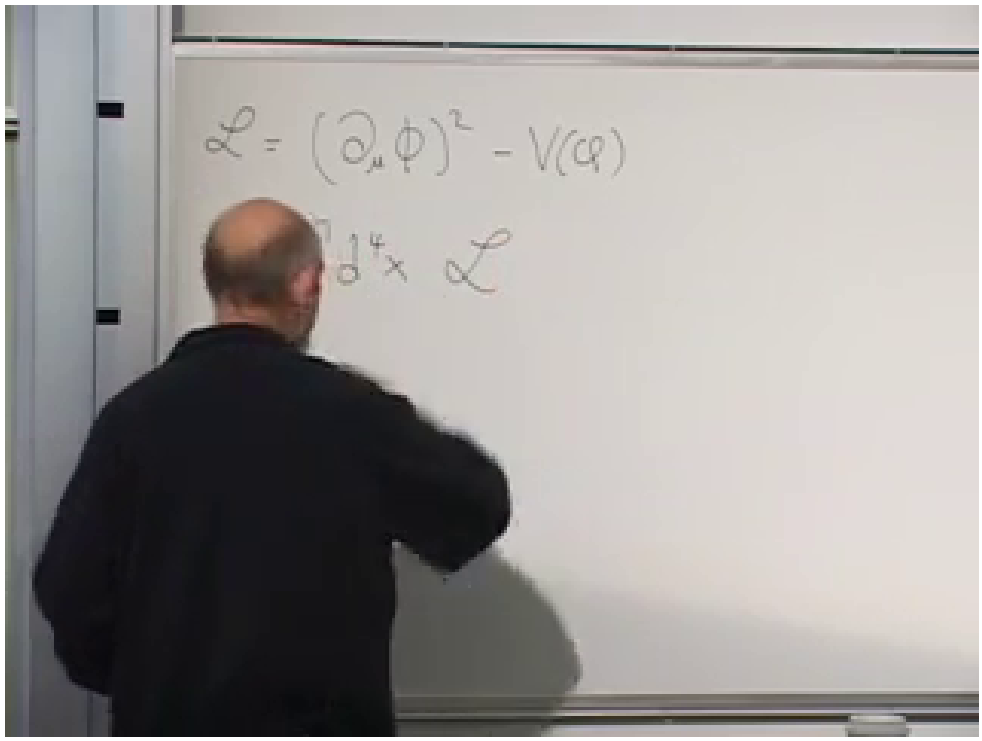

Figure 1.2: The lecture's two-step move from local Lagrangian density to spacetime action.

Remark 1.3. The board writes the scalar theory in a deliberately schematic normalization. We keep that form rather than silently inserting textbook factors of 1/2.

Because $\hbar = 1$, the action is dimensionless. Therefore the Lagrangian density must have dimensions L^{-4}. The kinetic term then fixes the dimension of the field itself:

$$[S] = 1, \qquad S = \int d^4x \, \mathcal{L}, \tag{1.7}$$

$$[\mathcal{L}] = L^{-4}, \qquad [(\partial_\mu \phi)^2] = L^{-4}. \tag{1.8}$$

Since $[\partial_\mu] = L^{-1}$, we obtain

$$[\phi]^2 L^{-2} = L^{-4} \qquad \Longrightarrow \qquad [\phi] = L^{-1}. \tag{1.9}$$

This is exactly the kind of elementary dimensional fact that the lecture wants us to keep in our pocket.

Next the potential is expanded:

$$V(\phi) = \frac{m^2}{2}\phi^2 + g\phi^3 + \lambda\phi^4 + \cdots . \tag{1.10}$$

Since V must also have dimensions L^{-4}, we read off

$$[g] = L^{-1}, \qquad [\lambda] = 1. \tag{1.11}$$

The point is immediate. The coefficients in the potential become the coefficients of Feynman vertices. The cubic term gives a three-point vertex weighted by g. The quartic term gives a four-point vertex weighted by λ. The quadratic term gives a two-point insertion: the lecture draws it as a particle that comes in, is absorbed, and is emitted again from essentially the same point.

The second ingredient is the propagator. Susskind defines it operationally before he defines it algebraically. It is the amplitude to create a particle at one spacetime point and detect it at another. In the lecture this is written as

$$\langle 0|\phi(y)\phi(x)|0\rangle. \tag{1.12}$$

Now the vacuum has no dimensions, but each ϕ has dimensions L^{-1}, so the whole object has dimensions L^{-2}. If the scalar is massless, there is no intrinsic length scale in the problem except the separation of the two points themselves. Therefore the propagator must scale schematically as

$$\langle 0|\phi(y)\phi(x)|0\rangle \propto \frac{1}{(x-y)^2}. \tag{1.13}$$

Here one should read $(x-y)^2$ schematically as the Lorentz-invariant squared separation. The lecture is not deriving a fully normalized Green's function. It is isolating the short-distance behavior. And that behavior is what matters: the propagator blows up as the points approach one another. That is the source of ultraviolet divergence.

Now the lecture asks what the mass term really means in an effective theory. At the microscopic level the bare $m^2\phi^2/2$ term is the simplest local two-point insertion. But if our experimental microscope cannot resolve distances smaller than some cutoff δ, then any short-distance process that imitates a particle coming in and going out again from essentially the same place must be counted as part of the mass term. This is where renormalization enters.

The first example uses the quartic coupling. A particle comes in, goes out, and at the same point a short virtual line is emitted and reabsorbed. If the separation is smeared only over distances of order δ, the propagator contributes $1/\delta^2$, so

$$\delta m^2 \sim \frac{\lambda}{\delta^2}. \tag{1.14}$$

The next example is a higher-order diagram with two quartic vertices. Here we must integrate over the relative separation of the two nearby points. To keep the notation straight, let Δ^μ denote the integration variable, while δ denotes the cutoff distance. Then the lecture's dimensional estimate is

$$\lambda^2 \int d^4\Delta \, \frac{1}{\Delta^6} \sim \frac{\lambda^2}{\delta^2}. \tag{1.15}$$

The logic is simple: three propagators give Δ^{-6}, the four-dimensional measure gives $d^4\Delta$, and the answer can depend only on the lower cutoff δ. So it must again scale like δ^{-2}.

The cubic interaction gives something slightly subtler. Two cubic vertices and two propagators produce

$$g^2 \int d^4\Delta \, \frac{1}{\Delta^4} \sim g^2 \log(\delta \, \mu_{\text{ref}}), \tag{1.16}$$

with a reference scale μ_{ref} inserted only to make the logarithm dimensionless. The lecture treats such logarithms cautiously and loosely; what matters for the present discussion is that logarithms are much weaker than power divergences. When δ is very small, δ^{-2} dominates $\log \delta$.

Putting the structure together, we are led to the schematic relation

$$m_{\text{phys}}^2 \sim m_0^2 + \frac{c_1 \lambda}{\delta^2} + \frac{c_2 \lambda^2}{\delta^2} + c_3 g^2 \log(\delta \, \mu_{\text{ref}}) + \cdots . \tag{1.17}$$

The numerical coefficients and signs are not tracked in the lecture; the dominant message is that the scalar mass receives an infinite series of ultraviolet-sensitive corrections. And the same is true of the other couplings. The coefficient λ itself is renormalized by four-point diagrams, typically logarithmically, although Susskind stops short of turning that into a full study of beta functions.

1.4.1 Question & Answer

Question. What are we actually measuring: the input parameter or the renormalized one?

Answer. We measure the renormalized one. The point of the cutoff description is precisely that we refuse to resolve all of the very short-distance structure. Once we do that, every unresolved process that imitates a local two-point interaction is lumped into the coefficient we call the mass term. The laboratory therefore does not see the pristine m_0^2. It sees the effective coefficient after all the short-distance fluctuations we have chosen not to resolve have been summed into it.

1.5 From Scalar Renormalization to the Higgs Hierarchy Problem

At this point the lecture turns the scalar example into a crisis. For a scalar particle the additive corrections are present even if the bare mass vanishes. That is what makes the Higgs field dangerous. The lecture therefore asks us to recall the Higgs potential in its broken-symmetry form:

$$V(\phi) = -\frac{\mu^2}{2}\phi^2 + \lambda\phi^4. \tag{1.18}$$

Because the board normalization is schematic, the minimum is quoted in the lecture in the convention used there as

$$\phi_{\text{min}}^2 = f^2 = \frac{\mu^2}{\lambda}. \tag{1.19}$$

The key point is the scale, not the harmless convention-dependent factors. The vacuum expectation value is already known from electroweak physics:

$$f \sim 200\,\text{GeV}. \tag{1.20}$$

So unless λ is made implausibly tiny or huge, μ is also of electroweak order.

Now compare that with the ultraviolet-sensitive scalar corrections. If we trust quantum field theory all the way down to a scale as short as the Planck length, then in GeV units the lecture estimates

$$\frac{1}{\delta^2} \sim 10^{38}\,\mathrm{GeV}^2. \tag{1.21}$$

This is the point where the tone of the lecture changes. The scalar story is no longer just a technical example of renormalization. It becomes bad behavior. We want a Higgs parameter of electroweak size, but the short-distance contributions are incomparably larger. The precise number of decimal places shifts a bit in the spoken lecture, and the signs of the various contributions are left untracked, but the central point is stable: a huge collection of ultraviolet contributions must cancel to extraordinary precision in order to leave behind a weak-scale mass parameter.

That is the hierarchy problem, or fine-tuning problem, of the Higgs sector. It is not just that the weak scale is small compared with the Planck scale. It is that, in a scalar theory, the small physical parameter appears as the residue of a delicate cancellation among individually huge pieces.

1.5.1 Question & Answer

Question. Must the bare mass term cancel all the short-distance corrections?

Answer. If we insist on keeping a very short-distance cutoff while also insisting that the physical Higgs parameter remain at the electroweak scale, then yes: the bare term must conspire with the radiative corrections so that the large ultraviolet-sensitive pieces almost exactly cancel. That is what the lecture means by fine-tuning. The problem is not that a correction exists. The problem is that a very small measured answer seems to require a cancellation among many much larger terms.

1.6 Why Fermions Behave Better

The lecture then deliberately changes tempo and asks the obvious question: if scalar masses are so unstable, why is the electron not equally disastrous? The answer comes from the structure of the fermion mass term,

$$m\bar{\psi}\psi. \tag{1.22}$$

This term is linear in the fermion mass, not quadratic. More importantly, it flips chirality. It takes a left-chiral fermion into a right-chiral one, or vice versa. Susskind sometimes phrases this in helicity language in the lecture, but the mathematically safer language here is chirality.

The lecture now contrasts three kinds of vertices. Gauge-boson emission preserves chirality: $L \to L$ and $R \to R$. Scalar emission flips chirality: $L \leftrightarrow R$. The mass insertion $m\bar{\psi}\psi$ also flips chirality. The right-hand chalk sketches are mnemonic rather than literal Feynman graphs, but the underlying logic can be summarized schematically as follows.

Now suppose the electron begins massless. Can photon emission and reabsorption generate a mass? No. Every gauge interaction preserves chirality, so no chain of such processes can turn left into right. Can scalar emission and reabsorption do it? Again no. A scalar emission flips chirality, but the corresponding reabsorption flips it back. Since the emitted scalar must be reabsorbed, the total number of scalar vertices is even, and the net effect cannot be a left-right transition. So neither gauge-boson loops nor scalar loops generate an additive fermion mass from zero.

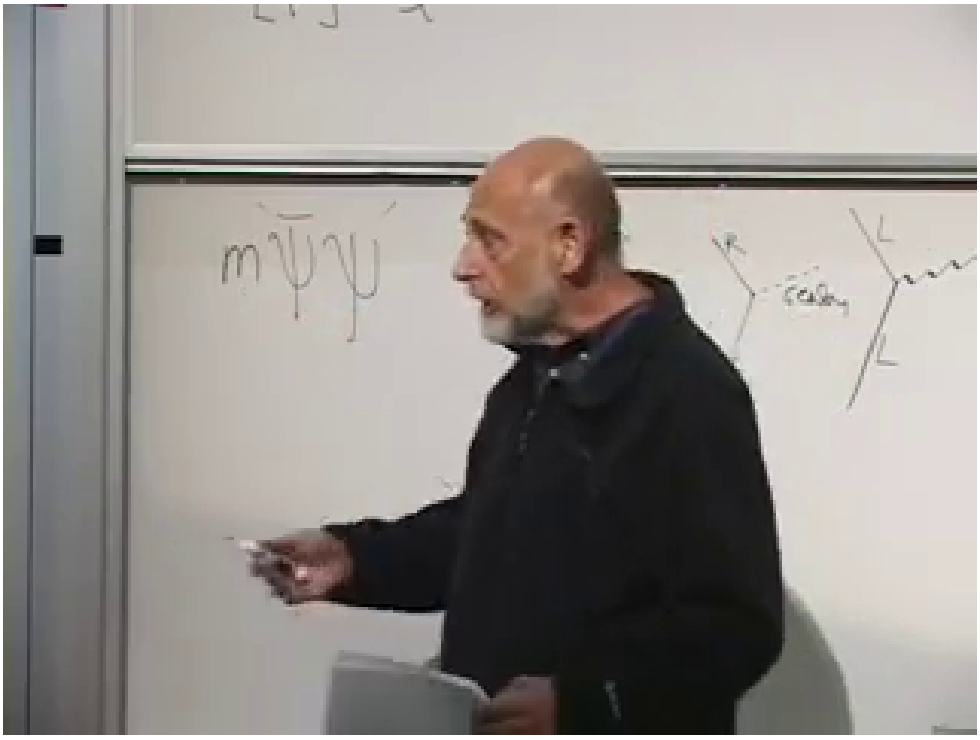

Figure 1.3: The fermion mass term on the board, together with the lecture's chirality mnemonics.

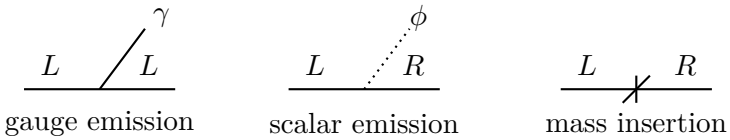

Figure 1.4: Schematic chirality mnemonics: gauge emission preserves chirality, while scalar emission and mass insertion flip it.

To get a correction to the fermion mass, one must already insert a mass somewhere in the diagram. Then the correction is proportional to the starting mass itself. That is the lecture's crucial point. Schematically,

$$m_e = m_0 + O(e^2 m_0) + O(e^4 m_0) + \cdots . \tag{1.23}$$

Therefore if $m_0 = 0$, the electron stays massless. If m_0 is small, all the corrections remain small. One may still ask why the electron mass is numerically small compared with some deeper scale, but one is not forced into the same kind of catastrophic cancellation that afflicted the scalar.

Remark 1.4. The lecture slides back and forth between chirality and helicity in a physically intuitive way. For the purpose of writing the argument cleanly, chirality is the better variable. In the relativistic limit the lecture's helicity-based intuition remains a useful picture.

Gauge bosons behave in the same protected way for the purpose at hand: if they begin massless, they remain massless to all orders. That is why the lecture insists that the Standard Model does not contain a separate fine-tuning problem for every light particle. The truly severe fine-tuning is the one tied to the Higgs field. If that one is solved, the rest of the light spectrum falls into place.

1.6.1 Question & Answer

Question. Why doesn't the electron have the same fine-tuning problem?

Answer. Because the electron mass is protected against additive renormalization. Starting from zero, neither gauge interactions nor scalar emission-and-reabsorption can generate the chirality-flipping two-point structure that defines a fermion mass. Any nonzero correction already contains one explicit mass insertion, and is therefore proportional to the original mass itself. A small fermion mass may remain a fact in need of explanation, but it is not produced by canceling huge ultraviolet contributions against each other.

1.7 Vacuum Energy and the Supersymmetry Outlook

The lecture closes by widening the frame once more. There is another quantity that renormalization shifts: the energy of the vacuum itself. At first one is tempted to say, with Susskind, "who cares?" In ordinary nongravitational physics only energy differences matter. If the vacuum energy is shifted by a constant amount, that seems irrelevant so long as adding a particle adds the right extra amount of energy.

But gravity does care. Empty space gravitates if it carries energy. So the renormalization of the vacuum energy becomes physically significant once the gravitational field is allowed to respond to it.

In ϕ^4 theory the simplest vacuum-to-vacuum diagram contains no external particles. It consists only of a quartic vertex together with short propagators that begin and end within the unresolved region. The lecture's estimate is then

$$\delta\rho_{\text{vac}} \sim \frac{\lambda}{\delta^4}. \tag{1.24}$$

More complicated vacuum diagrams typically preserve the same δ^{-4} behavior in this schematic discussion. So vacuum energy is also highly sensitive to short-distance physics.

The lecture therefore ends with two fine-tuning problems, not one. The Higgs mass is one. The vacuum energy is the other, and it is even more severe once gravity is included. Supersymmetry is introduced as a possible response to the first of these. It is not presented here as a cure for the cosmological-constant problem.

1.7.1 Question & Answer

Question. Does the fine-tuning depend on the arbitrary cutoff δ?

Answer. The contribution does depend on how far down in distance we trust the theory, but that is not an empty arbitrariness. We choose δ to be the shortest scale at which we still believe the present quantum field theory makes sense. Then the accumulated contribution from all scales above that point is already there and must be accounted for. In that sense the estimate is a lower bound on the size of the problem. If the theory remains valid to even shorter distances, the scalar fine-tuning can only become worse. The lecture mentions scales as small as unification, around 10^{16} GeV, as a conservative benchmark, with the Planck scale making the tension sharper still.

Two late clarifications from the lecture belong here. First, a scalar particle is a spin-zero particle; the Higgs is the central example in this discussion. Gauge bosons, by contrast, are spin-one particles. Second, although the simplest fermion and boson vacuum diagrams come with opposite signs, cancellation is not automatic. To cancel the vacuum energy naturally, masses and couplings would have to be matched with extraordinary precision. That is the kind of matching supersymmetry tries to impose in the Higgs sector, which is why it arrives here as the next subject of the course.

1.8 Summary

The lecture begins by teaching us how to think about renormalization before teaching us how to compute with it. We first integrate out fast electrons in a molecular problem and replace them by an effective potential for slow nuclei. We then carry the same idea into quantum field theory, where short-distance modes renormalize the coefficients in the Lagrangian. Dimensional analysis shows why scalar propagators are singular at short distance and why scalar mass terms are especially vulnerable to large additive corrections. That lesson becomes the Higgs hierarchy problem once the cutoff is taken seriously. Fermions and gauge bosons behave better because their masses are protected when they start from zero. Finally the lecture widens from the Higgs problem to the renormalization of vacuum energy, and ends by presenting supersymmetry not as a finished solution, but as the natural next attempt to tame the Higgs-sector fine-tuning.

Chapter 2

Supersymmetry and Grand Unification: Lecture 2

This lecture does not begin by defining supersymmetry. It does something more characteristic. It backs up. Before bosons and fermions can be related to one another, we are asked to remember what is already peculiar about fermions. Before that, we are asked to remember something peculiar about rotations themselves. The chapter should therefore be read in the same order in which the lecture unfolds: rotations, signs, exchange, loops, cancellations, broken supersymmetry, and only then the question of what an exactly supersymmetric world would look like.

The payoff is twofold. On the mathematical side, opposite signs for bosonic and fermionic loops create the possibility of cancellation. On the physical side, exact supersymmetry would radically alter ordinary matter. Between those two observations lies the lecture's real theme: supersymmetry is interesting precisely because exact supersymmetry is not the world we live in.

2.1 Rotations, 2π, and the Topological Prehistory of Supersymmetry

"Actually before we talk about supersymmetry," Susskind says, "we should talk again about fermions and bosons." Then he backs up once more: before fermions, rotations in space. The familiar statement that a rotation by 2π is the same as no rotation at all is, in the topological sense relevant here, too naive. He demonstrates the point physically with his arm, and then translates it into the box-and-ball picture.

We imagine a ball in a box, connected to the walls by strings. The walls represent spatial infinity, so the box is held fixed. A rotation of the ball by 2π tangles the strings, and the tangle cannot be removed if the ball is not rotated further. A rotation by 4π, however, can be undone by moving the strings around the ball. He jokingly says he has already proved the theorem "by hand-waving." That is enough for the lecture. The point is not a formal homotopy proof; the point is the qualitative fact that 2π is not topologically trivial, while 4π is.

This is the geometric memory that later reappears in fermions. In one of the discussion remarks Susskind even says, roughly, that bosons behave as though there were no strings attached to the wall, whereas fermions remember that a 2π rotation has taken place.

Now we freeze the position of a particle and look only at its spin state. Following the lecture, we may temporarily call that state $|S\rangle$; more generally we write $|\psi\rangle$. The simplest possibility is that a 2π rotation acts by multiplication with a number:

$$R(2\pi)\,|\psi\rangle = \xi\,|\psi\rangle\,. \tag{2.1}$$

If we perform the same rotation twice, then

$$R(4\pi)\,|\psi\rangle = \xi^2\,|\psi\rangle\,. \tag{2.2}$$

But the topological statement was precisely that 4π is equivalent to doing nothing. Therefore

$$R(4\pi)\,|\psi\rangle = |\psi\rangle\,, \qquad \xi^2 = 1, \qquad \xi = \pm 1. \tag{2.3}$$

So the state may come back to itself after a 2π rotation either with a plus sign or with a minus sign:

$$R(2\pi)\,|\psi_{\text{boson}}\rangle = +\,|\psi_{\text{boson}}\rangle\,, \qquad R(2\pi)\,|\psi_{\text{fermion}}\rangle = -\,|\psi_{\text{fermion}}\rangle\,. \tag{2.4}$$

That already separates bosons from fermions in a very stark way. The lecture emphasizes that such a rotation is not merely formal. One can implement it in the laboratory by letting a spin precess in a magnetic field, or by rotating the magnetic field itself. The mathematical distinction is therefore physically meaningful, if only we can find the right way to see it.

2.2 Spin States Under Rotation and the Interference Test of a Sign Change

At first sight a minus sign seems like no physics at all. If we rotate every copy of a wavefunction by 2π and the result is just $-\Psi$, then every observable constructed from $\Psi^*\Psi$ is unchanged. That is exactly the paradox the lecture wants to raise before it resolves it.

2.2.1 Question & Answer

Can the minus sign from a 2π rotation ever be observed?

Yes, but only as a relative sign.

The lecture answers with a two-path interference experiment. A single electron is prepared in a superposition of two branches, one going through an upper path and one through a lower path. If the two branches are ψ_1 and ψ_2, then before anything is rotated the total wavefunction is

$$\Psi = \psi_1 + \psi_2. \tag{2.5}$$

Now imagine that while the electron is trapped in one of the two regions, we rotate only one branch by 2π. For a fermion, that branch picks up a minus sign. If it is the second branch that is rotated, the recombined state becomes

$$\Psi' = \psi_1 - \psi_2. \tag{2.6}$$

So the physical question is not whether Ψ differs from $-\Psi$. The physical question is whether

$$\psi_1 + \psi_2 \qquad \text{or} \qquad \psi_1 - \psi_2 \tag{2.7}$$

emerges after recombination.

If the two returning wave packets are symmetric by reflection about the center of the screen, then the odd combination must vanish there:

$$\Psi_{\text{odd}}(0) = 0. \tag{2.8}$$

That is the cleanest way to state the lecture's argument. A relative minus sign shifts the interference pattern; maxima become minima, and the electron is never found at the symmetry point. On the other hand, if both branches are rotated together, the full state changes only by an overall sign, and then nothing observable happens:

$$|\Psi|^2 = (-\Psi)^*(-\Psi). \tag{2.9}$$

This is why the sign under 2π is real physics but only in a relational way. Rotating half the experiment relative to the other half is like rotating the ball relative to the box: memory can then be detected. Rotating the whole experiment together is like rotating ball and box together: the effect disappears. And if the relative rotation were 4π, the extra sign would disappear as well.

The lecture also makes a useful contrast here. The special role of 2π is that it changes only the sign of a fermion state while leaving the spin orientation effectively the same. A generic rotation changes the spin state more substantially. Then the two branches are no longer the same object up to a sign, and the interference pattern is partially washed out or even destroyed. A 180° flip, for example, destroys it completely.

2.3 Bosons, Fermions, and Exchange Symmetry

Having extracted one distinction between bosons and fermions from rotations, the lecture turns to a second one: exchange symmetry. This shift is not accidental. The next target is a sign rule for Feynman loops, and that rule depends on what happens when identical particles are interchanged.

Let $\Psi(1,2)$ be the two-particle wavefunction, and let P_{12} denote the operator that exchanges particles 1 and 2. Then the standard statements are

$$P_{12}\Psi(1,2) = +\Psi(1,2), \qquad \Psi(1,2) = +\Psi(2,1) \qquad \text{for bosons,} \tag{2.10}$$

$$P_{12}\Psi(1,2) = -\Psi(1,2), \qquad \Psi(1,2) = -\Psi(2,1) \qquad \text{for fermions.} \tag{2.11}$$

The lecture phrases this more directly: bosonic wavefunctions are symmetric, fermionic wavefunctions are anti-symmetric.

The board evidence is especially useful here because Susskind is speaking in ket language at this moment, not only in coordinate language.

The only symbol that is really clear on the board is $|S\rangle$. So the cleaned equations have to be treated as careful reconstructions rather than verbatim board transcriptions. The mathematically appropriate statements are

$$\Psi(1,2) = -\Psi(2,1), \tag{2.12}$$

$$P_{12}\,|\psi\rangle = -\,|\psi\rangle\,. \tag{2.13}$$

The second line is the lecture's ket-language version of the same fact. It is this minus sign under exchange, not some abstract slogan about statistics, that is about to reappear in the sign of a closed fermion loop.

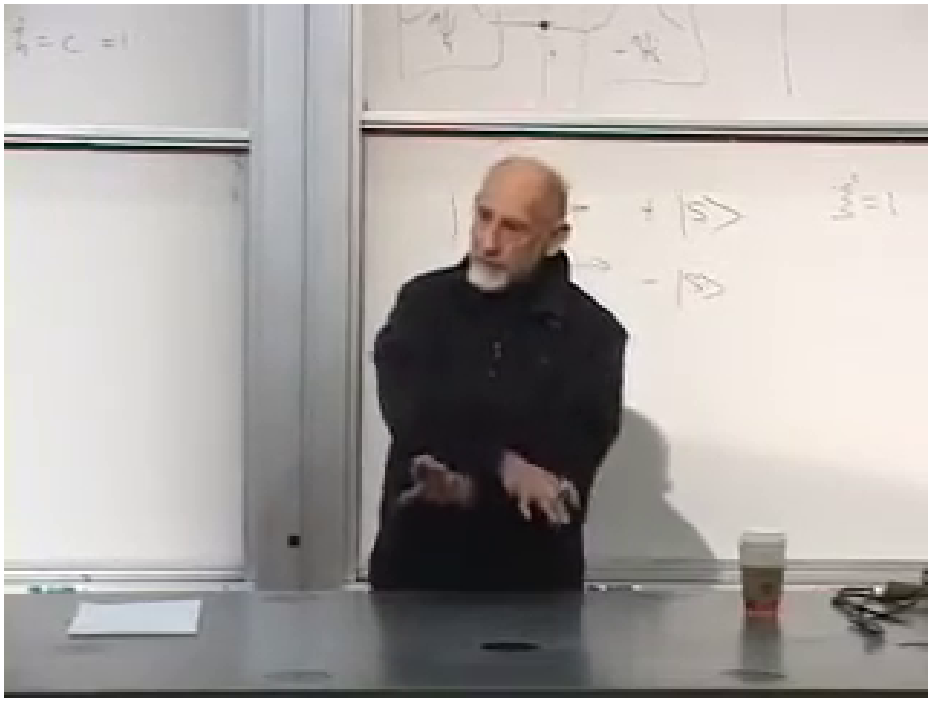

Figure 2.1: Fermion anti-symmetry and state-label sketch. The screenshot is kept because it preserves the board layout and the ket-style presentation, even though only $|S\rangle$ is securely legible.

2.4 Why a Closed Fermion Loop Carries a Minus Sign

We now come to what Susskind calls "yet another question": a question about very simple Feynman diagrams. These are vacuum diagrams with no external lines, just a closed loop in empty space. One may think of the loop either as a particle moving around a closed curve in spacetime or, more physically, as pair creation followed by annihilation. The precise integral associated with the diagram is not the point here. The point is the sign.

2.4.1 Question & Answer

Why does one closed fermion loop come with a minus sign?

The lecture gives an elegant proof that uses nothing heavier than exchange symmetry.

Begin with two disconnected loops. Since disconnected diagrams multiply, the sign of the two-loop diagram is the square of the sign of the one-loop diagram:

$$\text{sign}(\text{two disconnected loops}) = \big(\text{sign}(\text{one loop})\big)^2 = +1. \tag{2.14}$$

So two disconnected loops always carry an overall plus sign, regardless of whether one loop by itself is positive or negative.

Now cut both loops at a common time slice. At that instant we see two identical particles. There are then two possibilities: reconnect them without exchanging the particles, or interchange the two particles before reconnecting them. The first operation gives back the original two disconnected loops. The second changes the amplitude by the exchange sign:

1. For bosons, exchanging the two particles contributes no extra sign.
2. For fermions, exchanging the two particles contributes a minus sign.

But after this interchange and reconnection, what had been two loops has become one single loop. Therefore the sign of the one-loop diagram must be

$$\text{sign(one closed boson loop)} = +1, \qquad \text{sign(one closed fermion loop)} = -1. \tag{2.15}$$

That is the whole proof. It is not a path-integral derivation, and the lecture does not want one here. It wants the elementary mechanism to be visible: the extra minus sign comes from exchanging identical fermions. The consequences are immediate. Bosonic and fermionic loops contribute with opposite signs to quantum corrections. We have finally reached the first real mathematical doorway into supersymmetry.

2.5 Vacuum Energy, Boson Self-Energy, and the Zeroth-Order Supersymmetry Idea

The first place Susskind uses the loop-sign result is the vacuum energy. A closed boson loop contributes with one sign; a closed fermion loop contributes with the opposite sign. Since the observed vacuum energy is astonishingly small compared with the naive quantum field theoretic estimate, the lecture pauses over the fact that cancellation is at least possible.

All things being equal, meaning the same mass and the same relevant quantum numbers, a boson and a fermion would give equal contributions in magnitude and opposite contributions in sign. Schematically,

$$m_{\text{partner}} = m_{\text{particle}}, \qquad Q_{\text{partner}} = Q_{\text{particle}}, \tag{2.16}$$

and then one may hope for

$$\Delta E_{\text{vac}}^{(b)} + \Delta E_{\text{vac}}^{(f)} = 0. \tag{2.17}$$

The lecture is careful not to oversell this. It does not say that the cosmological constant problem has been solved. It says only that the sign structure creates room for cancellation, which is already something important.

Then comes one of Susskind's characteristic pivots: "another thought to keep in mind." Vacuum energy is not the main issue. The sharper question is whether the same logic can help with the self-energy of bosons, especially Higgs-like scalars. Bosonic loop corrections by themselves drive scalar masses upward. Fermions, by contrast, can be protected against such enormous corrections. So if a theory forced bosons and fermions to travel in closely matched pairs, one could imagine transferring that protection to the bosons. In schematic form,

$$\Delta m_B^2\big|_{\text{boson loop}} + \Delta m_B^2\big|_{\text{fermion loop}} \approx 0. \tag{2.18}$$

Exact partner matching would give exact cancellation. Near matching would give only approximate cancellation, but approximate cancellation is enough if the goal is merely to keep the Higgs sector from running away to enormous scales.

After the break, the lecture rephrases the whole idea "to zeroth order," not in the strict mathematical sense of an expansion, but as a rough first picture. The rough picture is that every particle in nature comes with a superpartner of the opposite statistics. Here "statistics" is exactly Susskind's word for

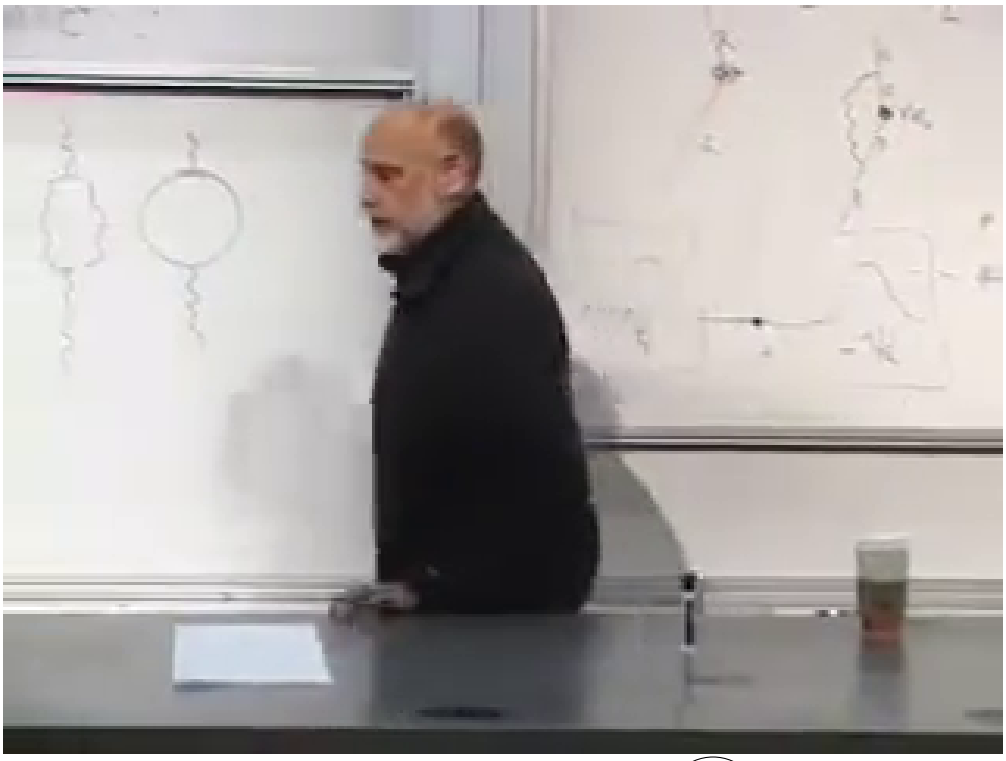

Figure 2.2: Loop-diagram comparison in the supersymmetry discussion. The redraw is intentionally schematic: the left-hand side shows simple loop topologies, while the right-hand side suggests loop corrections to propagating particles.

fermion-ness versus boson-ness. Supersymmetry, at this stage of the lecture, is the hypothesis that the spectrum is doubled in just this way.

The board at this point is valuable mainly as layout evidence: simple loops on the left, more elaborate loop-corrected structures on the right.

The lecture still has not defined the supersymmetry algebra. It is doing something more primitive and, pedagogically, more effective. It is giving us a reason to want such a symmetry in the first place.

2.6 Broken Supersymmetry, Naturalness, and Experimental Reach

At this point the lecture turns from motivation to reality. Exact mass matching would give exact cancellation, but exact mass matching is not how nature looks. We know this at once from experiment. If there were a bosonic partner of the electron with the same charge and the same mass, it would already have shown up in ordinary production processes. An electron-positron pair can annihilate into a virtual photon, and if the partner were truly degenerate with the electron, the

virtual photon could produce the partner pair just as easily:

$$e^-e^+ \to \gamma^* \to \tilde{e}^-\tilde{e}^+. \tag{2.19}$$

We do not see that. So exact supersymmetry is not the world we live in.

2.6.1 Question & Answer

If supersymmetry helps, why have we not seen the partners?

Because the symmetry, if it exists in nature, must be broken.

The lecture insists on two points at once. First, the masses of particles and superpartners are certainly not equal. Second, the splittings cannot be arbitrarily large if supersymmetry is supposed to do the job for which it was invented. Exact degeneracy would produce exact cancellation. Near degeneracy gives approximate cancellation. But if the partner masses were pushed all the way up to some enormous scale, then the cancellations would be poor and the Higgs self-energy problem would simply return in a slightly disguised form.

This is the naturalness argument in the form the lecture wants it. If superpartners were as heavy as the unification scale or the Planck scale, their loops would drag the Higgs sector upward almost as badly as if there were no supersymmetry at all. So a supersymmetry that protects the Higgs mass requires that the particle-superpartner splittings not be too large. Susskind gives a lecture-era estimate of order a few hundred GeV as the scale beyond which the cancellation becomes too poor to be convincing. The exact number should be taken historically rather than timelessly; the logical point is the important one.

The lecture also rejects a merely aggregate cancellation, where the sum of all fermionic effects accidentally cancels the sum of all bosonic effects. That would be just another fine-tuning. The supersymmetric hope is a structured cancellation, enforced by a symmetry, not a miraculous bookkeeping accident.

Here the line-style contrast between ordinary loops and partner loops becomes visually explicit.

The lecture then sharpens the matching requirement. It is not enough that masses be comparable. Couplings must match as well. Once external photons are attached to the loop, cancellation requires that the partner carry the same electric charge as the original particle:

$$Q_{\text{partner}} = Q_{\text{particle}}. \tag{2.20}$$

Otherwise the two diagrams do not have the same numerical coefficient and the cancellation fails.

Experimentally, there are then two fronts. One may look directly for superpartner production in sufficiently energetic collisions. Or one may look indirectly for their virtual effects in high-precision electroweak processes. The lecture emphasizes that electron-positron machines are cleaner for the second purpose than hadron colliders, because protons are composite and the collision environment is correspondingly messy. The high-precision data do constrain supersymmetry, but in the lecture's discussion they do not yet rule it out.

Susskind also remarks that most superpartners would not be expected to sit around forever. They should decay, perhaps cascading down to a lightest superpartner. In many models that lightest state is neutral and may be stable. But the main phenomenological message of the lecture is simpler: if supersymmetry is to protect the Higgs sector, the new particles must lie not too far beyond experimental reach.

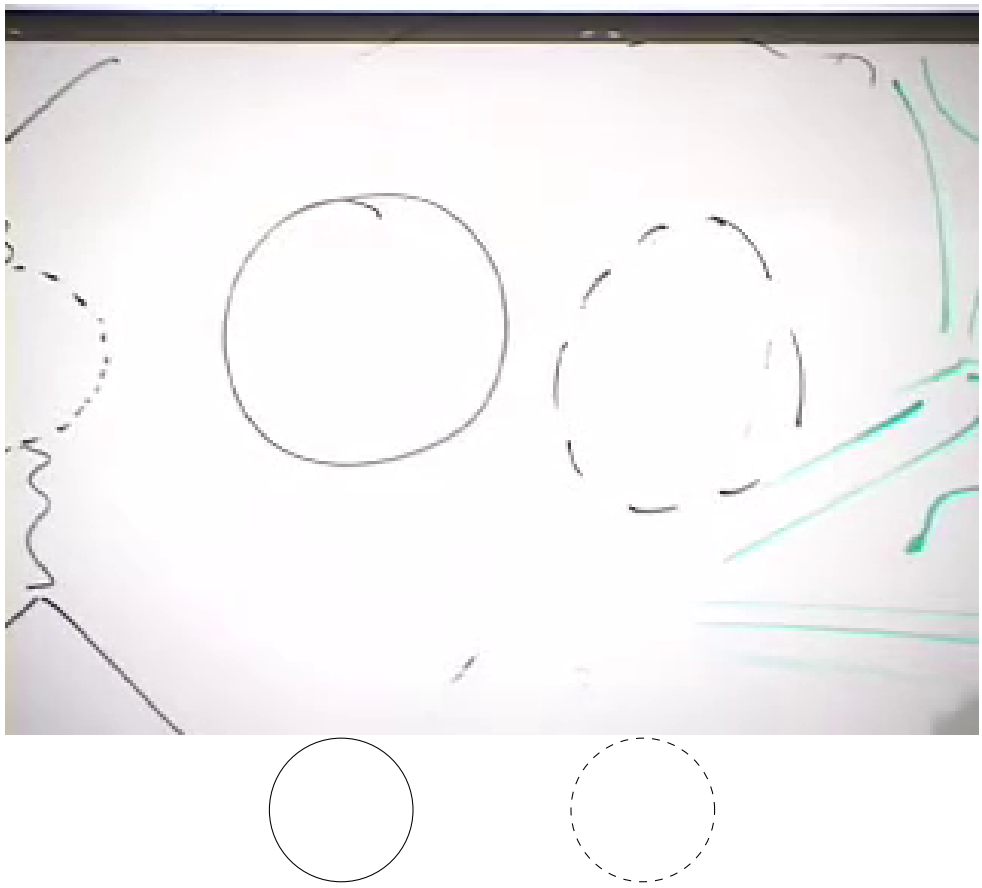

Figure 2.3: Solid and dashed loop topologies. The essential content is the side-by-side comparison of ordinary and superpartner loop contributions with different statistics and therefore opposite signs.

2.7 Exact Supersymmetry Would Destroy Ordinary Chemistry

Before closing, the lecture fixes the relation between the spins of particles and their superpartners. Bosons are paired with fermions, fermions with bosons, and the spin changes by half a unit:

$$j_{\text{superpartner}} = j \pm \frac{1}{2}. \tag{2.21}$$

Thus a gauge boson has a spin-$\frac{1}{2}$ partner, while an ordinary spin-$\frac{1}{2}$ fermion such as the electron has a scalar partner. This is also where the familiar names appear. Bosonic particles acquire fermionic partners with "-ino" names such as photino and Higgsino, whereas fermions acquire bosonic partners such as the selectron and the squark. The lecture also notes that extra scalar fields raise the danger of additional spontaneous symmetry breaking, which would generally be unwelcome and must be controlled.

There is a brief aside about lower dimensions: in two spatial dimensions the boson-fermion dichotomy is generalized, and one finds anyons with more general spin-statistics behavior. The lecture does not pursue that topic further. Instead it asks the question with which it really wants to end.

2.7.1 Question & Answer

What would the world be like if supersymmetry were exact?

Very different. In particular, ordinary chemistry would fail.

Here "exact" means exact degeneracy between particles and superpartners. If that were true, the electron would have a bosonic partner with exactly the same mass and charge, and the photon would have a massless fermionic partner, the photino:

$$m_\gamma = 0 \qquad \Rightarrow \qquad m_{\tilde{\gamma}} = 0. \tag{2.22}$$

Now consider an atom with more than two electrons. In hydrogen and helium the available electrons occupy the lowest orbit. In lithium and heavier atoms, outer electrons sit farther out because the low-lying states are blocked by the Pauli exclusion principle. The outer electron wants to fall inward, but ordinary Fermi statistics forbids it.

In an exactly supersymmetric world there is a new escape route. The outer electron can emit a photino and turn into a selectron:

$$e^-_{\text{outer}} \to \tilde{e}^- + \tilde{\gamma}. \tag{2.23}$$

That is the key move in the lecture's final argument. The emitted particle is a fermion, so what remains must be a boson. Once the electron has turned into a selectron, Pauli blocking no longer applies. The selectron can drop to the lowest orbit.

The force of the argument is cumulative. It is not only one outer electron in a lithium-like atom that would do this. Valence electrons more generally would have the same route available to them. They would convert into bosonic selectrons and fall inward. Instead of ordinary shell structure, one would have something closer to a Bose condensate of selectrons gathered near the nucleus. Whatever such a world might be, it would not support familiar atoms, familiar valence structure, or familiar chemistry.

The lecture even anticipates the obvious objection: perhaps the coupling for this transition is tiny. But here there is no escape of that sort. The relevant coupling is electromagnetic. In the

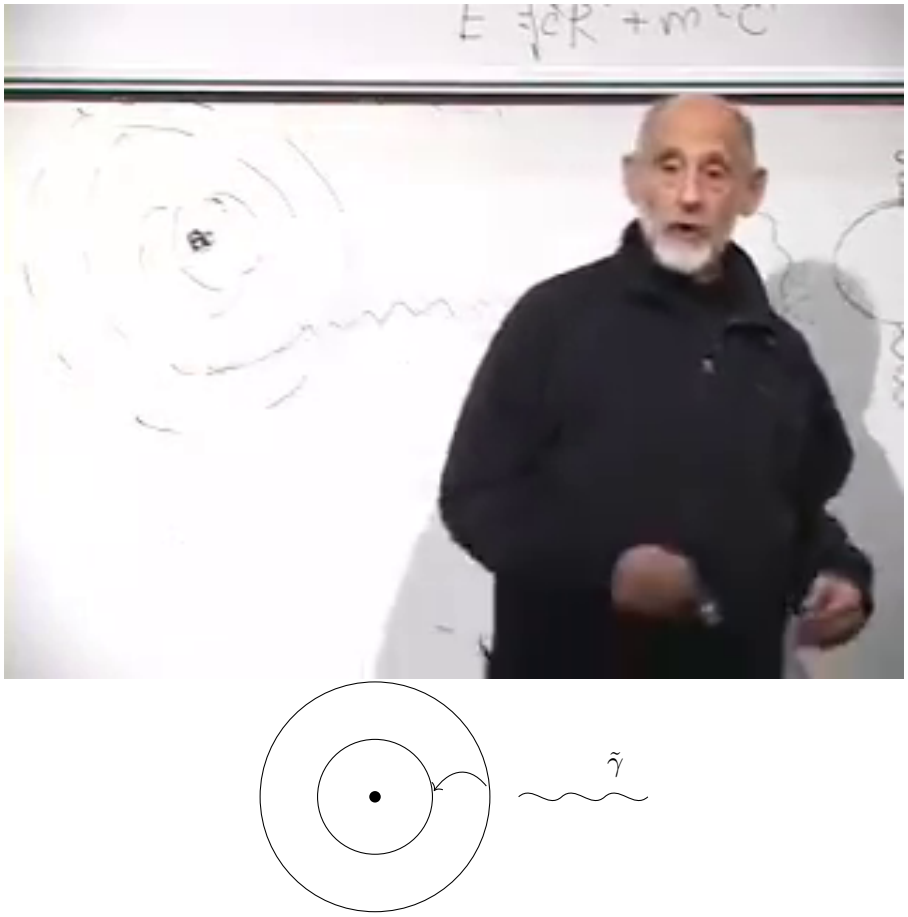

Figure 2.4: Radiative drop to a lower orbit. The board shows emitted radiation and an inward transition; the photino label in the redraw is supplied from the transcript rather than from a legible board inscription.

lecture's rough estimate, a lithium-like atom would not linger in its ordinary form for long at all; the transition would be rapid on atomic timescales.

So the closing lesson is not merely that exact supersymmetry is experimentally excluded. It is that exact supersymmetry would make a world unlike ours. Susskind's answer to the question "why is the symmetry broken?" is correspondingly blunt: if it were not broken, we would not be here to ask. That is not a complete theory of supersymmetry breaking, but it is the right final physical punch.

2.8 Summary

The lecture reaches supersymmetry by refusing to define it too early. First we learn that a 2π rotation is not topologically as innocent as it seems, whereas a 4π rotation is. Then we learn that a spin state may therefore come back under 2π with a sign, and that fermions pick up -1 while bosons do not. Then the sign is made observable through interference, where relative minus signs shift the pattern even though global ones do not.

Only after that does the lecture turn to exchange symmetry and then to Feynman loops. Antisymmetry under exchange implies that a single closed fermion loop comes with a minus sign, whereas a boson loop comes with a plus sign. From there the supersymmetric idea becomes natural: if bosons and fermions occur in sufficiently matched pairs, their quantum corrections can cancel. That possibility first appears in the vacuum energy and then, more importantly for the lecture, in the self-energy of bosons.

Nature, however, does not realize the exact version of the idea. The partners are not degenerate with the known particles, so supersymmetry, if it exists, must be broken. But it cannot be broken too badly if it is to protect the Higgs sector from enormous radiative corrections. And if it were not broken at all, ordinary chemistry would collapse. That is the balance the lecture leaves us with: supersymmetry is interesting because it can cancel dangerous quantum effects, and it must be broken because an exactly supersymmetric world would not look anything like ours.

Chapter 3

Divergences, Propagators, and the Supersymmetric Motive

This lecture was supposed to begin supersymmetry itself. Instead, it does something more deliberate: it backs up and rebuilds the need for supersymmetry from the problem of ultraviolet divergences, especially the renormalization of masses. That is the right order. Before we talk about a powerful new symmetry, we should understand what sort of damage needs repairing. The chapter therefore unfolds in the same order as the lecture: first effective Lagrangians and propagators, then the short-distance behavior of scalar and fermion fields, then the simplest divergent loop corrections, and only at the end the first real glimpse of what supersymmetry is supposed to accomplish.

3.1 Why We Are Still Talking About Divergences

A great deal of particle physics is the computation of Feynman diagrams. To say that, however, is not yet to say what Feynman diagrams are for. The lecture begins by reminding us that they serve at least two rather different purposes.

First, they compute scattering amplitudes. Particles come in, particles go out, and we sum the diagrams that contribute to the probability amplitude. Second, and for the present lecture more importantly, they help us construct an effective Lagrangian. In that setting the point is not to write the exact microscopic theory in a different notation. The point is to write an approximate theory that is effective because it has been stripped of short-distance complications that we do not yet know how to calculate.

That is why a cutoff enters almost immediately. An effective Lagrangian is typically accompanied by either

- a smallest distance, or
- a largest momentum.

The lecture keeps both descriptions alive from the start, because the same ultraviolet problem will soon be read in both languages.

3.2 Propagators in Position Space and Momentum Space

The propagator is the bridge between those two descriptions. If a particle is inserted at one spacetime point and later detected at another, the corresponding amplitude depends on the separation between the two points. To keep the notation unambiguous on the page, we shall write

$$\Delta^\mu = y^\mu - x^\mu \tag{3.1}$$

for the spacetime separation, reserving δ for the ultraviolet cutoff length.

For a scalar field, the propagator is used in the lecture schematically as both an amplitude from x to y and as a vacuum expectation value:

$$G_\phi(x, y) \sim \langle 0|\phi(x)\phi(y)|0\rangle. \tag{3.2}$$

The exact time-ordering conventions are not the point here. What matters is that the propagator depends on x and y only through their difference.

Now the lecture makes a move that is conceptually simple but structurally crucial. A closed loop may be regarded in either of two ways:

- as a propagator from a point back to itself;
- as an integral over all momentum that can circulate around the loop.

That is the same diagram seen from position space and momentum space. Once one sees this, the later discussion of divergences becomes almost inevitable.

3.2.1 Question & Answer

Question. Why are short-distance divergences the same thing as high-momentum divergences?

Answer. Because momentum is inverse wavelength. Very large momentum means very short wavelength, and very short wavelength means very fine spatial resolution. Thus a singularity when $\Delta \to 0$ in position space is the same ultraviolet phenomenon as a divergence from integrating over arbitrarily large loop momentum in momentum space. The lecture insists on this early because every later estimate can be translated back and forth between the two pictures.

3.3 Scalar Fields, Dimensional Analysis, and the Massless Propagator

Having set up the general language, the lecture narrows to the first case that matters physically here: scalar fields. There are not many fundamental scalars in particle physics, but the Higgs boson is enough to make the problem urgent.

The argument begins in a very characteristic way: before computing anything in detail, let us ask what dimensional analysis already forces on us. For a scalar field ϕ, the simplest action is

$$S_\phi \sim \int d^4x\, (\partial_\mu \phi)^2. \tag{3.3}$$

The exact normalization is not important for the present argument. What matters is that in units with $\hbar = c = 1$, the action is dimensionless. Therefore

$$1 \sim [S_\phi] \sim [d^4x]\,[\partial_\mu \phi]^2 \tag{3.4}$$

$$\sim L^4 \left(L^{-1}[\phi]\right)^2. \tag{3.5}$$

Hence

$$[\phi] = L^{-1}. \tag{3.6}$$

The lecture then checks the other familiar terms one by one. If we add a mass term $m^2\phi^2$, dimensional consistency requires

$$[m] = L^{-1}. \tag{3.7}$$

If we add a quartic interaction $\lambda\phi^4$, then

$$[\lambda] = L^0, \tag{3.8}$$

so the quartic scalar coupling is dimensionless in four dimensions.

Now let us return to the propagator. In the massless case there is no scale in the problem other than the separation Δ. Translation invariance says that the propagator depends only on $x - y$, Lorentz invariance says that it depends only on the invariant interval, and dimensional analysis says that it must carry the dimensions of two scalar fields. There is therefore essentially no choice:

$$G_\phi(\Delta) \propto \frac{1}{\Delta^2}, \qquad \Delta^2 = \Delta_\mu \Delta^\mu. \tag{3.9}$$

Numerical factors are present, but no new scale is.

The lecture then asks what happens when the field has a mass. The point is not to write the exact massive propagator. The point is to understand its short-distance behavior. For this the dispersion relation is enough:

$$E = \sqrt{p^2 + m^2}, \qquad E = |p| \quad \text{for a massless particle.} \tag{3.10}$$

When $|p| \gg m$, the massive expression forgets the mass and approaches the massless one. Large momentum means short distance, so the short-distance propagator must likewise forget the mass.

At larger separations the mass suppresses propagation. The lecture writes this schematically as an exponential cutoff, so one may think of the massive scalar propagator as behaving roughly like

$$G_{\phi,m}(\Delta) \approx \frac{1}{\Delta^2} e^{-m|\Delta|}. \tag{3.11}$$

This should be read as a guide, not an exact formula. Its purpose is to make one point clear: at large separation the mass matters, but near $\Delta = 0$ the singularity is still the same $1/\Delta^2$.

3.4 The Simplest Scalar Self-Energy Diagram and the Fine-Tuning Problem

Now the lecture cashes out all of this preparatory work. We turn from the abstract question "what does a propagator look like?" to the concrete question "what does the simplest loop correction do to a scalar mass?"

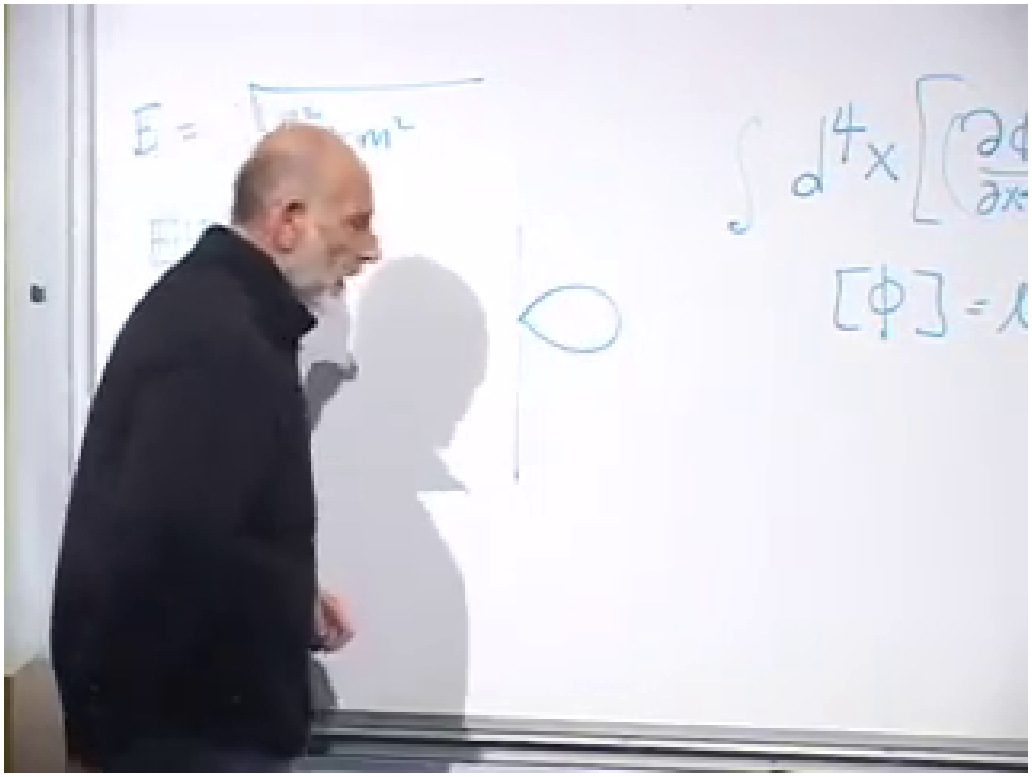

Figure 3.1: Blackboard moment at the transition from propagators and dimensional analysis to the simplest loop correction. The visible kinematic formulas and the partly visible action term are part of the same chain of reasoning as the loop diagram in the middle.

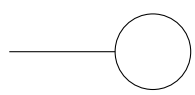

Figure 3.2: Minimal redraw of the central loop correction. The exact geometry on the board is slightly ambiguous, so the schematic form is kept deliberately spare.

The preserved blackboard frame is useful here because it shows the transition exactly as it happened in the lecture: the earlier discussion has not yet disappeared. One still sees the relativistic energy formula, its massless limit, part of the scalar-field action, and now in the middle the simplest loop correction.

Cleaned up, and with the partly visible action term written in the standard form suggested by the immediately preceding discussion, the board content may be summarized schematically as

$$E = \sqrt{p^2 + m^2}, \qquad E = |p|, \qquad S_\phi \sim \int d^4x\, (\partial_\mu \phi)^2, \qquad [\phi] = L^{-1}. \tag{3.12}$$

So the loop diagram is not appearing from nowhere. It is the direct consequence of what we have just learned about short-distance propagation.

In a $\lambda\phi^4$ theory this diagram is a correction to the scalar mass term. The lecture reads it precisely that way: it modifies the amplitude for a scalar particle to be absorbed at a point and re-emitted from that same point, which is exactly what the $m^2\phi^2$ term already does. Its size is therefore of

order

$$\delta m_\phi^2 \sim \lambda\, G_\phi(0). \tag{3.13}$$

But the short-distance form of the propagator is $G_\phi(\Delta) \sim 1/\Delta^2$. At coincident points that is singular. If we now admit what the lecture keeps insisting on—that our theory should not be trusted down to arbitrarily small distances—then we replace the ill-defined coincidence limit by a shortest-distance cutoff δ. The estimate becomes

$$\delta m_\phi^2 \sim \frac{\lambda}{\delta^2}. \tag{3.14}$$

This is the quadratic divergence.

3.4.1 Question & Answer

Question. Why does the simplest scalar loop correction behave like λ/δ^2, and why is that such a serious problem?

Answer. The diagram contributes one factor of the quartic coupling λ, and the rest of the diagram is just a scalar propagator evaluated at zero separation. Since the propagator behaves like $1/\Delta^2$ at short distance, the only cutoff-regulated estimate is $1/\delta^2$. The seriousness lies in the scale. If δ is extremely small, the correction is extremely large, typically much larger than the scalar mass one hoped to explain. A small low-energy scalar mass must then come from a delicate cancellation between a bare parameter and a huge ultraviolet correction. That is the fine-tuning problem.

The lecture then says what one should not miss: this is the Higgs problem, but it is first a scalar-field problem. The Higgs boson is simply the physically urgent scalar.

3.5 Fermions: Dirac Lagrangian, Field Dimension, and Short-Distance Propagator

Only now, after the scalar divergence has been made unmistakable, does the lecture pivot to fermions. The logic is parallel to the scalar case, and the parallelism is part of the point.

The Dirac action is written schematically as

$$S_\psi \sim \int d^4x\, \bar\psi\, \gamma^\mu \partial_\mu \psi, \tag{3.15}$$

with the mass term

$$\int d^4x\, m\, \bar\psi\psi. \tag{3.16}$$

The gamma matrices carry no dimensions, and there is only one derivative. So the kinetic term gives

$$1 \sim [S_\psi] \sim [d^4x]\,[\bar\psi]\,[\partial_\mu]\,[\psi] \tag{3.17}$$

$$\sim L^4 [\psi]^2 L^{-1}. \tag{3.18}$$

Therefore

$$[\psi] = L^{-3/2}. \tag{3.19}$$

This point is stated almost playfully in the lecture: the half-integer power looks odd, but fermions are full of halves. A student remarks that the mass term would furnish the same result more quickly. Susskind agrees, but keeps the kinetic term in front because it displays the structure more transparently. One may then check from the mass term that

$$[d^4x]\,[m]\,[\bar{\psi}\psi] \sim L^4 \cdot L^{-1} \cdot L^{-3} \sim 1, \tag{3.20}$$

so m really does have the dimension of mass.

Now repeat the propagator argument. The fermion propagator has the dimensions of two fermion fields, so in the massless case

$$S_\psi(\Delta) \sim \frac{1}{\Delta^3}. \tag{3.21}$$

This is the lecture's preferred shorthand for dimensional reasoning. If one restores the spinor structure that the lecture temporarily suppresses, then one inserts a gamma matrix and one power of the separation vector upstairs:

$$S_{ij}(\Delta) \propto \frac{(\gamma_\mu \Delta^\mu)_{ij}}{\Delta^4}. \tag{3.22}$$

That more structured form is useful as a consistency check, but for the ultraviolet counting the compact rule $1/\Delta^3$ is enough.

The lecture then makes exactly the same ultraviolet point as in the scalar case: a nonzero fermion mass suppresses propagation at large distance, but near $\Delta = 0$ the short-distance behavior is governed by the same massless estimate.

3.6 Closed Fermion Loops, Yukawa Couplings, and the Supersymmetry Motive

Now we come to the real turning point of the lecture. We have a bad scalar loop. Can a fermionic diagram cancel it?

Before estimating any such diagram, the lecture first settles the sign. This is not a secondary technicality. It is the first reason to think a cancellation could even exist.

3.6.1 Question & Answer

Question. Why does a closed fermion loop come with a minus sign, and how can that possibly help with the scalar divergence?

Answer. The lecture gives a neat exchange argument. Begin with two disconnected identical loops. Their product is positive no matter what sign each one-loop factor has. Now cut and reconnect them by exchanging the two identical particles. For bosons, exchange does nothing to the sign because the two-particle wavefunction is symmetric:

$$\Psi_B(x_1, x_2) = +\Psi_B(x_2, x_1). \tag{3.23}$$

For fermions, exchange produces a minus sign because the wavefunction is antisymmetric:

$$\Psi_F(x_1, x_2) = -\Psi_F(x_2, x_1). \tag{3.24}$$

But after the exchange, the reconnected object is just a one-loop diagram. Therefore the one-loop fermion diagram has the opposite sign from the corresponding bosonic loop. That still does not solve the problem. But it opens the only door through which the problem might be solved: a fermion loop can carry the same ultraviolet scaling with the opposite sign.

The simplest scalar–fermion coupling is a Yukawa term,

$$g\,\phi\,\bar{\psi}\psi. \tag{3.25}$$

Its dimension is fixed just as before:

$$1 \sim [d^4x]\,[g]\,[\phi]\,[\bar{\psi}\psi] \tag{3.26}$$

$$\sim L^4[g]L^{-1}L^{-3}, \tag{3.27}$$

so

$$[g] = L^0. \tag{3.28}$$

The Yukawa coupling is dimensionless.

Now estimate the loop correction. The scalar is absorbed at one point and re-emitted at another. There are two Yukawa vertices, hence a factor g^2, and two fermion propagators, each of order $1/\Delta^3$. Holding one point fixed and integrating over the separation of the other, the lecture arrives at

$$\mathcal{A}_f \sim -g^2 \int d^4\Delta\, \frac{1}{\Delta^6}. \tag{3.29}$$

The integral contributes four powers of length upstairs and six downstairs, so by dimensional analysis

$$\mathcal{A}_f \sim -\frac{g^2}{\delta^2}. \tag{3.30}$$

The integrand itself is positive; the overall minus sign is the closed-fermion-loop sign.

At this point the lecture makes an important sober observation. It does not yet look as though we have solved anything. We still have a number multiplying $1/\delta^2$. But now, at minimum, we have found a possible source of a minus sign. The scalar loop contributes

$$+\frac{\lambda}{\delta^2}, \tag{3.31}$$

while the fermion loop contributes

$$-\frac{g^2}{\delta^2}. \tag{3.32}$$

So a cancellation of the dangerous divergence becomes imaginable.

But it would be a miracle if nature simply chose the two couplings to satisfy

$$g^2 = \lambda \tag{3.33}$$

for no reason at all. And even that is not quite enough. For exact cancellation one would want the bosonic and fermionic propagators to be matched closely enough in the ultraviolet, which in practice means matching masses. If the masses differ, the full answers need not cancel exactly; but as the lecture emphasizes, the divergent part can still cancel provided the short-distance behavior remains sufficiently close.

Now we can finally say why supersymmetry enters. Supersymmetry is not yet introduced here as an abstract algebraic game. It is introduced as precisely the sort of mathematical structure one would want if one hoped to relate bosonic and fermionic couplings and masses strongly enough to remove the nasty scalar divergences. The formal mathematics is postponed to the next lecture. The motive has now been established.

3.7 Summary

The lecture begins by refusing to rush into supersymmetry. Instead it reopens the ultraviolet problem that makes supersymmetry worth wanting in the first place. We first recast loop divergences in both position-space and momentum-space language, then use dimensional analysis to determine the short-distance propagators for scalar and fermion fields. From there the central scalar self-energy estimate follows almost immediately: $\delta m_\phi^2 \sim \lambda/\delta^2$. That is the fine-tuning problem in its sharpest form.

Only after the problem is clear do fermions enter as possible rescuers. Their propagator scales as $1/\Delta^3$, a closed fermion loop comes with a minus sign, and the simplest Yukawa loop contributes $-g^2/\delta^2$. Thus the lecture does not yet give us supersymmetry, but it does give us the reason to seek it: a symmetry capable of tying bosons and fermions together so tightly that the dangerous ultraviolet sensitivity of scalar masses is no longer a numerical accident.

Chapter 4

Symmetry, Supersymmetry, and Grassmann Numbers

This lecture begins by stepping back before it goes forward. Before supersymmetry can be introduced as a new and very strange symmetry, we are first reminded what an ordinary symmetry is, how it is represented in quantum mechanics, how infinitesimal generators encode the structure of a continuous group, and why a genuine symmetry must leave the energy unchanged. Only after that long preparation does the lecture make its sharp turn: every symmetry discussed so far preserves bosons as bosons and fermions as fermions, whereas supersymmetry mixes the two. These companion notes follow Leonard Susskind's Lecture 4 closely and are curated by LazyingArt LLC.

4.1 Symmetries As Operations On States

A symmetry relates things. That is the lecture's starting point, and it is worth keeping it in exactly that elementary form. Two configurations may look different, but if one is obtained from the other by a symmetry transformation, then the symmetry ties them together and forces equalities between their physical properties. In particular, states related by symmetry generally have the same mass or the same energy.

This is why the lecture begins with an almost homely example: left-right symmetry. The point is not the example itself. The point is that symmetry is not a slogan about invariance; it is a relation among transformed configurations. Once that reminder is in place, the lecture moves immediately to the mathematics.

Quantum mechanically, a symmetry is represented by an operator acting on a state vector. We begin with a state $|\psi\rangle$, whatever physical system it may describe, and we apply a transformation U. The result is another state,

$$U|\psi\rangle, \tag{4.1}$$

which represents the transformed physical configuration. Rotation is the main example in the lecture, though translation is mentioned at once as an equally basic possibility.

The operator U must be unitary:

$$U^\dagger U = 1. \tag{4.2}$$

The surviving board image is partially blocked, so we should be cautious about pretending to read

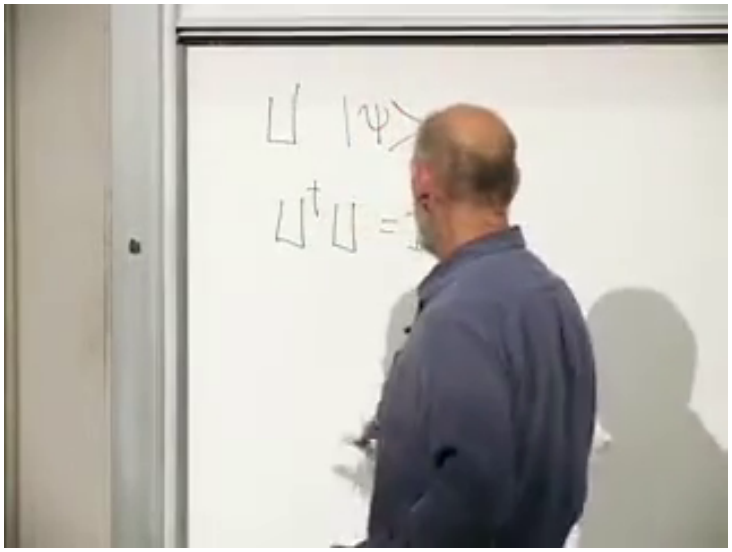

Figure 4.1: Early unitary-operator notation on the board. The upper line is partially occluded, but the transformed ket and the unitarity condition are both clear in intent.

more than is really visible. But the intended equation is secure, and the lecture explains at once what it means:

$$U^\dagger = U^{-1}. \tag{4.3}$$

This is the statement that norms are preserved, and hence that probabilities are preserved. In Susskind's phrasing, the vector must keep the same length in Hilbert space.

Before going further, the lecture makes a small but important remark. Supersymmetry does not fit very comfortably into the ordinary classification of symmetries as continuous or discrete. Precisely because of that, he first reviews the familiar continuous case carefully.

4.1.1 Question & Answer

Are we rotating ordinary space or Hilbert space? The lecture pauses here over a natural question. If a state vector lives in Hilbert space, then when we speak about rotation are we rotating ordinary three-dimensional space or are we rotating Hilbert space?

The answer is: both, but not in the same sense. The physical operation is a rotation in ordinary space. The mathematical representation of that physical operation is a unitary transformation acting in Hilbert space. So U is not "a rotation of states" in the naive geometric sense; it is a Hilbert-space operator that represents the act of rotating the physical system.

4.2 Infinitesimal Transformations And Generators

Once the operator U is on the table, the lecture narrows to very small transformations. This is the first genuinely technical step. No transformation at all is represented by

$$U = 1, \tag{4.4}$$

not by $U = 0$. So an infinitesimal transformation must differ from the identity by a small correction. If the small parameter is ϵ, then

$$U(\epsilon) = 1 + i\epsilon L + O(\epsilon^2). \tag{4.5}$$

To preserve unitarity at leading order, L must be Hermitian:

$$L^\dagger = L. \tag{4.6}$$

For rotations the generator must be labeled by a direction in space. Thus we have L_x, L_y, and L_z, corresponding to infinitesimal rotations about the three coordinate axes. A rotation about an arbitrary axis is then a linear combination of these three basic generators. In this sense there are only three independent infinitesimal rotational generators; every other small rotation is built from them.

At this stage the lecture is still speaking about ordinary symmetries, and in the case of rotations these generators are just angular momentum operators. But the notation L_i is deliberately more general, because the real point is not angular momentum itself. The point is that continuous symmetry is carried by infinitesimal generators.

4.3 Why Commutators Encode The Group

Now comes the long algebraic beat of the lecture. One of the most important facts about a continuous group is its commutator algebra. The lecture first states this abstractly and only then gives it geometric meaning.

For rotations one writes, first in components,

$$[L_x, L_y] = iL_z, \qquad [L_y, L_z] = iL_x, \qquad [L_z, L_x] = iL_y, \tag{4.7}$$

and then in condensed form,

$$[L_i, L_j] = i\epsilon_{ijk} L_k. \tag{4.8}$$

What matters is that the commutator of any two generators is again proportional to a generator. The set closes under commutation.

But the lecture does not want us to take that on authority. It wants us to see what a commutator means.

A small-rotation example. Take a small rotation about the x-axis and a small rotation about the y-axis:

$$U_x(\epsilon) = 1 + i\epsilon L_x, \qquad U_y(\delta) = 1 + i\delta L_y. \tag{4.9}$$

Now perform the loop

$$U_x(\epsilon)\, U_y(\delta)\, U_x^{-1}(\epsilon)\, U_y^{-1}(\delta). \tag{4.10}$$

A useful side remark, made in the lecture because a student interrupts to insist on it, is that unitarity lets us rewrite "dagger" as "inverse." Undoing a small transformation means replacing $+$ by $-$:

$$U_x^{-1}(\epsilon) = 1 - i\epsilon L_x, \qquad U_y^{-1}(\delta) = 1 - i\delta L_y. \tag{4.11}$$

If both rotations were about the same axis, the whole loop would collapse to the identity. The interesting case is that the axes are different. Then the first-order terms in ϵ and δ cancel, but the

mixed term survives. The spoken combinatorics become messy at this point, and the lecture briefly struggles with a factor of two before settling down. For the written notes it is best to keep the narrative struggle and typeset only the cleaned final result:

$$U_x(\epsilon)\, U_y(\delta)\, U_x^{-1}(\epsilon)\, U_y^{-1}(\delta) = 1 + i\epsilon\delta L_z + O(\epsilon^2, \delta^2). \tag{4.12}$$

So the failure of cancellation is itself another small rotation, here about the z-axis, by an angle $\epsilon\delta$. This is the geometric meaning of the commutator. The commutator keeps track of the non-cancellation of operations performed in different orders.

The lecture then broadens the discussion from rotations to translations. A small translation along the x-axis or y-axis is generated by momentum:

$$T_x(\epsilon) = 1 + i\epsilon P_x, \qquad T_y(\epsilon) = 1 + i\epsilon P_y. \tag{4.13}$$

Here the parameter ϵ is now a distance rather than an angle. The lecture also pauses to correct another possible misunderstanding: these are not literal motions of a system requiring forces and accelerations. They are mathematical transformations of a state, asking what the world would look like if everything were displaced.

Two translations commute. Translate in x, then in y, then undo x, then undo y, and one comes back to the same place. Therefore

$$[P_i, P_j] = 0. \tag{4.14}$$

Translations and rotations do not commute. Translate a point, rotate it, translate back along the original axis, and then undo the rotation, and one does not return to the original point. The lecture gives the corresponding algebra in rough spoken notation; the standard normalized form is

$$[L_i, P_j] = i\epsilon_{ijk} P_k. \tag{4.15}$$

The exact index bookkeeping is not the important thing here. What matters is the idea Susskind emphasizes repeatedly: the whole local structure of the group is encoded in these commutation relations. The commutator algebra is the multiplication table of the group written in infinitesimal form.

4.4 Symmetry And The Hamiltonian

Only after this algebraic recap does the lecture make its central dynamical pivot. A true symmetry should not change the energy of a state. That sounds obvious in words, but the lecture takes pains to derive its operator content carefully, and even runs through the derivation twice once a student asks for it again.

The first sparse board image in this sequence isolates the transformed energy eigenstate:

The object of interest is simply

$$U|E\rangle. \tag{4.16}$$

The second board image isolates the defining eigenvalue equation:

We start from

$$H|E\rangle = E|E\rangle. \tag{4.17}$$

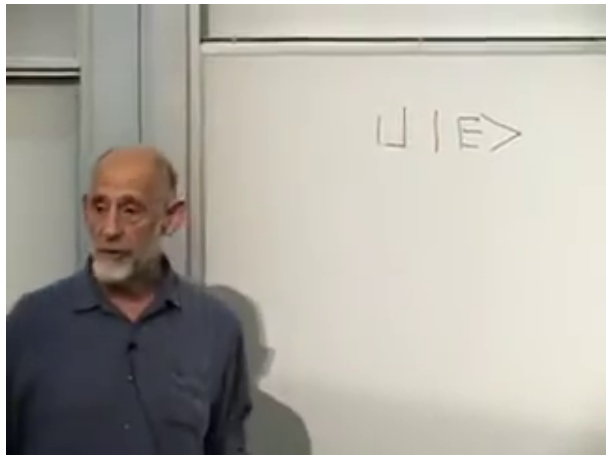

Figure 4.2: Symmetry operator acting on an energy eigenstate.

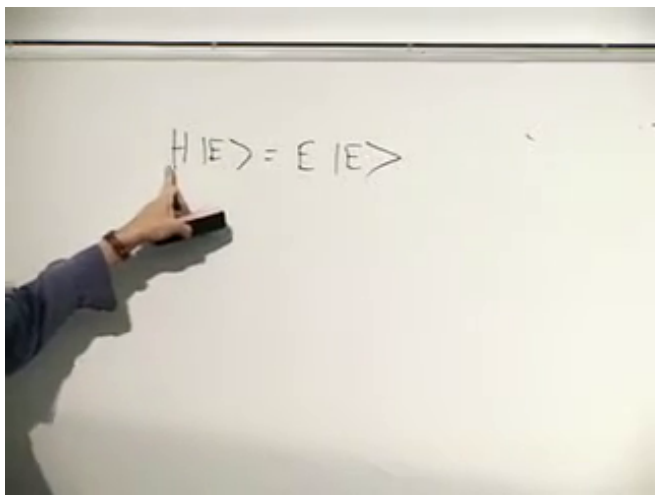

Figure 4.3: Hamiltonian eigenvalue equation.

The lecture uses E both as the scalar energy eigenvalue and as the label of the ket $|E\rangle$. We will keep the notation because it is the lecture's notation, but we must keep the roles distinct in words: the E multiplying the ket is a number, while $|E\rangle$ is the state.

Now suppose U is a symmetry. Then $U|E\rangle$ must have the same energy. That is the whole dynamical content of symmetry in this context. Therefore

$$H|E\rangle = E|E\rangle, \tag{4.18}$$
$$H\,U|E\rangle = E\,U|E\rangle, \tag{4.19}$$
$$= U\,E|E\rangle, \tag{4.20}$$
$$= U\,H|E\rangle. \tag{4.21}$$

The middle step is the one the lecture lingers over. Since E is just a scalar, it can be moved through the operator U. Then, because $E|E\rangle = H|E\rangle$, the scalar may be replaced by the Hamiltonian acting on the original eigenket.

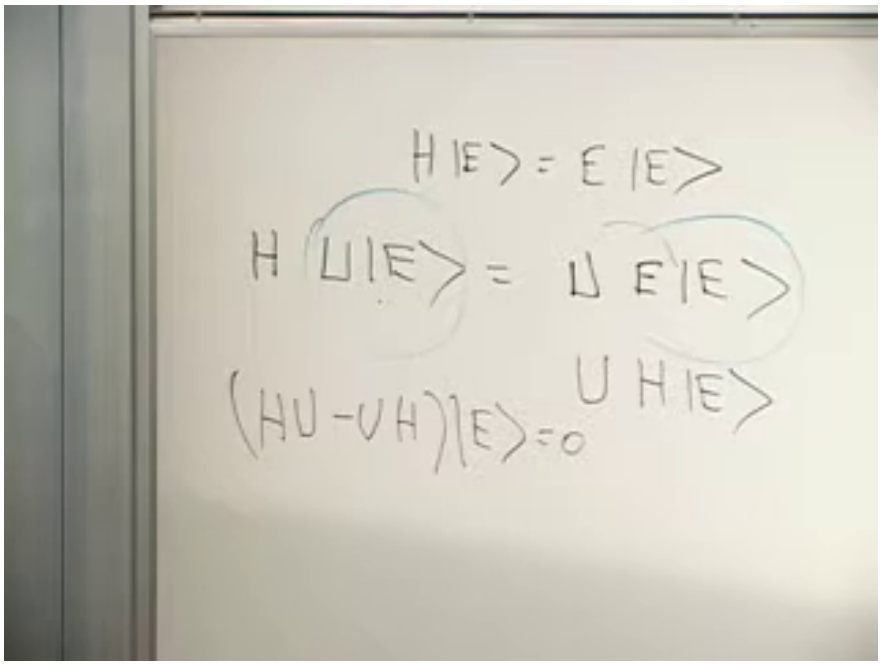

Figure 4.4: From the transformed eigenstate to a commutator condition. The curved board marks are part of the derivational staging, not decorative flourishes.

$$H|E\rangle = E|E\rangle$$

$$H\,U|E\rangle = E\,U|E\rangle = U\,E|E\rangle = U\,H|E\rangle$$

$$(HU - UH)|E\rangle = 0$$

Figure 4.5: A clean companion reconstruction of the board derivation; the curved strokes only echo the lecture's guide marks.

Subtracting the first and last expressions gives

$$(HU - UH)|E\rangle = 0. \tag{4.22}$$

Because that board layout is so important pedagogically, it is useful to keep a clean companion rendering nearby:

If this annihilates every energy eigenstate, and if the energy eigenstates form a complete basis, then the operator itself must vanish:

$$[U, H] = 0. \tag{4.23}$$

That is the mathematical criterion for a symmetry transformation.

Now return to the infinitesimal form

$$U(\epsilon) = 1 + i\epsilon L_i + O(\epsilon^2). \tag{4.24}$$

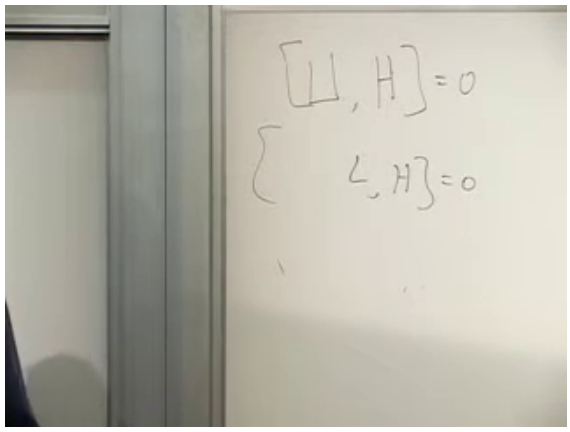

Figure 4.6: Symmetry operators and generators commuting with the Hamiltonian. The upper relation is fully legible; the lower one is standardized cautiously as $[L_i, H] = 0$.

Since the identity commutes with everything, the same statement tells us that the generators commute with the Hamiltonian:

$$[L_i, H] = 0. \tag{4.25}$$

The lower board label in the surviving frame is a little soft, but the normalized form $[L_i, H] = 0$ is the safest and most coherent way to write the lecture's conclusion.

This is more than a static condition. If an operator commutes with the Hamiltonian, then in the Heisenberg sense it is time-independent:

$$[O, H] = 0 \qquad \Longrightarrow \qquad \dot{O} = 0. \tag{4.26}$$

Thus symmetry and conservation law stand very close to each other.

The lecture immediately cashes this out in the familiar example of atomic states. Fix the total angular momentum l. The different magnetic quantum numbers m are rotated into one another, so they must all have the same energy:

$$E_{l,m} = E_l. \tag{4.27}$$

That is precisely the content of rotational symmetry, provided there is no external magnetic or electric field to spoil it.

4.4.1 Question & Answer

What counts as a true symmetry? This is where the lecture corrects a very common mistake. Suppose we rotate a spin in empty space. Its energy does not change. Good: rotation is a symmetry.

Now put that same spin in a magnetic field and rotate only the spin. Its energy changes. Does that mean rotation is not a symmetry? No. It means we did not perform the symmetry transformation. We rotated one part of the setup relative to another part that we held fixed. A genuine symmetry operation would rotate the spin and the magnetic field together.

The same point appears again for translation. An electron in the field of a proton does not keep the same energy if we translate only the electron. But that is not the symmetry operation either. Translating the whole electron-proton system is the symmetry. So when we say that symmetry

does not change the energy, we mean a transformation of the whole physical configuration, not an isolated manipulation of one subsystem inside a fixed background.

4.5 Why Supersymmetry Is Different

Only after the entire review of ordinary symmetry does the lecture introduce the new idea. All symmetries discussed so far have one common feature: they preserve particle type. Rotating an electron gives an electron. A color transformation turns a quark into another quark. A scalar goes to a scalar; a spin-$\frac{1}{2}$ particle goes to another spin-$\frac{1}{2}$ particle. The symmetries may mix internal labels, spatial orientations, or magnetic quantum numbers, but they do not turn bosons into fermions.

Supersymmetry breaks exactly that pattern. It is a symmetry whose generators take bosons into fermions and fermions into bosons. The lecture presents this as a real escalation, not as one more example on the same list. Ordinary symmetry multiplets are already familiar: rotational symmetry forces the states with different m-values inside a fixed l-multiplet to have the same energy. Supersymmetry proposes a new kind of multiplet, one that crosses the boson-fermion divide.

The physical motivation is also kept in view. If a theory truly has supersymmetry, then bosons and fermions come in matched partners of equal mass. That is precisely the structure one would want if bosonic and fermionic contributions are to cancel the large renormalization effects discussed earlier in the course. So the lecture does not motivate supersymmetry by elegance alone. It motivates it by the possibility of a dynamical payoff.

4.6 Supersymmetry Generators And Statistics

The lecture now asks the central constructive question: what sort of operator could take a boson into a fermion and a fermion into a boson?

Let A and $A^\dagger$ be bosonic annihilation and creation operators, and let C and $C^\dagger$ be the corresponding fermionic operators. Then a simple prototype is

$$Q = C^\dagger A + A^\dagger C. \tag{4.28}$$

The first term annihilates a boson and creates a fermion. The second annihilates a fermion and creates a boson. Schematically,

$$Q|B\rangle \sim |F\rangle, \qquad Q|F\rangle \sim |B\rangle. \tag{4.29}$$

A student asks whether this means we are somehow creating a mysterious boson-fermion superposition as a single state. The lecture corrects that immediately. Q is an operator, not a state. Acting on a general superposition it may of course produce another superposition, but that is just the ordinary linearity of operators. The distinctive property is simpler: acting on a boson it gives a fermion, and acting on a fermion it gives a boson.

4.6.1 Question & Answer

What happens when Q acts on the vacuum? This is the natural place to make a mistake, so the lecture stops and checks it explicitly.

Apply

$$Q = C^\dagger A + A^\dagger C \tag{4.30}$$

to the vacuum $|0\rangle$. The operator A tries to annihilate a boson, but no boson is present. The operator C tries to annihilate a fermion, but no fermion is present either. Therefore

$$Q|0\rangle = 0. \tag{4.31}$$

This is not the vacuum again. It is the zero vector. The lecture is explicit about that distinction.

The next question is about spin. If a spin-zero boson is mapped to a fermion, which fermion appears? A spin-$\frac{1}{2}$ fermion has two components. So a single unindexed generator cannot be enough. The lecture's answer is that supersymmetry generators must carry a spinor index:

$$Q_\alpha. \tag{4.32}$$

Schematically one should think of

$$Q_\alpha|B\rangle \sim |F_\alpha\rangle, \qquad Q_\alpha|F_\beta\rangle \sim \delta_{\alpha\beta}|B\rangle. \tag{4.33}$$

The precise coefficients are not the point here. The point is that the generator itself carries half-integer spin because it changes spin by half a unit.

This already marks supersymmetry out as different from ordinary observables. Supersymmetry generators are fermionic operators: they contain an odd number of fermion operators. That is exactly why they can change bosons into fermions and fermions into bosons.

To see how radical this is, the lecture reviews the algebra of bosonic and fermionic creation and annihilation operators. For bosons we normalize the relations as

$$[A_i, A_j] = 0, \qquad [A_i^\dagger, A_j^\dagger] = 0, \qquad [A_i, A_j^\dagger] = \delta_{ij}. \tag{4.34}$$

Hence bosonic creation operators commute:

$$A_i^\dagger A_j^\dagger = A_j^\dagger A_i^\dagger. \tag{4.35}$$

So a two-boson state is symmetric under interchange of the labels i and j.

For fermions we replace commutators by anticommutators:

$$\{C_i, C_j\} = 0, \qquad \{C_i^\dagger, C_j^\dagger\} = 0, \qquad \{C_i, C_j^\dagger\} = \delta_{ij}. \tag{4.36}$$

In particular,

$$(C^\dagger)^2 = 0, \tag{4.37}$$

which is the Pauli exclusion principle in operator form: two fermions cannot occupy the same state. And for distinct labels,

$$C_i^\dagger C_j^\dagger = -\,C_j^\dagger C_i^\dagger. \tag{4.38}$$

So fermionic two-particle states are antisymmetric under exchange.

This is the point at which the lecture broadens again. Bosonic field amplitudes can become large and classical-looking. Fermionic objects cannot, because their squares vanish. That is what forces us toward a new kind of arithmetic.

4.7 Grassmann Numbers

The lecture introduces Grassmann numbers as the bookkeeping language appropriate to fermionic quantities and, in particular, to supersymmetry generators. Ordinary numbers commute:

$$\alpha_i \alpha_j = \alpha_j \alpha_i. \tag{4.39}$$

Grassmann numbers anticommute:

$$\theta_i \theta_j = -\theta_j \theta_i. \tag{4.40}$$

Setting $i = j$ gives

$$\theta_i^2 = 0. \tag{4.41}$$

And ordinary numbers commute with Grassmann numbers:

$$\alpha\theta = \theta\alpha. \tag{4.42}$$

The lecture is careful not to overinterpret these objects. They are not directly measurable quantities. They are a new algebraic device. It is also careful about parity. A single θ is odd. A product such as $\theta_1\theta_2$ is even, in the graded sense that it commutes with other even quantities, but it need not be an ordinary scalar in the naive sense, because it may still be nilpotent.

With one Grassmann variable θ, the most general function is

$$f(\theta) = \alpha + \beta\theta. \tag{4.43}$$

There is no quadratic term because $\theta^2 = 0$. With two Grassmann variables,

$$f(\theta_1, \theta_2) = \alpha + \beta_1\theta_1 + \beta_2\theta_2 + \gamma\theta_1\theta_2. \tag{4.44}$$

That is already the most general polynomial. The lecture lingers over how little room there is in this algebra.

Differentiation is then introduced with just one twist. For one variable we define

$$\partial_\theta(\alpha + \beta\theta) = \beta. \tag{4.45}$$

For two variables we obtain

$$\partial_{\theta_1} f = \beta_1 + \gamma\theta_2, \qquad \partial_{\theta_2} f = \beta_2 - \gamma\theta_1. \tag{4.46}$$

The minus sign in the second formula is the whole point. The derivative operator with respect to a Grassmann variable is itself Grassmann-odd, so when it passes through another Grassmann quantity it picks up a sign. The lecture emphasizes that this is precisely what is needed to preserve the usual product rule of calculus.

The one-variable exponential remains simple:

$$e^{a\theta} = 1 + a\theta. \tag{4.47}$$

Therefore

$$e^{a\theta} e^{b\theta} = e^{(a+b)\theta}. \tag{4.48}$$

Up to this point the algebra behaves exactly as the lecture hopes.

But then the lecture tries two Grassmann variables and the tone changes from exposition to investigation. One certainly computes

$$e^{\theta_1} e^{\theta_2} = (1 + \theta_1)(1 + \theta_2) = 1 + \theta_1 + \theta_2 + \theta_1\theta_2. \tag{4.49}$$

The tempting next step is to compare this with $e^{\theta_1+\theta_2}$. Here the lecture runs into a real puzzle. Because $(\theta_1 + \theta_2)^2$ brings in $\theta_1\theta_2 + \theta_2\theta_1$, the naive exponential law becomes slippery. The discussion does not settle the matter. It is important not to flatten this into a polished theorem that the lecture itself never reached. The honest endpoint is that Grassmann arithmetic exists, is useful, and is essential, but the multi-variable exponential is left here as an unresolved question.

The lecture closes this part by noting that integration over Grassmann variables can also be defined, and that because the space of functions is so small the whole integration table will be correspondingly short. But that development is postponed. What matters for now is the existence of a consistent algebraic framework in which fermion operators and supersymmetry generators naturally live.

4.8 Summary

The lecture unfolds in a very deliberate order. We first rebuild the ordinary notion of symmetry: a unitary operator acting on states, with infinitesimal generators that close under commutation. We then learn to read the whole local structure of a continuous group from those commutators. Next comes the dynamical condition that turns transformation theory into symmetry theory:

$$[U, H] = 0, \qquad [L_i, H] = 0. \tag{4.50}$$

Only after that groundwork is laid does supersymmetry appear, not as one more ordinary symmetry, but as a new sort of transformation that mixes bosons and fermions. That step forces us beyond ordinary commuting quantities and into Grassmann arithmetic. The lecture therefore ends, quite appropriately, not with a closed formal system, but with the sense that a new algebra has opened up and that supersymmetry lives at its center.

Chapter 5

Supersymmetry as Odd Symmetry and Superspace

Supersymmetry enters this lecture with an explicit warning. It is not presented as a tidy embellishment of familiar symmetry, but as one of the most abstract constructions in physics. Susskind therefore refuses to begin with the final algebra. He first backs up and reviews the odd variables out of which the subject is built. These notes follow that order on purpose. We first learn how Grassmann quantities behave, then recall the ordinary language of continuous symmetries, then build the simplest supersymmetry algebra, and only afterward turn to the geometric interpretation in terms of superspace. The key surprise is that the odd generators do not close on another internal generator. Their anticommutator closes on the Hamiltonian itself.

5.1 Why Supersymmetry Begins with Grassmann Numbers

Supersymmetry is described as a symmetry of space-time, but of a very bizarre kind. Rotations, translations, and the ordinary internal symmetries of particle physics are familiar enough: they act either on ordinary coordinates or on particle labels. Supersymmetry does something stranger. It mixes bosonic and fermionic states, and in order to write it naturally one is driven to odd variables that are not ordinary numbers at all.

That is the reason for the apparently backward beginning. Before saying what the symmetry is, the lecture teaches us the language in which it will be expressed. Grassmann variables are not introduced as decorative formalism. They are the minimal machinery needed for the later algebra. Without them, the odd generators Q and $\bar{Q}$ would look unmotivated; with them, the later manipulations become a continuation of rules we have already learned.

5.2 Grassmann Numbers, Functions, and Calculus

Odd and even elements. Let ϵ_i denote Grassmann variables. They are odd elements of the algebra, and they anticommute:

$$\epsilon_i \epsilon_j + \epsilon_j \epsilon_i = 0. \tag{5.1}$$

In particular,

$$\epsilon_i^2 = 0. \tag{5.2}$$

Ordinary real or complex numbers are even elements. The parity rules are

$$\text{odd} \cdot \text{odd} = \text{even}, \qquad \text{odd} \cdot \text{even} = \text{odd}, \qquad \text{even} \cdot \text{even} = \text{even}. \tag{5.3}$$

Even elements commute with everything. The only genuine anticommutation comes from moving one odd element past another.

This much already explains why the subject looks unusual. The square of an odd Grassmann variable vanishes, but the variable itself need not vanish. That single fact drastically truncates the algebra.

Functions of Grassmann variables. A function of one Grassmann variable cannot be an infinite power series in the usual sense, because the square of the variable is zero. Thus

$$f(\epsilon) = a + b\,\epsilon. \tag{5.4}$$

With two independent odd variables the most general form is

$$f(\epsilon_1, \epsilon_2) = a + b_1\epsilon_1 + b_2\epsilon_2 + c\,\epsilon_1\epsilon_2. \tag{5.5}$$

Nothing higher survives. If we pass to a complex pair $\theta, \bar{\theta}$, the same structure becomes

$$f(\theta, \bar{\theta}) = a + \bar{b}\,\theta + \bar{\theta}\,b + c\,\bar{\theta}\theta, \tag{5.6}$$

up to the usual sign conventions associated with ordering the odd variables.

The lecture makes a practical point here. There are not very many functions of finitely many Grassmann variables. One is working with a small polynomial algebra, not with a large analytic zoo. That is why Susskind keeps describing the formalism as bookkeeping. It is a strange bookkeeping, but still bookkeeping.

Even and odd functions. A function built from Grassmann variables is usually taken to have definite parity. If f is even, then the coefficients must alternate in parity so that each term is even. If f is odd, the pattern reverses. One can invent mixed expressions, but in the lecture the natural objects are always taken to be either even or odd.

That point will matter later. Ordinary observables are even. Supersymmetry generators will be odd.

Differentiation and sign bookkeeping. Let θ be a Grassmann variable. Differentiation with respect to θ is simple once the ordering is handled correctly. If a term does not depend on θ, then

$$\partial_\theta a = 0. \tag{5.7}$$

If the θ-dependence is written in standard position, then

$$\partial_\theta(a + \theta b) = b. \tag{5.8}$$

But one must be careful when odd coefficients sit on the other side. If b is odd, then

$$b\theta = -\theta b, \tag{5.9}$$

so before differentiating we bring θ next to the derivative and keep the sign:

$$\partial_\theta(b\theta) = -b. \tag{5.10}$$

This is exactly the kind of local obstacle the lecture wants us to feel. The calculus is not difficult in itself; the difficulty is remembering that every time one odd element passes another, the sign changes.

For two odd variables θ and $\bar{\theta}$, the same issue appears in a slightly richer example. If

$$f(\theta, \bar{\theta}) = a + \bar{b}\,\theta + \bar{\theta}\,b + c\,\bar{\theta}\theta, \tag{5.11}$$

then differentiating with respect to θ requires us to move each θ into the correct place before acting. In an even example, this produces the sign pattern

$$\partial_\theta f = -\,\bar{b} - c\,\bar{\theta}, \tag{5.12}$$

while in an odd example the signs come out differently. The lecture's lesson is not the specific formula but the method: reorder first, then differentiate.

A bridge to operator algebra. Susskind now compares Grassmann calculus with ordinary calculus. For an ordinary coordinate x,

$$[\partial_x, x] = 1. \tag{5.13}$$

The corresponding Grassmann identity is obtained by acting on a test function

$$f(\theta) = a + \theta b. \tag{5.14}$$

We compute

$$\theta\,\partial_\theta f = \theta b, \tag{5.15}$$

$$\partial_\theta(\theta f) = \partial_\theta(\theta a + \theta^2 b) = a, \tag{5.16}$$

because $\theta^2 = 0$. Adding the two orders gives

$$(\partial_\theta\theta + \theta\partial_\theta) f = a + \theta b = f. \tag{5.17}$$

Since this works for an arbitrary function of one Grassmann variable,

$$\{\partial_\theta, \theta\} = 1. \tag{5.18}$$

This identity is one of the lecture's turning points. We are no longer merely listing Grassmann rules. We have reached an operator relation of the same flavor as the commutators that organize symmetry generators in ordinary quantum mechanics.

Integration. The integral calculus is only stated, not derived in detail, but the statement is striking:

$$\int d\theta\, f(\theta) = \partial_\theta f(\theta). \tag{5.19}$$

The lecture motivates this by saying that if one insists on integration by parts, the Grassmann integral must coincide with the Grassmann derivative. Once again the point is that the whole calculus is tiny compared with the ordinary one.

5.2.1 Question & Answer

At this stage a natural question is whether the order of θ and $\bar{\theta}$ carries any deep significance. Suppose we write $c\,\bar{\theta}\theta$. Could we not just as well write $c\,\theta\bar{\theta}$?

Yes, but only at the price of a sign. Since both variables are odd,

$$\bar{\theta}\theta = -\theta\bar{\theta}. \tag{5.20}$$

So we may rewrite

$$c\,\bar{\theta}\theta = -\,c\,\theta\bar{\theta}, \tag{5.21}$$

or absorb the sign into the coefficient. If c is also odd, moving the variables past c produces further sign changes. What does not change is the parity. These manipulations may flip a sign, but they do not turn odd into even or even into odd. In the lecture's own register, this is still just bookkeeping for fermions. The deeper significance will come later.

5.3 Ordinary Symmetry as the Comparison Case

Only after the Grassmann prelude does the lecture return to the familiar language of continuous symmetry. Let G_i denote the infinitesimal generators of a continuous group. Their algebra closes under commutators:

$$[G_i, G_j] = i\,c_{ijk}G_k, \tag{5.22}$$

where c_{ijk} are the structure constants. This closure is the infinitesimal version of the multiplication law of the group.

The generators of sensible physical symmetries are Hermitian. One may combine them into useful complex linear combinations without losing the underlying Hermitian interpretation. The usual angular momentum example is

$$L_\pm = L_x \pm iL_y. \tag{5.23}$$

The point of recalling this is that later we will also use complex combinations of generators, but now the generators themselves will be odd.

For an actual symmetry of a physical system, the Hamiltonian must commute with the generators:

$$[H, G_i] = 0. \tag{5.24}$$

This is another way of saying that the symmetry does not change the energy. It also says that the corresponding charge is conserved.

Susskind emphasizes one more contrast before passing to supersymmetry. Ordinary measurable quantities are even. They are represented by ordinary numbers, or by operators whose eigenvalues are ordinary numbers. The ordinary symmetries of nature therefore live in an even algebra and are organized by commutators. Supersymmetry changes exactly this assumption: the generators themselves are odd. Once that happens, the natural algebraic operation is no longer the commutator but the anticommutator.

5.4 A Minimal Supersymmetry Algebra from Oscillators

To see what such a symmetry might look like, the lecture constructs the simplest possible toy model. We do not begin with a field theory in full space-time. We begin with one bosonic oscillator and one fermionic oscillator, both thought of as particles at rest. In that setting the bosonic creation and annihilation operators satisfy

$$[a_+, a_+] = [a_-, a_-] = 0, \qquad [a_-, a_+] = 1, \tag{5.25}$$

and the fermionic operators satisfy

$$\{c_+, c_+\} = \{c_-, c_-\} = 0, \qquad \{c_-, c_+\} = 1. \tag{5.26}$$

In particular,

$$c_+^2 = c_-^2 = 0. \tag{5.27}$$

The a's are even, the c's are odd, and every a commutes with every c.

Now define the odd generators

$$Q = \sqrt{m}\, a_+ c_-, \qquad \bar{Q} = Q^\dagger = \sqrt{m}\, a_- c_+. \tag{5.28}$$

The lecture uses the notations $Q^\dagger$ and $\bar{Q}$ interchangeably; we will use $\bar{Q}$ in the main flow. The action is direct and intuitive. Q removes a fermion and replaces it by a boson; $\bar{Q}$ does the reverse. We are therefore looking at symmetry operations that exchange bosonic and fermionic states.

The crucial question is the analog of the Lie-algebra closure relation: what is $\{Q, \bar{Q}\}$? We write

$$\{Q, \bar{Q}\} = Q\bar{Q} + \bar{Q}Q \tag{5.29}$$

$$= m(a_+ c_- a_- c_+ + a_- c_+ a_+ c_-). \tag{5.30}$$

Because the even a's commute with the odd c's, we may regroup:

$$\{Q, \bar{Q}\} = m(a_+ a_- c_- c_+ + a_- a_+ c_+ c_-). \tag{5.31}$$

Now use

$$c_- c_+ = 1 - c_+ c_-, \qquad a_- a_+ = a_+ a_- + 1. \tag{5.32}$$

Then

$$\{Q, \bar{Q}\} = m\Big[a_+ a_- (1 - c_+ c_-) + (a_+ a_- + 1) c_+ c_-\Big] \tag{5.33}$$

$$= m(a_+ a_- + c_+ c_-). \tag{5.34}$$

If we define

$$N_B = a_+ a_-, \qquad N_F = c_+ c_-, \tag{5.35}$$

this becomes

$$\{Q, \bar{Q}\} = m(N_B + N_F). \tag{5.36}$$

At this point the physical interpretation enters. In the toy model the particles are at rest, so the total energy is just the common mass times the total number of particles:

$$H = m(N_B + N_F). \tag{5.37}$$

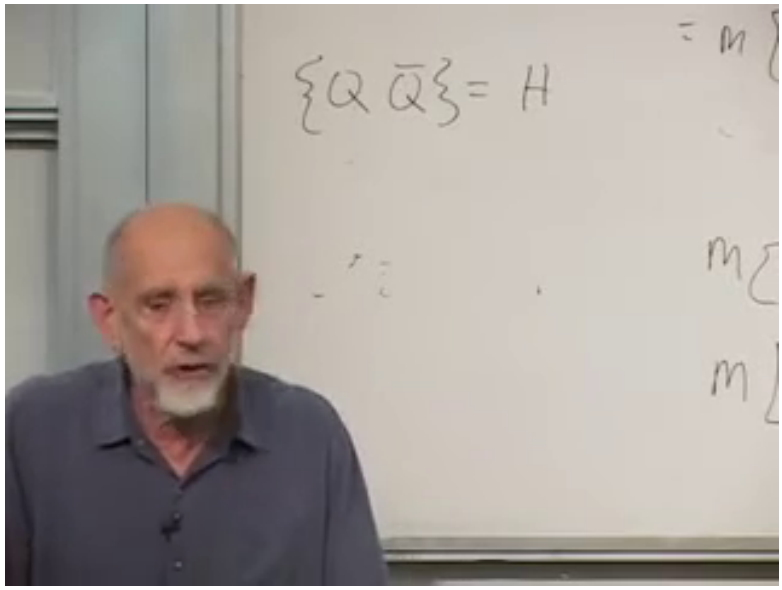

Figure 5.1: Supersymmetry anticommutator yielding the Hamiltonian.

Thus the algebra closes as

$$\{Q, \bar{Q}\} = H. \tag{5.38}$$

This is the first real climax of the lecture. We expected that anticommuting two odd generators might return some even element of the algebra. What we actually get is the Hamiltonian, the generator of time translations. The symmetry is therefore not merely an internal boson–fermion exchange. It is already entangled with space-time.

The remaining relations are immediate:

$$\{Q, Q\} = 0, \qquad \{\bar{Q}, \bar{Q}\} = 0, \tag{5.39}$$

because Q^2 contains c_-^2 and $\bar{Q}^2$ contains c_+^2.

5.4.1 Question & Answer

Why should Q commute with H, and why do the boson and fermion masses have to be equal?

The lecture first gives the result and then pauses to justify it carefully. Let $|E\rangle$ be an eigenstate of the Hamiltonian:

$$H|E\rangle = E|E\rangle. \tag{5.40}$$

Assume now that the boson and fermion have the same mass. Acting with Q removes one particle and inserts another of exactly the same mass. So $Q|E\rangle$ has the same energy E. Therefore

$$HQ|E\rangle = E\,Q|E\rangle, \tag{5.41}$$

$$QH|E\rangle = Q(E|E\rangle) = E\,Q|E\rangle. \tag{5.42}$$

Subtracting the two,

$$[Q, H]|E\rangle = 0. \tag{5.43}$$

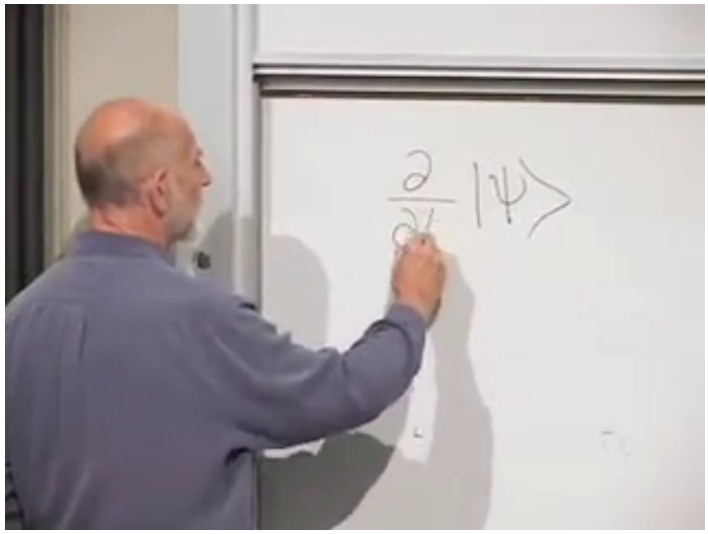

Figure 5.2: Time derivative acting on a state vector.

If the energy eigenstates form a complete basis, it follows that

$$[Q, H] = 0. \tag{5.44}$$

The same reasoning applies to $\bar{Q}$.

This is the point in the lecture where one hears the usual symmetry logic returning in a new form. A symmetry generator commutes with the Hamiltonian precisely because it does not change the energy. Here the new odd generator has that property only because boson and fermion masses have been matched. If the masses were different, the transformed state would not have the same energy, the commutator with H would fail to vanish, and the algebra would cease to close neatly. That is why the equal-mass condition is not an ornament. It is the condition under which the toy supersymmetry works at all.

5.5 From the Hamiltonian to Superspace

Once the Hamiltonian appears on the right-hand side of the supersymmetry algebra, the lecture takes one more backward step. If $\{Q, \bar{Q}\} = H$, then we had better remind ourselves what H does. In this simple example there is no space, only time. But that is enough: if there is energy, there is time evolution.

For a state vector, the standard infinitesimal relation is

$$i\,\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle. \tag{5.45}$$

The board evidence at this stage shows only the time-derivative operator acting on the ket, not the full equation, so we keep the partial board expression nearby:

$$\frac{\partial}{\partial t}|\psi\rangle. \tag{5.46}$$

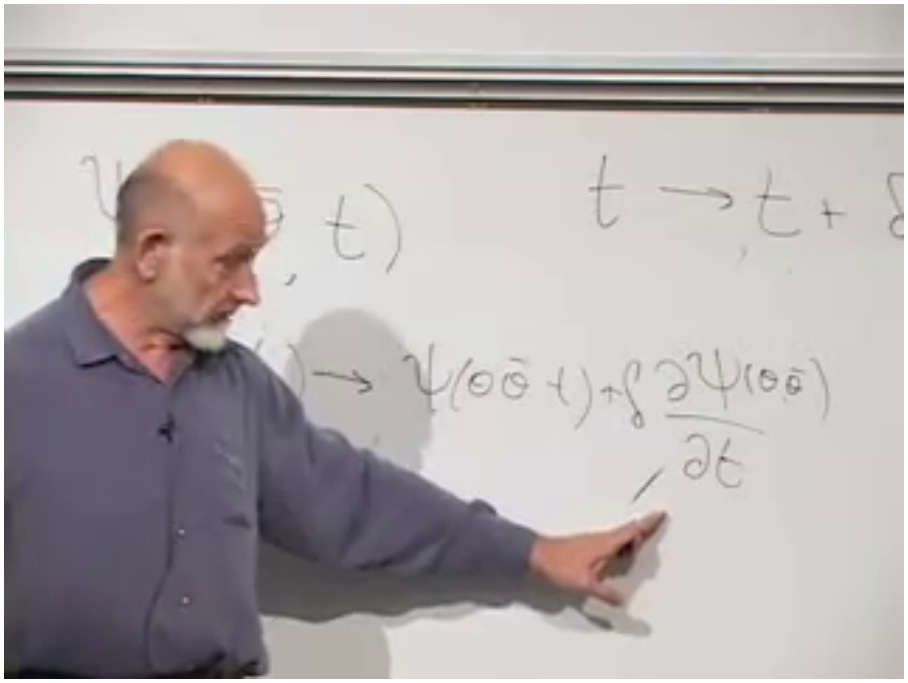

Figure 5.3: First-order time translation of the wavefunction.

The same time evolution may be written in finite form:

$$e^{-iH\delta}|\psi(t)\rangle = |\psi(t+\delta)\rangle. \tag{5.47}$$

The lecture now turns the same idea into a statement about a function. Suppose we consider

$$\psi(\theta, \bar{\theta}, t), \tag{5.48}$$

a wavefunction or field-like object depending not only on time but also on Grassmann coordinates. An ordinary time translation is

$$t \to t + \delta, \tag{5.49}$$

and to first order it induces

$$\psi(\theta, \bar{\theta}, t) \to \psi(\theta, \bar{\theta}, t) + \delta\, \frac{\partial \psi(\theta, \bar{\theta}, t)}{\partial t}. \tag{5.50}$$

This is still ordinary quantum mechanics. The radical move comes next. If space translations are generated by derivatives with respect to spatial coordinates, could there be an analogous symmetry associated with shifts of Grassmann coordinates? Susskind proposes that we consider

$$\theta \to \theta + \epsilon, \qquad t \to t + i\,\bar{\theta}\,\epsilon, \tag{5.51}$$

where ϵ is itself an odd Grassmann parameter.

This is a very strange transformation, and the lecture says so explicitly. But it is strange in a controlled way. Since $\epsilon^2 = 0$, everything is automatically first order. If we shift θ by ϵ, then

$$\delta\psi = \epsilon\, \partial_\theta \psi \tag{5.52}$$

from the change in the Grassmann argument alone. The simultaneous change in t contributes another first-order term proportional to $\bar{\theta}\,\partial_t\psi$. In a conventional normalization, one may summarize the total variation as

$$\delta_\epsilon\psi = \epsilon\Big(\partial_\theta - i\,\bar{\theta}\,\partial_t\Big)\psi. \tag{5.53}$$

This suggests an odd differential operator

$$q = \partial_\theta - i\,\bar{\theta}\,\partial_t. \tag{5.54}$$

Its conjugate partner is then written as

$$\bar{q} = \partial_{\bar{\theta}} + i\,\theta\,\partial_t. \tag{5.55}$$

Here we should be careful, because the lecture itself is careful. Susskind says outright that he may have the sign wrong and that a factor of two may have slipped somewhere into the normalization. So the exact formulas above should be read as a standard reconstruction of the intended structure, not as a claim that every sign is visible on the board.

What is solid is the algebraic mechanism. When we take the anticommutator $\{q, \bar{q}\}$, the derivative-derivative term vanishes, just as ordinary derivatives commute. The odd-coordinate–odd-coordinate term vanishes as well. The surviving terms are the mixed ones, and they use precisely the relations

$$\{\partial_\theta, \theta\} = 1, \qquad \{\partial_{\bar{\theta}}, \bar{\theta}\} = 1. \tag{5.56}$$

Thus the anticommutator is proportional to the time-translation generator:

$$\{q, \bar{q}\} \propto -\,i\,\partial_t. \tag{5.57}$$

Up to the sign and factor-of-two uncertainty flagged in the lecture, this is the differential-operator version of the earlier relation $\{Q, \bar{Q}\} = H$.

Now the earlier surprise becomes geometric. In the oscillator model the algebra closed on the Hamiltonian. In superspace the same fact is reinterpreted as follows: translations in the odd directions θ and $\bar{\theta}$, when anticommuted, produce an ordinary translation in time. Supersymmetry is therefore not merely a device for relabeling particles. It enlarges the geometry on which the theory lives.

5.6 Summary

The lecture unfolds in a very deliberate order. First we are taught how to manipulate odd variables: anticommuting Grassmann numbers, their truncated functions, their derivatives, and the basic identity

$$\{\partial_\theta, \theta\} = 1. \tag{5.58}$$

Only then are we reminded of the ordinary algebra of continuous symmetries and invited to change one decisive feature: let the generators themselves be odd.

In the minimal boson–fermion oscillator model this leads to

$$\{Q, \bar{Q}\} = H, \tag{5.59}$$

with $\{Q, Q\} = \{\bar{Q}, \bar{Q}\} = 0$ and $[Q, H] = 0$ when the boson and fermion masses are equal. The remarkable point is that the algebra closes only when the Hamiltonian is included. That is what ties supersymmetry to space-time.

The lecture then pushes this one step further. Since H generates time translations, the same algebra may be viewed geometrically. A shift in odd coordinates, together with an induced shift in time, leads to differential supercharges whose anticommutator reproduces time translation. In one sense this remains a bookkeeping device for bosons and fermions. In another sense it is a deep enlargement of the very notion of space-time.

Chapter 6

Vacuum Energy, Unification, and Grassmann Superspace

These notes follow Leonard Susskind's sixth lecture in the supplementary course on supersymmetry and grand unification, with curation by LazyingArt LLC. We begin exactly where the lecture begins: not with a formal definition of superspace, but with a student question about the Casimir effect. From that concrete entry point the argument unfolds in a deliberate chain. First we ask what vacuum energy means and what would count as honest evidence for it. Then we clarify what a supersymmetry generator actually does. Only after that do we ask why low-energy supersymmetry is worth taking seriously, why cosmology sharpens the whole question of energy, and finally what kind of Grassmann calculus is needed if supersymmetric theories are to be written in a controlled way.

6.1 Vacuum Energy, Casimir Effect, and Honest Evidence

Susskind reenters the subject by answering unfinished business from the previous lecture. What is the Casimir effect? Take two reflecting plates a distance L apart. In empty space there are vacuum processes. One may picture them as virtual particle pairs, or as zero-point oscillators associated with the quantum fields. In the lecture those are presented as two descriptions of the same thing. Once the plates are introduced, the vacuum fluctuations are modified by the boundary conditions. Virtual loops that would have occupied the region freely are interrupted, reflected, and reorganized by the plates.

The result is that the vacuum energy depends on the plate separation. That is the first mathematical point, and it is all we need. If the energy is a function of L, then we may read it as a potential energy, and a force follows:

$$F(L) = -\frac{dE_{\text{Casimir}}(L)}{dL}. \tag{6.1}$$

This is the lecture's real use of the Casimir effect. It is not a place to derive a famous formula. Susskind even hesitates over the exact power law, and we should not pretend the lecture settled it. What matters here is the mechanism: altered vacuum energy means a force between the plates.

For the electromagnetic vacuum that force is ordinarily attractive. Susskind adds, with a characteristic caution, that a fermionic loop would likely reverse the sign. The details are not the point. The

point is that the force is generated because the vacuum energy changes when the plates are moved.

But the lecture then immediately becomes more severe about interpretation. The Casimir force is measurable, but it is a very small effect, and one may still think of it as a force between nearby plates without insisting that it is the cleanest possible evidence for the gravitating energy of empty space. The stronger question is: what do we really mean by energy? Energy gravitates. Mass gravitates. If the vacuum really carries energy, then the honest evidence should be gravitational.

That is the pivot of the section. The lecture moves from laboratory boundary conditions to cosmology. The real evidence for vacuum energy, Susskind says, is the accelerated expansion of the universe. The Casimir effect is suggestive. Cosmic acceleration is the direct gravitational evidence.

6.1.1 Question & Answer

Question. What is actually mysterious about vacuum energy?

Answer. Not its existence. This distinction is one of the lecture's central clarifications. Quantum field theory predicts zero-point energy for every field. In that sense vacuum energy is not an unknown substance waiting to be discovered; it is a structural consequence of the theory. The mystery is that the observed vacuum energy is so tiny.

The lecture is especially good on the psychology of this. For a long time physicists convinced themselves that vacuum energy must vanish exactly, because it appeared to be absent to very high precision and because naive quantum-field-theoretic estimates were so much larger. Then better observations showed that vacuum energy was not absent after all. The resulting surprise was real, but the real surprise was never that empty space could carry energy. It was that the amount of energy that gravitates is so extraordinarily small.

That distinction will matter again when we return to cosmology. We are not looking for a reason why vacuum energy exists at all. We are looking at a world in which it exists, gravitates, and yet is remarkably suppressed.

6.2 What a Supersymmetry Generator Really Does

The next question comes from a possible misunderstanding about many-particle states. If bosons can pile into the same state, what happens when supersymmetry acts? Do we create a whole population of fermions in that state? The lecture slows down here, and it is worth following that slowdown.

The supersymmetry transformation is generated by an operator Q. Susskind compares it with an ordinary symmetry generator, such as angular momentum, but only to emphasize the difference. A rotation generator makes an infinitesimal rotation, and repeated action can be compounded into a finite one. A supersymmetry generator is not like that. It is fermionic, and its action is local in fermion number:

$$Q\,|B\rangle \propto |F\rangle, \qquad Q\,|F\rangle \propto |B\rangle, \qquad Q^2 = 0. \tag{6.2}$$

The lecture's verbal formulation is even more concrete: Q does not turn a system of bosons into a system of fermions. It removes one boson and replaces it by one fermion, or vice versa. On

a many-body state it changes the occupation only by one unit. That is why Susskind says that supersymmetry changes us by one unit of fermion number.

The nilpotency $Q^2 = 0$ is the algebraic reason this does not build up into a macroscopic conversion. One act of Q changes a boson into a fermion. Act again with the same fermionic generator and the result vanishes. The lecture is careful to linger over this because it clears away a conceptual obstacle before any discussion of motivation.

6.2.1 Question & Answer

Question. Why can repeated action of Q not turn a Bose population into a Fermi population?

Answer. Because Q is not an ordinary bosonic generator that can be compounded over and over again. Its square vanishes:

$$Q^2 = 0. \tag{6.3}$$

So the action of supersymmetry is not "convert the whole system." It is "change one boson into one fermion, or one fermion into one boson," and then stop. The analogy with an infinitesimal rotation is useful only up to the point where it breaks: rotations can be compounded indefinitely, but fermionic generators cannot.

6.3 Why Low-Energy Supersymmetry Is Worth Taking Seriously

Only now does the lecture ask directly what supersymmetry may buy us. The student puts the question bluntly: is there motivation other than cancelling vacuum energy? Susskind's answer is, roughly, yes, a little bit, and then he organizes the rest of the middle lecture around three motivations. He also fixes the scale of the discussion: by supersymmetry he means supersymmetry low enough to matter for experiments not too far beyond present reach, with the LHC very much in mind.

The list is:

1. control of the Higgs self-energy;
2. stable new particles, especially dark-matter candidates;
3. more precise gauge-coupling unification.

The first two are developed here; the third leads naturally into the next section.

The Higgs self-energy problem. The lecture phrases the issue qualitatively, not through a loop integral but through the physical effect of self-energy diagrams. The Higgs mass is shifted by vacuum fluctuations attached to the Higgs line. Left to itself, that shift would naturally drive the Higgs mass to a very high scale. Since the ordinary particle masses are responses to the Higgs phenomenon, everything else would be dragged upward with it.

This is the naturalness problem in the register of the lecture. The dangerous diagrams are not merely large; they threaten to relocate the whole low-energy world. Supersymmetry is attractive because it pairs bosonic and fermionic contributions in such a way that the most dangerous pieces

cancel. The lecture does not derive the cancellation quantitatively here, and we should not clutter the argument with formulas it did not motivate. The structural point is enough: if the symmetry survives to low enough energies, the Higgs self-energy is controlled.

Stable superpartners and dark matter. The second motivation comes from decay chains. The lecture divides the world into ordinary particles and their superpartners. When a superpartner decays, it decays into an ordinary particle plus another superpartner. Schematically,

$$\tilde{X} \to X + \tilde{Y} \to X + Y + \tilde{Z} \to \cdots \tag{6.4}$$

and the superpartner never disappears from the chain. If one starts with a superpartner, one ends with a superpartner. Therefore the chain terminates at a state of minimum superpartner energy, the lightest superpartner or LSP. In the simple version of the lecture's story, the LSP is stable.

That makes supersymmetry immediately interesting for dark matter. Dark matter is believed to be particulate, long-lived on cosmological time scales, and weakly coupled to ordinary matter. A stable superpartner is exactly that kind of object. Susskind also takes time to explain why the dark particle must be weakly interacting. If it were strongly coupled, electrically charged, or too readily scattered by ordinary matter, it would lose energy and collapse into the visible galactic disk instead of remaining in a diffuse halo. So a viable candidate should be weakly interacting and, very likely, neutral.

The lecture then adds a useful cosmological beat. In the hot early universe thermal equilibrium would populate ordinary particles and superparticles comparably. As the temperature fell, unstable heavy states would decay away. If the LSP is stable, a relic abundance could remain. No relic-density formula is derived here, but the physical logic is exactly the one Susskind wants on the page: supersymmetry naturally provides the sort of particle one needs.

He also keeps the theory modest. The lecture does not claim we know which superpartner is lightest. Depending on the version of the theory, it might be photino-like, Higgsino-like, gravitino-like, or something mixed. The point is not the exact identity; it is the structural existence of a stable endpoint.

6.4 Running Couplings and Higher-Precision Unification

The third motivation arrives through renormalization. Susskind returns to a familiar phenomenon: electric charge is screened. At shorter distances, or equivalently higher energies, the effective electromagnetic coupling is larger. He then introduces the notation α for the fine-structure constant and observes that it is more convenient to plot α^{-1}, since the inverse coupling runs roughly linearly in the approximation he has in mind. In that language the electromagnetic inverse coupling slopes downward with energy.

The lecture then moves, in this order, to the electroweak coupling and only afterward to the strong coupling. The electroweak inverse coupling also runs roughly linearly, with a different slope, so the two weakly coupled sectors seem to head toward a crossing at high energy. Only then does the strong coupling enter. Because of asymptotic freedom, quantum chromodynamics runs the other way: the strong coupling gets smaller at short distance, so its inverse gets larger.

The lecture's image of this running is not abstract. It is tied directly to Feynman diagrams. A photon exchanged between charged particles may contain an electron-positron loop. At large distances that

loop contributes to screening; at short distances there is effectively no room for it. The variation of couplings with scale is thus read physically from the loops that can dress the exchange process. This is why Susskind insists that running couplings are defined by Feynman diagrams.

At the level of the curves, the lecture wants us to remember only the broad pattern:

$$\alpha_{\rm em}^{-1}(E) \text{ and } \alpha_{\rm ew}^{-1}(E) \text{ decrease with } E, \qquad \alpha_s^{-1}(E) \text{ increases with } E. \tag{6.5}$$

If one extrapolates these curves using the plain Standard Model, they show a tempting tendency to meet, but they do not really join. They miss each other by too much.

That is where supersymmetry matters. Once superpartners are added into the brew, new loops appear. The electron loop that affects the photon can be replaced by a loop of the electron's superpartner; analogous modifications occur throughout the other sectors. The slopes change. The lecture's punchline is that with suitably light superpartners the three inverse couplings extrapolate to a common point with unexpectedly good accuracy:

$$\alpha_{\rm em}^{-1}(E_U) \approx \alpha_{\rm ew}^{-1}(E_U) \approx \alpha_s^{-1}(E_U), \qquad E_U \sim 10^{15}\text{–}10^{16}\,\text{GeV}. \tag{6.6}$$

No one has measured physics there. What is measured is the low-energy data, after which the theory is extrapolated upward by calculating the relevant diagrams. But that is precisely why Susskind calls this higher-precision unification: getting two lines to cross is not very impressive, while getting three to meet cleanly at one point is much harder to dismiss as accident.

The lecture then sharpens the phenomenology. This is not exact supersymmetry. The superpartners need not have the same mass as the ordinary particles. But they cannot be too heavy. If they are split far upward, their loops only switch on too late, the three couplings fail to meet, the Higgs self-energy problem reappears, and the dark-matter scale becomes less compelling. Thus the three motivations are tied together by the same rough requirement: if supersymmetry is to do us any good in these domains, it had better set in not too far above experimentally accessible energies.

Two further remarks in the lecture belong here. First, superpartners do not define new interactions. They carry the same gauge quantum numbers as the particles whose partners they are, and precisely for that reason they enter the same loop corrections. Second, Susskind adds a cautious comment about gravity: because gravitational charge is energy, a gravitational coupling would strengthen with energy and its inverse would fall. That suggests a common high scale, though he is careful not to oversell the comparison between gravity and the three gauge forces.

6.5 Vacuum Energy in Cosmology and the Friedmann Balance

From unification the lecture swings, by way of a student question, into a long cosmological detour. The question is not idle. What is the difference between the Higgs divergence and the vacuum-energy divergence? Susskind answers it before he ever writes a cosmological equation, and the order matters.

6.5.1 Question & Answer

Question. What is the difference between the Higgs divergence and the vacuum-energy divergence?

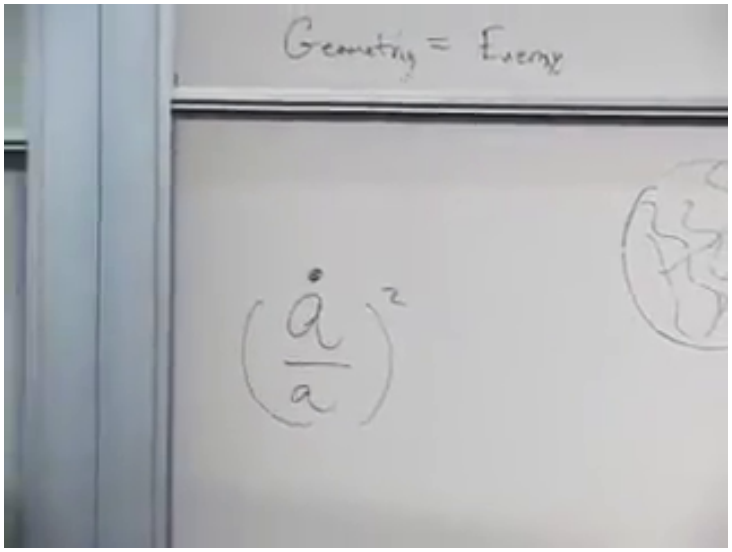

Figure 6.1: Geometry equals energy in the cosmological equation.

Answer. Vacuum-energy diagrams are just virtual processes in empty space. They contribute to the energy density of the vacuum itself. Higgs self-energy diagrams are different because they are connected to the Higgs line. The presence of the Higgs particle acts, in Susskind's phrase, like an impurity in the vacuum. That impurity modifies the surrounding vacuum fluctuations and thereby shifts the Higgs mass. The two phenomena are related, but one is the energy of empty space and the other is the modification of that vacuum by the presence of a specific particle.

That answer then opens the cosmology detour. What is special about vacuum energy, as opposed to ordinary matter or radiation, is how it behaves as the universe expands.

For ordinary non-relativistic matter the energy density falls because the number of particles in a comoving volume stays fixed while the volume grows. Radiation falls faster still, because the expansion not only dilutes the quanta but stretches their wavelengths and lowers the energy of each one. Vacuum energy does not dilute at all. The lecture states this in words; the clean mathematical summary is

$$\rho_{\text{matter}} \propto a^{-3}, \qquad \rho_{\text{radiation}} \propto a^{-4}, \qquad \rho_{\text{vacuum}} = \text{constant}. \tag{6.7}$$

Here $a(t)$ is the scale factor, which Susskind treats informally as the radius or expansion factor of the universe.

This means that a component that was initially negligible eventually dominates. Radiation dominates first, then matter, and finally vacuum energy. In the lecture's rough contemporary numbers, the universe today is about 70% vacuum energy and 30% everything else. One hears the live lecture thinking aloud here: the precise crossover time is not the point; the ordering is.

At the climax of this detour Susskind writes the slogan that organizes the rest of the discussion: geometry equals energy. The first term he isolates on the board is the expansion term

$$\left(\frac{\dot{a}}{a}\right)^2, \tag{6.8}$$

the first genuinely geometric contribution in the time-time Einstein equation.

The wider board then shows the intended balance. One term comes from the rate of expansion, another from spatial curvature, and the right-hand side is decomposed into matter, radiation, and

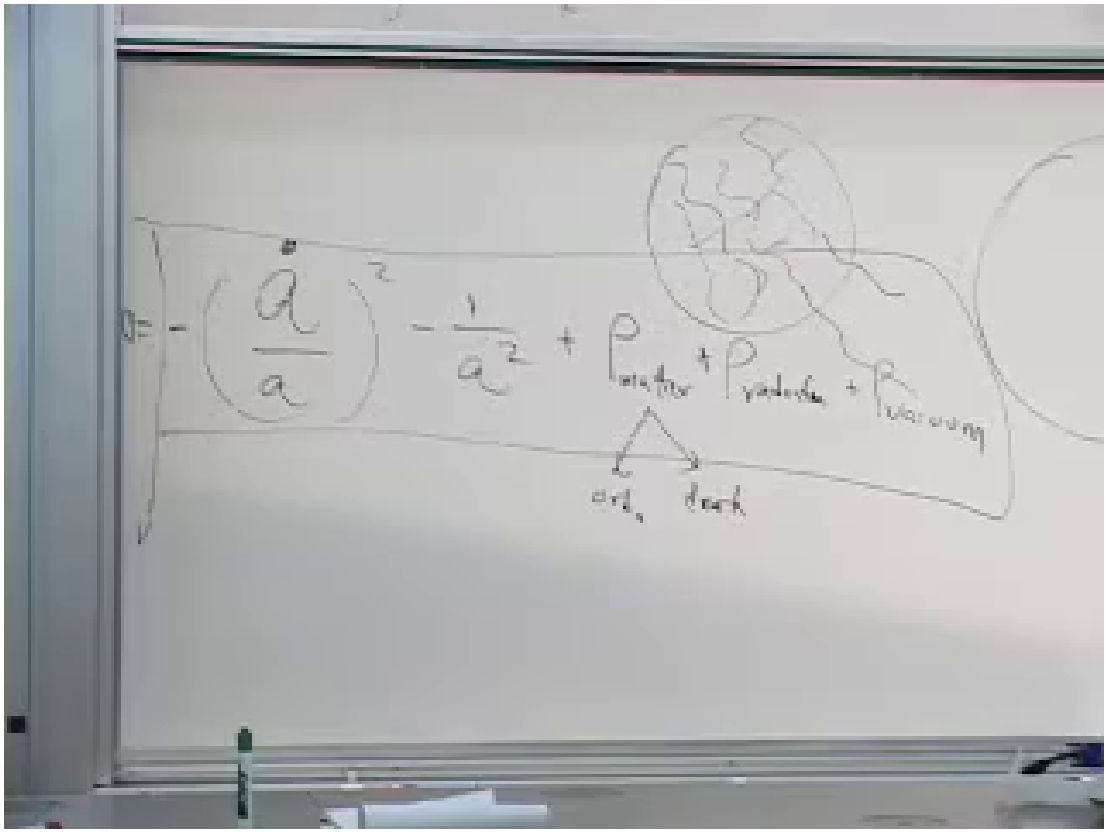

Figure 6.2: Friedmann balance of expansion, curvature, and energy content.

vacuum energy, with matter itself split into ordinary and dark components. A cautious cleaned-up reconstruction of the board's mathematical content is

$$\left(\frac{\dot{a}}{a}\right)^2 \pm \frac{1}{a^2} \sim \rho_{\text{matter}} + \rho_{\text{radiation}} + \rho_{\text{vacuum}}, \tag{6.9}$$

together with

$$\rho_{\text{matter}} = \rho_{\text{ordinary}} + \rho_{\text{dark}}. \tag{6.10}$$

Remark 6.1. The structure of this equation is secure in the lecture and in the selected frames; the overall normalization is not. We therefore keep the Friedmann balance at the level of a careful structural reconstruction. The curvature term is written as $\pm 1/a^2$ because the sign depends on whether the universe is open, closed, or flat.

This is the point at which the lecture itself pauses and almost asks, how did we get into this? We got into it because a question about vacuum-energy diagrams versus Higgs self-energy forced us to ask what vacuum energy really does in an expanding universe. Once that question is asked, the cosmology is not optional.

6.5.2 Question & Answer

Question. How can energy be conserved if vacuum energy density does not dilute as the universe expands?

$$\boxed{\begin{array}{c}\left(\dfrac{\dot{a}}{a}\right)^2 \\ \pm\dfrac{1}{a^2}\end{array}} \quad = \quad \boxed{\begin{array}{c}\rho_{\text{matter}} + \rho_{\text{radiation}} + \rho_{\text{vacuum}} \\ \rho_{\text{matter}} = \rho_{\text{ordinary}} + \rho_{\text{dark}}\end{array}}$$

geometry energy

Figure 6.3: Schematic reconstruction of the board layout: geometry on the left, cosmic contents on the right.

Answer. Susskind's answer is deliberately bizarre: in this cosmological setting the total energy is exactly conserved and exactly zero. One must frame that sentence correctly. It is not a blanket statement about every spacetime in general relativity. It is the content of the cosmological Hamiltonian constraint in the setting being discussed.

The lecture first describes the expansion term as a kind of kinetic energy of the expanding universe and says that, in the energy language one is trying to force onto the problem, that contribution is negative. Then it writes the Einstein equation as geometry on the left and energy density on the right. The next move is the decisive one: put everything on one side. The result is no longer merely an equality between two sides of a board calculation. It is a constraint:

$$H_{\text{tot}} = 0. \tag{6.11}$$

Schematically,

$$\left(\frac{\dot{a}}{a}\right)^2 \pm \frac{1}{a^2} - (\rho_{\text{matter}} + \rho_{\text{radiation}} + \rho_{\text{vacuum}}) = 0. \tag{6.12}$$

The lecture interprets the quantity on the left as the canonical Hamiltonian of the cosmological system. So if vacuum energy does not dilute, that does not violate the relevant conservation law. It changes which terms are balancing which. Matter and radiation may decrease, vacuum energy may stay fixed, and the compensating geometric part adjusts so that the total Hamiltonian constraint remains zero. In Susskind's phrase, that is the rule of the game.

6.6 Grassmann Numbers, Superspace, and Superfields

Only after this cosmological excursion does the lecture consciously return to supersymmetry proper. A student asks whether Grassmann numbers actually appear in the Lagrangian. Susskind answers yes, in two different ways.

First, fermion fields themselves are Grassmann-valued. Bosonic fields take ordinary numerical values; fermionic fields anticommute. That is already part of standard field theory. The second appearance is the one he wanted to reach all along: in a supersymmetric theory there are additional coordinates, Grassmann coordinates θ, and fields may depend on them as well as on the ordinary spacetime coordinates.

The lecture describes this in an exploratory way rather than a heavily formal one. We have the ordinary coordinates x^μ, but we also have two complex Grassmann coordinates, or equivalently four real ones. These are not large extra dimensions. They are tiny in the sense that they are nilpotent:

$$\theta^2 = 0. \tag{6.13}$$

That is why Susskind says we cannot move around in them the way we move around in ordinary space. But they are enough to organize the component fields of a supersymmetric multiplet into a single object.

Ordinary fields are functions of x. A superfield is a function of both x and θ:

$$\Phi = \Phi(x, \theta). \tag{6.14}$$

Because the θ's square to zero, the theta-expansion terminates after finitely many terms. In the four-real-theta schematic form used in the lecture,

$$\Phi(x, \theta) = \phi(x) + \theta_\alpha \psi_\alpha(x) + \theta_\alpha \theta_\beta z_{\alpha\beta}(x) + \cdots + \theta_1 \theta_2 \theta_3 \theta_4 \, D(x). \tag{6.15}$$

This should be read as bookkeeping, not as a finalized convention. The important point is that a superfield packages a finite set of ordinary bosonic and fermionic fields. Terms with an odd number of θ's are Grassmann-valued; terms with an even number are ordinary-number-valued.

Once we accept that fields and Lagrangian densities can depend on θ, the structure of the action changes as well. We no longer integrate only over spacetime. We integrate over superspace:

$$S = \int d^4x \, d^4\theta \, \mathcal{L}(x, \theta). \tag{6.16}$$

That is where the lecture deliberately stops and asks the next question. What could such an integral possibly mean?

6.7 Grassmann Integration as the Calculus of Supersymmetry

The answer begins from an ordinary formula. Susskind writes a definite integral of a derivative on the board before he says anything about Grassmann variables. That is not accidental. He wants a model.

For an ordinary function $F(x)$ that dies off at spatial infinity, we have

$$\int_{-\infty}^{\infty} \frac{dF}{dx} \, dx = F(\infty) - F(-\infty) = 0. \tag{6.17}$$

The lecture uses this as the prototype for the Grassmann rule: the definite integral of a derivative should vanish.

The other preliminary point is just as important. Grassmann integration is not an indefinite integral, and it is not a sum of infinitesimal pieces. It is a definite integral over the whole theta-space. In fact, the lecture goes even further: in the one-variable case, the integral turns out to be the same operation as the derivative.

6.7.1 Question & Answer

Question. What do we mean by integrating over Grassmann variables?

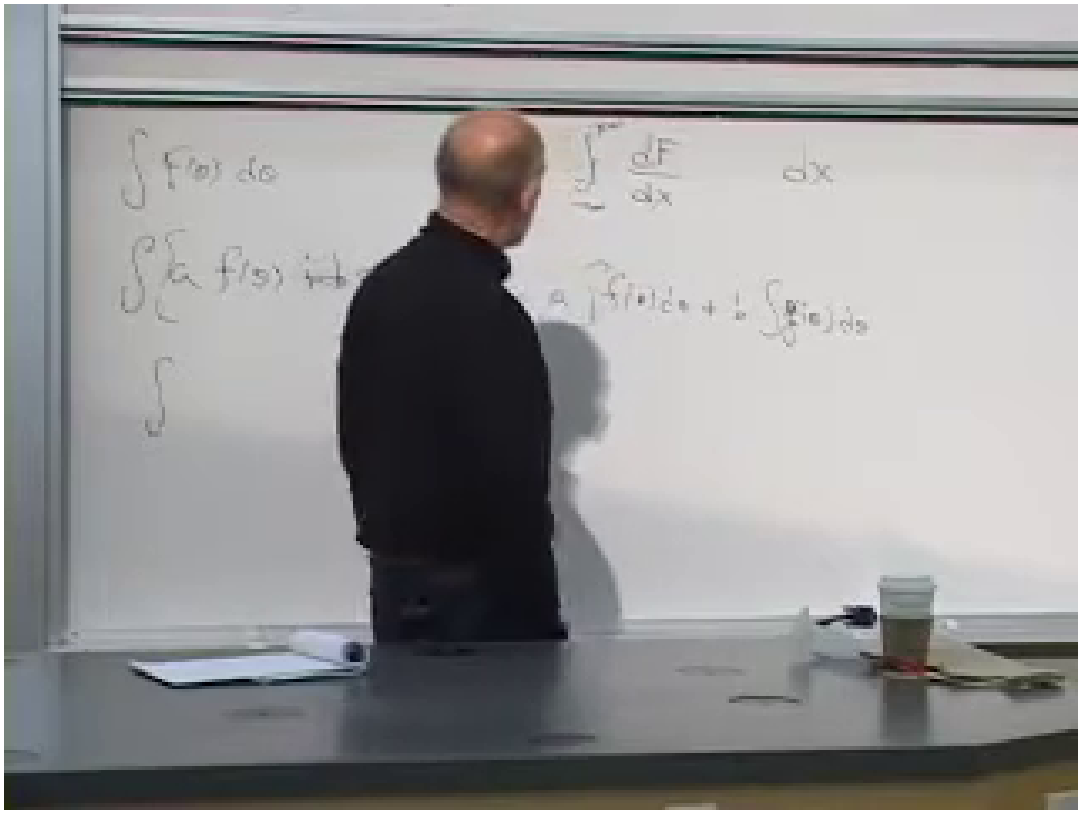

Figure 6.4: Definite integral of a derivative.

Answer. We mean a linear operation, defined so as to mimic the ordinary rule for definite integrals of total derivatives and fixed by a very small number of conventions.

For a single Grassmann variable θ, the lecture argues as follows:

1. Demand linearity.
2. Impose the analog of the ordinary total-derivative rule,

$$\int d\theta \, \frac{d}{d\theta} f(\theta) = 0. \tag{6.18}$$

3. Choose $f(\theta) = \theta$. Since $\frac{d}{d\theta}\theta = 1$, we obtain

$$\int d\theta \, 1 = 0. \tag{6.19}$$

4. Fix the remaining normalization by convention:

$$\int d\theta \, \theta = 1. \tag{6.20}$$

That is already the whole one-variable theory, because every function of one Grassmann variable has the form

$$f(\theta) = a + b\theta. \tag{6.21}$$

Therefore

$$\int d\theta\, f(\theta) = \int d\theta\, (a + b\theta) = b. \tag{6.22}$$

The integral simply extracts the coefficient of the highest theta term. In this one-variable case it is identical to the derivative:

$$\int d\theta\, (a + b\theta) = b = \frac{d}{d\theta}(a + b\theta). \tag{6.23}$$

This is why the lecture says there is no analog of an indefinite integral. One cannot ask for a function whose derivative is θ, because that function would have to contain θ^2, and θ^2 vanishes.

The two-variable case is only slightly richer. Writing the variables as θ and $\bar{\theta}$, the most general function is

$$f(\theta, \bar{\theta}) = a + \bar{\theta}\psi + \bar{\psi}\,\theta + b\,\bar{\theta}\theta. \tag{6.24}$$

There are only four terms because $\theta^2 = \bar{\theta}^2 = 0$. If we adopt the measure ordering

$$\int d\theta\, d\bar{\theta}\, \bar{\theta}\theta = 1, \tag{6.25}$$

then every term except the quadratic top component integrates to zero, and we get

$$\int d\theta\, d\bar{\theta}\, f(\theta, \bar{\theta}) = b. \tag{6.26}$$

Remark 6.2. The sign depends on ordering, and the lecture corrects itself on this point in real time. With the convention used here,

$$\int d\bar{\theta}\, d\theta\, \bar{\theta}\theta = -1.$$

Reversing the order of the measure or of the Grassmann variables changes the sign.

This is where the lecture ends and where the conceptual payoff becomes visible. The rule sounds trivial: integrate over theta and you extract the top component. But that trivial-looking rule is what makes superspace useful. A supersymmetric Lagrangian can be built from superfields and their derivatives, and integrating over the theta's picks out exactly the component that turns into the ordinary action while preserving the boson-fermion structure of the theory. That is why Susskind keeps saying, in effect, that this is the point of it. We are not learning a curiosity. We are assembling the calculus by which supersymmetric theories are actually constructed.

6.8 Summary

The lecture begins with the Casimir effect, but only as a way of asking what vacuum energy really means. From there it sharpens the distinction between the existence of vacuum energy and the mystery of its tiny size, clarifies that a supersymmetry generator changes fermion number one unit at a time and satisfies $Q^2 = 0$, lays out three reasons to care about low-energy supersymmetry, detours through cosmology to explain why vacuum energy is special and how the Friedmann equation becomes a zero-Hamiltonian constraint, and finally returns to supersymmetry by introducing superfields and the Berezin calculus. The chapter therefore ends exactly where the lecture ends: with the formal machinery ready, and with the expectation that concrete supersymmetric examples can now be built on top of it.

Chapter 7

Lecture 7: Supersymmetry, Superfields, and Chirality in the One-Mode Toy Model

These companion notes follow Leonard Susskind's lecture, with curation by LazyingArt LLC. The lecture proceeds by a deliberate retreat to the simplest possible supersymmetric system: one bosonic mode, one fermionic mode, both taken at rest, with all spatial dependence temporarily suppressed. From that toy model Susskind rebuilds the subject in stages. We first rewrite the bosonic mode as an ordinary harmonic oscillator, then discover why the fermionic Lagrangian must be first order in time, then review the supercharge algebra, and only after that pass to superspace, superfields, and the first recipe for constructing supersymmetric Lagrangians. The lecture is mathematically serious, but it is also exploratory; where a sign is corrected live, or where the blackboard calculation breaks off, the notes should preserve that rhythm.

7.1 Bosons, Fermions, and the Zero-Momentum Toy System

Susskind begins by saying, in effect, let us come back to fermions and bosons for a moment, and let us forget motion. The particles are either present or absent; momentum plays no role. In that setting the natural description is in terms of creation and annihilation operators. He uses the older $+$ and $-$ notation on the board, but in normalized form the Hamiltonian is

$$H = m\left(a^\dagger a + c^\dagger c\right), \tag{7.1}$$

up to the usual ignored ground-state constant. Here $a^\dagger, a$ create and annihilate the bosonic quantum, and $c^\dagger, c$ do the same for the fermion.

This opening matters. We are not reviewing the algebra for its own sake. The point is to isolate the most elementary boson–fermion pairing before we add the machinery that will later guarantee that the pairing survives interactions. In this one-mode system the boson and fermion already have the same energy m, and supersymmetry is the statement that there is an operation that exchanges them without changing that energy.

7.2 The Bosonic Oscillator in Lagrangian Form

Once the operator language is in place, Susskind changes viewpoint. Instead of talking about occupation numbers, he invents a field, or in this stripped-down context simply a coordinate $\phi(t)$. The bosonic field theory with no space dependence is just a single harmonic oscillator.

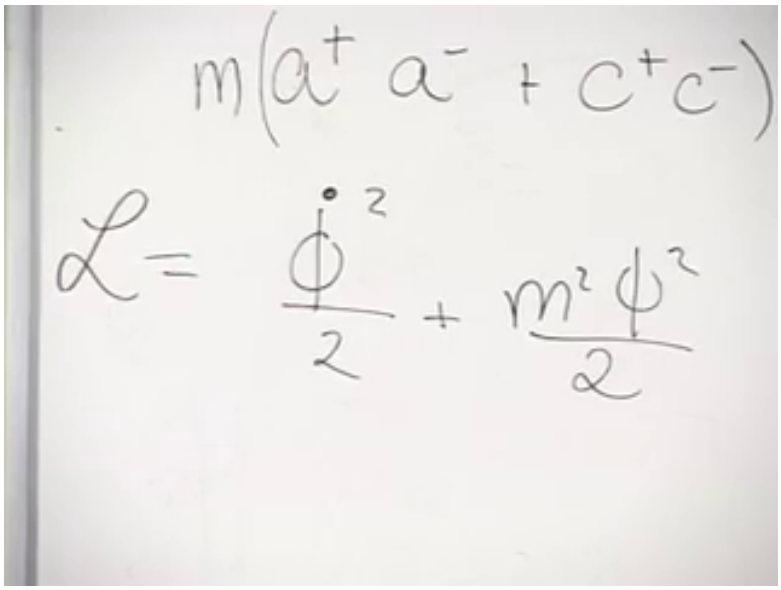

Figure 7.1: Bosonic oscillator Lagrangian from the one-mode field setup. The board still shows the pre-correction $T + V$ sign.

The board frame is worth keeping because it captures the exact transition: the operator-language Hamiltonian sits above, while the one-mode Lagrangian is written below. It also records a live correction. The board shows

$$L = \frac{\dot{\phi}^2}{2} + \frac{m^2\phi^2}{2},$$

but Susskind immediately stops himself: that is $T + V$, the energy, whereas the Lagrangian must be $T - V$. The corrected expression is therefore

$$L_b = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}m^2\phi^2. \tag{7.2}$$

Now the Euler–Lagrange equation is completely straightforward. We compute

$$\frac{\partial L_b}{\partial \dot{\phi}} = \dot{\phi}, \qquad \frac{d}{dt}\left(\frac{\partial L_b}{\partial \dot{\phi}}\right) = \ddot{\phi}, \tag{7.3}$$

$$\frac{\partial L_b}{\partial \phi} = -m^2\phi. \tag{7.4}$$

So

$$\ddot{\phi} + m^2\phi = 0. \tag{7.5}$$

This is the ordinary oscillator equation, and the frequency is m, not m^2. That point is stated explicitly in the lecture. In this one-dimensional world the bosonic field is nothing but the familiar harmonic oscillator in disguise.

7.3 Grassmann Variables and the Fermionic Lagrangian

The fermionic part begins with the first real puzzle of the lecture. Susskind does not start by writing down the final fermionic Lagrangian. Instead he deliberately asks what happens if we try the simplest first-order expression. Take

$$L = \phi\,\dot{\phi}. \tag{7.6}$$

If ϕ is an ordinary bosonic variable, this does nothing:

$$L = \phi\,\dot{\phi} = \frac{1}{2}\frac{d}{dt}\phi^2. \tag{7.7}$$

It is a total derivative, and the Euler–Lagrange equation is just the tautology $\dot{\phi} = \dot{\phi}$. Every differentiable trajectory satisfies it.

Now comes the surprise. Suppose the same symbol denotes a fermionic variable, that is, a Grassmann variable. At this point the lecture renames it ψ, precisely to avoid confusing bosons and fermions. We still have

$$L = \psi\,\dot{\psi}, \qquad \frac{\partial L}{\partial \dot{\psi}} = \psi. \tag{7.8}$$

But when we differentiate with respect to ψ, we must first move ψ past $\dot{\psi}$, and that produces a minus sign:

$$\frac{\partial L}{\partial \psi} = -\,\dot{\psi}. \tag{7.9}$$

So the equation of motion becomes

$$\dot{\psi} = -\,\dot{\psi}, \tag{7.10}$$

hence

$$\dot{\psi} = 0. \tag{7.11}$$

This is no longer trivial. It is too strong for what we want, but it has real dynamical content.

7.3.1 Question & Answer

Question. Why can a first-order term be trivial for bosons but dynamical for fermions?

Answer. For commuting variables, $L = \phi\dot{\phi}$ is literally a total time derivative, so the Euler–Lagrange equation collapses to a tautology. For Grassmann variables, the same algebraic expression is sensitive to order. The variation with respect to ψ forces us to move ψ through $\dot{\psi}$, and because both are anticommuting, that reordering contributes a minus sign. The lecture lingers over this point because it is the first moment when the Grassmann character of the variable does visible work in the equations of motion.

The story is not finished, because a single fermion is too naive. If we try to add a restoring term $m\psi^2$, it vanishes identically: $\psi^2 = 0$. This is exactly where Susskind repairs the model by passing to two real Grassmann variables ψ_1, ψ_2, or equivalently to one complex fermion. In the lecture he first writes a mixed first-order Lagrangian for ψ_1 and ψ_2, with a mass term proportional to $m\psi_1\psi_2$, and only afterwards repackages the result into complex notation.

The exact blackboard normalization wobbles, and the transcript is garbled for a short interval here, so the cleanest faithful reconstruction is to state the stabilized form of the answer:

$$L_f \sim \frac{i}{2}\left(\dot{\psi}^*\psi + \dot{\psi}\,\psi^*\right) + m\,\psi^*\psi. \tag{7.12}$$

The important structural points are the ones Susskind emphasizes aloud. The kinetic term contains only one time derivative, and the restoring term contains one power of m, not m^2. The resulting equation of motion is first order:

$$i\,\dot{\psi} = m\,\psi. \tag{7.13}$$

So the fermionic mode oscillates, just as the bosonic mode did, but now through a first-order equation rather than a second-order one. In this toy setting that is the one-dimensional shadow of the contrast between the Dirac equation and the Klein–Gordon equation.

Putting the bosonic and fermionic parts together, the lecture is aiming at a Lagrangian of schematic form

$$L \sim \frac{1}{2}\dot{\phi}^{\,2} - \frac{1}{2}m^2\phi^2 + \frac{i}{2}\left(\dot{\psi}^*\psi + \dot{\psi}\,\psi^*\right) + m\,\psi^*\psi, \tag{7.14}$$

with the explicit warning that the fermionic half-factors and ordering have been stabilized editorially rather than copied as exact board text.

7.4 Supercharges and the Basic Supersymmetry Algebra

With the toy bosonic and fermionic oscillators in hand, Susskind now returns to what supersymmetry means. The system has a symmetry that removes a fermion and inserts a boson, or vice versa, without changing the energy. The corresponding supercharges are

$$Q^\dagger = \sqrt{2m}\,a^\dagger c, \qquad Q = \sqrt{2m}\,a\,c^\dagger. \tag{7.15}$$

The factor $\sqrt{2m}$ is introduced purely by convention so that the algebra comes out in the standard form.

The key relation is the anticommutator:

$$\{Q, Q^\dagger\} = 2m\left(a^\dagger a + c^\dagger c\right) = 2H. \tag{7.16}$$

The nilpotence is equally important:

$$Q^2 = 0, \qquad (Q^\dagger)^2 = 0, \tag{7.17}$$

because every term in Q^2 or $(Q^\dagger)^2$ contains the square of a fermionic creation or annihilation operator. And finally,

$$[H, Q] = [H, Q^\dagger] = 0. \tag{7.18}$$

This last relation is interpreted carefully in the lecture. It means, first, that supersymmetry is a genuine symmetry of the system. But it also means that the Q's are conserved quantities. That is why they are called supercharges. Susskind spends a moment on this because it keeps the algebra from becoming empty symbolism: $[H, Q] = 0$ means that the operation of trading a boson for a fermion is not merely imaginable, but compatible with the dynamics.

7.5 From Supercharge Algebra to Superspace Geometry

Up to this point the lecture has spoken algebraically. The next move is to reinterpret the same algebra geometrically. The guiding thought is familiar from ordinary quantum mechanics: generators of symmetry are differential operators. The Hamiltonian is the generator of time translations,

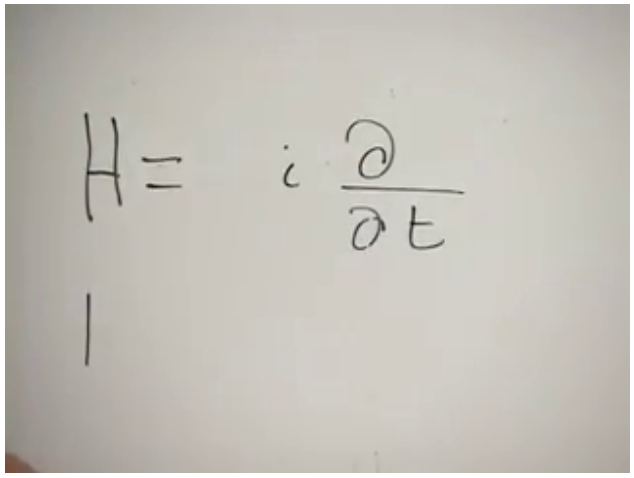

Figure 7.2: Hamiltonian as the generator of time translations.

$$H = i\,\frac{\partial}{\partial t}, \tag{7.19}$$

and momentum is the generator of spatial translations,

$$P = -\,i\,\frac{\partial}{\partial x}. \tag{7.20}$$

Angular momentum is likewise built from coordinates and derivatives. So the natural question is: if Q and $Q^\dagger$ are fermionic symmetry generators, might they be derivatives with respect to fermionic coordinates?

That is the entry into superspace. We enlarge the coordinate set from t to

$$(t, \theta, \bar{\theta}), \tag{7.21}$$

where θ is a Grassmann coordinate and $\bar{\theta}$ its conjugate. The lecture is explicit that the sign conventions are unstable on the board. Susskind corrects himself, debates the signs with the room, and finally settles only on the fact that the two signs inside the pair of operators must match. We therefore choose one consistent convention:

$$q = \frac{\partial}{\partial \theta} + i\,\bar{\theta}\,\frac{\partial}{\partial t}, \qquad \bar{q} = \frac{\partial}{\partial \bar{\theta}} + i\,\theta\,\frac{\partial}{\partial t}. \tag{7.22}$$

Then

$$\{q, \bar{q}\} = 2i\,\frac{\partial}{\partial t} = 2H. \tag{7.23}$$

This immediately explains why the time-derivative pieces are necessary. If q were only ∂_θ and $\bar{q}$ only $\partial_{\bar{\theta}}$, the anticommutator would vanish. The extra $i\bar{\theta}\partial_t$ and $i\theta\partial_t$ pieces are forced on us by the algebra.

In this language a supersymmetry transformation is a tiny shift in the Grassmann coordinates together with a compensating shift in time:

$$\theta \to \theta + \xi, \qquad \bar{\theta} \to \bar{\theta} + \bar{\xi}, \qquad t \to t + i\left(\xi\bar{\theta} + \bar{\xi}\theta\right), \tag{7.24}$$

again in a cleaned convention. The variation of a superfield Φ can then be written as

$$\delta\Phi = \left(\xi q + \bar{\xi}\,\bar{q}\right)\Phi. \tag{7.25}$$

That is the geometric heart of the lecture: supersymmetry is being re-read as motion in an enlarged space whose extra directions are Grassmann-valued.

7.6 Superfields, Superactions, and the D-Term

Once the geometric picture is in place, the lecture changes mood. The question is no longer what supersymmetry is, but how to build Lagrangians that keep it intact. For that we introduce a superfield.

In this one-dimensional superspace the most general bosonic superfield is finite, because $\theta^2 = \bar{\theta}^2 = 0$:

$$\Phi(t, \theta, \bar{\theta}) = \phi(t) + \bar{\theta}\,\psi(t) + \bar{\psi}(t)\,\theta + D(t)\,\bar{\theta}\theta. \tag{7.26}$$

The termination of the expansion is not incidental; it is one of the practical beauties of Grassmann variables, and Susskind remarks on it more than once. The coefficient $D(t)$ is another bosonic variable. At this stage it is simply the last coefficient in the power series.

Now we need an action. The superspace action is

$$S = \int dt\, d\theta\, d\bar{\theta}\, \Lambda, \tag{7.27}$$

where Λ is a super-Lagrangian. This is the central constructive move of the lecture. One now builds Λ from superfields and their allowed derivatives, and then integrates over superspace just as one integrates an ordinary Lagrangian over ordinary space-time.

Berezin integration is simple but decisive:

$$\int d\theta\, d\bar{\theta}\, \Lambda = (\Lambda)_{\bar{\theta}\theta}. \tag{7.28}$$

In words: integrating over θ and $\bar{\theta}$ picks out the coefficient of the highest Grassmann term. Susskind calls that surviving piece the D-term of the super-Lagrangian, though of course it need not be the auxiliary field $D(t)$ itself. The ordinary Lagrangian is precisely this $\bar{\theta}\theta$ coefficient.

That is why superspace is more than a bookkeeping device. It gives us a calculus for building candidate ordinary Lagrangians that are guaranteed, by construction, to fit inside a supersymmetric super-Lagrangian.

7.7 Powers of Superfields, Derivative Subtleties, and Chirality

Before confronting derivatives, the lecture takes up the easier operation: multiplication. The product of superfields is again a superfield, so the first nontrivial worked example is to square Φ.

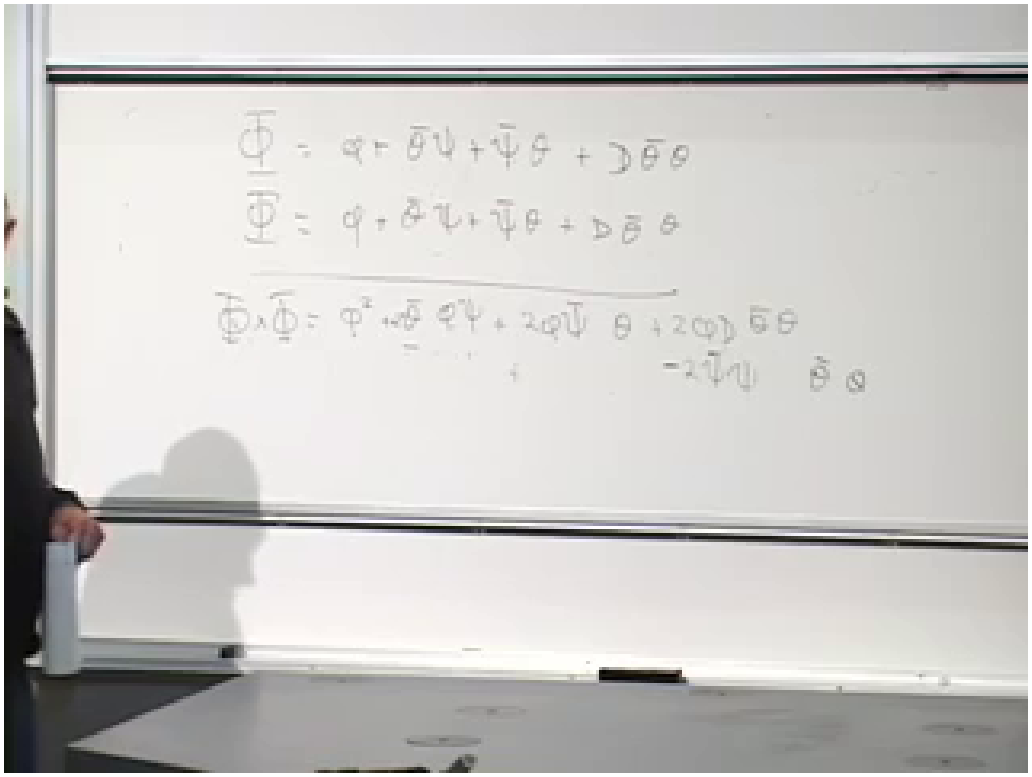

Figure 7.3: Product of superfields expanded in components. At this point in the lecture the safest cleaned interpretation is the square Φ^2.

The board work is valuable here because Susskind actually multiplies the terms out, line by line, and then gathers the $\bar{\theta}\theta$ coefficient. In cleaned form one finds

$$\Phi^2 = \left(\phi + \bar{\theta}\psi + \bar{\psi}\theta + D\,\bar{\theta}\theta\right)^2 \tag{7.29}$$

$$= \phi^2 + 2\phi\,\bar{\theta}\psi + 2\phi\,\bar{\psi}\theta + \left(2\phi D - 2\bar{\psi}\psi\right)\bar{\theta}\theta. \tag{7.30}$$

The ordering of the fermions has been normalized so that the bars remain on the left, exactly as Susskind says he prefers. Berezin integration then pulls out

$$\int d\theta\, d\bar{\theta}\,\Phi^2 = 2\phi D - 2\bar{\psi}\psi. \tag{7.31}$$

This is already a perfectly good supersymmetric contribution to the ordinary Lagrangian. More generally, the same logic applies to Φ^3, Φ^4, and indeed Φ^n: one computes only enough of the product to isolate the $\bar{\theta}\theta$ term.

At this point the lecture turns sharply. The expression above is supersymmetric, but it is not yet interesting dynamically, because it contains no time derivatives. So the next question becomes: what kind of derivative of a superfield is still the right sort of superspace object?

7.7.1 Question & Answer

Question. When is a constraint on a field compatible with the symmetry?

Answer. Susskind first answers this by an ordinary analogy. Suppose a field $f(x, y)$ is constrained by

$$\frac{\partial f}{\partial y} = 0. \tag{7.32}$$

Then f depends only on x. But if the theory is rotationally invariant, such a constraint is not stable under rotations, because the rotation generator L does not commute with ∂_y; schematically,

$$[\partial_y, L] \sim \partial_x \neq 0. \tag{7.33}$$

So the test is simple: a constraint is compatible with a symmetry only when the operator defining the constraint commutes with the symmetry generators. In the supersymmetric case the generators are fermionic, so the right condition is anticommutation rather than ordinary commutation.

That is exactly what motivates the covariant superspace derivatives. To avoid colliding with the auxiliary coefficient $D(t)$, we write them as $\mathcal{D}$ and $\bar{\mathcal{D}}$:

$$\mathcal{D} = \frac{\partial}{\partial\theta} - i\,\bar{\theta}\,\frac{\partial}{\partial t}, \qquad \bar{\mathcal{D}} = \frac{\partial}{\partial\bar{\theta}} - i\,\theta\,\frac{\partial}{\partial t}. \tag{7.34}$$

These are built to anticommute with the supercharges:

$$\{\mathcal{D}, q\} = \{\mathcal{D}, \bar{q}\} = \{\bar{\mathcal{D}}, q\} = \{\bar{\mathcal{D}}, \bar{q}\} = 0. \tag{7.35}$$

Now one may impose the chiral constraint

$$\bar{\mathcal{D}}\Phi = 0. \tag{7.36}$$

Because $\bar{\mathcal{D}}$ anticommutes with q, chirality is preserved by a supersymmetry variation:

$$\bar{\mathcal{D}}(q\Phi) = -\,q(\bar{\mathcal{D}}\Phi) = 0. \tag{7.37}$$

That is the superspace version of the ordinary compatibility test.

The lecture then solves the constraint by introducing the shifted time coordinate

$$\tau = t + i\,\bar{\theta}\theta. \tag{7.38}$$

A superfield that depends on τ and θ, but not on $\bar{\theta}$ independently, is automatically chiral:

$$\Phi(\tau, \theta) = \phi(\tau) + \bar{\psi}(\tau)\,\theta. \tag{7.39}$$

Indeed,

$$\bar{\mathcal{D}}\Phi(\tau, \theta) = \left(\frac{\partial}{\partial\bar{\theta}} - i\,\theta\,\frac{\partial}{\partial t}\right)\Phi(\tau, \theta) = i\,\theta\,\frac{\partial\Phi}{\partial\tau} - i\,\theta\,\frac{\partial\Phi}{\partial t} = 0, \tag{7.40}$$

since the dependence on t comes only through τ. Expanding back in t,

$$\phi(\tau) = \phi\Big(t + i\bar{\theta}\theta\Big) = \phi(t) + i\,\bar{\theta}\theta\,\dot{\phi}(t), \tag{7.41}$$

with no higher term because $(\bar{\theta}\theta)^2 = 0$. Thus the chiral superfield takes the explicit form

$$\Phi(t, \theta, \bar{\theta}) = \phi(t) + \bar{\psi}(t)\,\theta + i\,\bar{\theta}\theta\,\dot{\phi}(t). \tag{7.42}$$

Here the lecture reaches its most fragile point. Susskind now turns to the product of a chiral superfield with its complex conjugate, because that is the natural route to the kinetic super-Lagrangian. But at the board he begins to lose confidence in the ordering and the signs. He starts the conjugate expression, hesitates, and finally says that he is too tired to trust the derivation. That stopping point matters. We should not smooth it away. The lecture has earned the right to stop exactly where the speaker stops.

7.8 Summary

The chapter has unfolded in the same order as the lecture. We began with bosons and fermions at rest, described only by occupation numbers, and then rewrote the bosonic mode as an ordinary harmonic oscillator. We then met the first genuinely supersymmetric surprise: a first-order Lagrangian that is trivial for a commuting variable becomes dynamical for a Grassmann one. Repairing that naive fermionic model led us back to the matched bosonic and fermionic oscillators and to the supercharges that exchange them.

From there the lecture shifted perspective. The algebra of Q and $Q^\dagger$ was reinterpreted geometrically by enlarging time to superspace, with Grassmann coordinates θ and $\bar{\theta}$. Once that enlargement was in place, the problem became constructive: define superfields, define a superaction, use Berezin integration to extract the ordinary Lagrangian, and then build candidate invariant terms from powers of the superfield. The last step was subtler. Ordinary time derivatives were not enough; one needed covariant derivatives and then a constraint, $\bar{\mathcal{D}}\Phi = 0$, that was compatible with the symmetry. This produced the chiral superfield and brought the lecture to the edge of the kinetic construction.

The final punchline is the one Susskind insists on, even after the algebra becomes exhausting. A supersymmetric Lagrangian is not merely a convenient way of packaging fields. It enforces a matching between bosons and fermions strong enough to produce exact cancellation statements. That is why the machinery matters. At the same time, the lecture ends honestly: the clean $\bar{\Phi}\Phi$ computation is deferred, and the later breaking of supersymmetry is left for another discussion. The notes should preserve both facts.

Chapter 8

Supersymmetry in Four Dimensions

The previous lecture treated supersymmetry in a toy world with only time. Leonard Susskind's Lecture 8 begins by undoing that simplification. If energy is part of a four-vector, then a supersymmetry algebra involving only the Hamiltonian cannot be the full relativistic story. The lecture then does two things in sequence. First, it builds the four-dimensional algebra carefully enough that we see why spinorial Grassmann variables are unavoidable. Second, it constructs a simple supersymmetric field theory and follows the bookkeeping until ordinary bosons, fermions, masses, vertices, and cancellations appear. The present chapter is curated from the lecture by LazyingArt LLC, and it keeps the lecture's order on purpose: the formalism is introduced only when the preceding difficulty has made it necessary.

8.1 From Time-Only Supersymmetry to Lorentz Invariance

The starting point is the one-dimensional supersymmetry algebra from the previous lecture,

$$\{Q^\dagger, Q\} = 2H, \tag{8.1}$$

where the right-hand side involves only the Hamiltonian. In a purely time-dependent model this is natural. But now the lecture asks the objection that forces everything that follows: energy is only the time component of momentum, so how can a relativistic theory speak about H and ignore the spatial components?

That question is not merely cosmetic. Under a Lorentz transformation, the time component of a four-vector mixes with the spatial ones. So if the algebra is truly Lorentz covariant, the right-hand side of the anticommutator cannot remain a lone H. It must know about the whole four-momentum P_μ. The lecture therefore pivots at once from the toy model to the relativistic problem.

But instead of writing the finished superalgebra immediately, Susskind does something more instructive. He asks what kind of relativistic object ought to sit at the heart of the construction. To get a feel for that, he returns to the massless Dirac equation in its simplest two-component form.

8.2 Massless Chiral Fermions and the Hamiltonian $H = \sigma \cdot p$

A massive Dirac fermion comes with both handednesses. A massless fermion need not. This is the first physical fact the lecture wants to establish, because in four dimensions the supersymmetry

generators and the Grassmann coordinates themselves will have to transform as spinors, and the cleanest relativistic spinor on the table is the massless chiral one.

For such a particle we work with a two-component spinor,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \qquad \text{or, with an index, } \psi_\alpha. \tag{8.2}$$

The lecture discusses helicity and chirality in deliberately practical terms. Helicity is tied to $\sigma \cdot p$, chirality to the gamma-matrix structure; in the massless case under discussion the two coincide, and that is the case we need here.

8.2.1 Question & Answer

Question. Why can a massless particle be purely one-handed, whereas a massive particle cannot?

Answer. For a massive particle, helicity is frame-dependent. One can always boost to a frame that overtakes the particle. Its spin does not flip, but its momentum reverses relative to the observer, so a state that looked right-handed can look left-handed. A massive relativistic theory must therefore contain both handednesses. A genuinely massless particle moves at the speed of light, so it cannot be overtaken. In the massless case, where chirality and helicity coincide, a purely one-handed description is consistent.

With that picture in mind, the lecture writes the Hamiltonian

$$H = \boldsymbol{\sigma} \cdot \mathbf{p}. \tag{8.3}$$

Here $\boldsymbol{\sigma} = (\sigma^1, \sigma^2, \sigma^3)$ are the Pauli matrices. Susskind immediately interprets the sign of $\sigma \cdot p$: the relative orientation of spin and momentum determines the sign of the energy, and one then proceeds to the usual relativistic reinterpretation of negative-energy states.

The next move is concrete. Replace H by $i\partial_t$ and p_i by $-i\partial_i$. Then the Hamiltonian equation becomes

$$i\frac{\partial \psi}{\partial t} = -\, i\, \sigma^i \partial_i \psi. \tag{8.4}$$

The lecture then rewrites this first in a partially covariant form,

$$i\frac{\partial \psi}{\partial x^0} + i\, \sigma^i \partial_i \psi = 0, \tag{8.5}$$

and only afterward introduces the final compact notation. Defining

$$\sigma^0 = \mathbf{1}, \tag{8.6}$$

we may combine the four pieces into

$$i\, \sigma^\mu \partial_\mu \psi = 0. \tag{8.7}$$

This is the chiral Dirac equation in the lecture's form. The important narrative point is that the equation was not introduced as abstract notation. It arose by starting with the chiral Hamiltonian $H = \sigma \cdot p$, turning it into a wave equation, and only then packaging it into a Lorentz-looking formula. That rhythm matters because it prepares the next leap: if the relativistic fermion is a two-component anticommuting object, then the Grassmann coordinates of superspace must be as well.

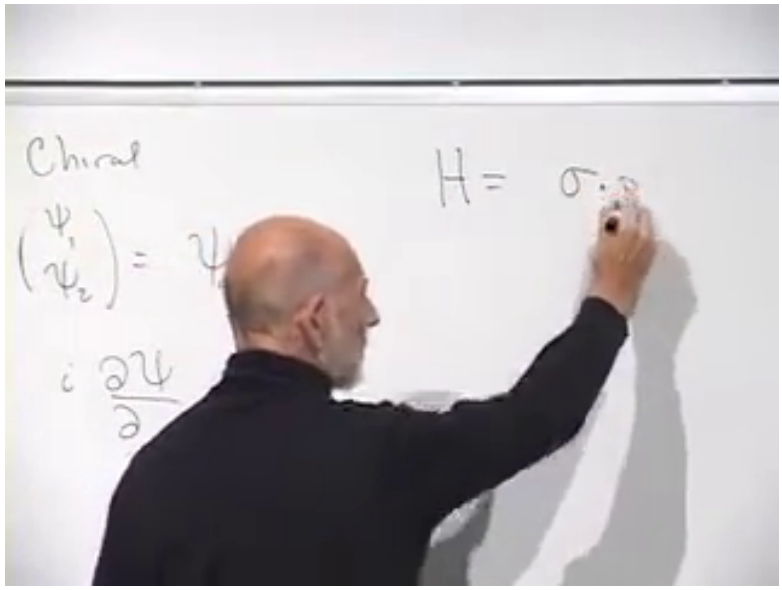

Figure 8.1: Chiral-spinor setup and the Hamiltonian $H = \sigma \cdot p$.

8.3 The Four-Dimensional Superalgebra

We now return to supersymmetry proper. The chiral spinor has shown us what kind of object Lorentz invariance wants. So the Grassmann coordinates cannot remain a single θ; they must be θ_α and their conjugates $\bar{\theta}_\alpha$. There are therefore four real Grassmann directions altogether. A general superfield is a function of

$$x^\mu, \qquad \theta_\alpha, \qquad \bar{\theta}_\alpha,$$

and because the Grassmann variables anticommute, its expansion terminates at fourth order. The lecture emphasizes this bookkeeping point before doing anything more sophisticated.

Now we ask what replaces the one-dimensional relation $\{Q^\dagger, Q\} = 2H$. The right-hand side must involve P_μ, but that immediately creates an index mismatch. The left-hand side carries spinor indices α, β; the momentum carries a Lorentz index μ. The sigma matrices enter because something has to soak up μ and return the pair of spinor indices. This is one of the lecture's most characteristic moves: the next mathematical object is introduced because the previous formula quite visibly does not fit together.

In schematic lecture form, the four-dimensional algebra is

$$\{Q^\dagger_\alpha, Q_\beta\} = 2\,\sigma^\mu_{\alpha\beta} P_\mu = 2i\,\sigma^\mu_{\alpha\beta}\partial_\mu, \tag{8.8}$$

$$\{Q_\alpha, Q_\beta\} = 0, \qquad \{Q^\dagger_\alpha, Q^\dagger_\beta\} = 0, \tag{8.9}$$

$$[Q_\alpha, P_\mu] = 0, \qquad [Q^\dagger_\alpha, P_\mu] = 0, \qquad [P_\mu, P_\nu] = 0. \tag{8.10}$$

The precise ordering of spinor indices is not stable in the lecture itself; Susskind discusses and revises it on the board. We therefore keep the index structure schematic where that is enough to express the logic.

The supercharges act as differential operators on superspace. In the lecture their detailed indices

are soon suppressed because, as he says, the indexology gets awful. In that same spirit we write

$$Q = \frac{\partial}{\partial\theta} + i\,\bar{\theta}\sigma^\mu\partial_\mu, \qquad \bar{Q} = \frac{\partial}{\partial\bar{\theta}} + i\,\sigma^\mu\theta\,\partial_\mu, \tag{8.11}$$

again with spinor contractions understood schematically. These operators generate supersymmetry transformations:

$$\delta\Phi = (\epsilon Q + \bar{\epsilon}\,\bar{Q})\Phi. \tag{8.12}$$

The lecture's point is simple and geometric. Just as P_μ generates infinitesimal shifts in x^μ, the supercharges generate infinitesimal shifts in the Grassmann directions, accompanied by the compensating motion in space-time demanded by the algebra.

8.4 Chiral Superfields as a Consistent Constraint

At this stage the lecture changes register. It does not yet want component transformation laws. It wants manageable superfields. A completely general superfield depending on all of $x, \theta, \bar{\theta}$ is too elaborate for the present purpose. So the local problem becomes: can we impose a simplifying constraint and still remain inside a supersymmetric class of fields?

Susskind motivates this through ordinary symmetry. The condition

$$v_x = 0$$

is not rotationally invariant: rotate the vector and v_x generally reappears. By contrast,

$$v^2 = 1$$

is preserved by rotations. The lesson is that a constraint must be tested against the symmetry. If the symmetry maps the constrained configurations back into constrained configurations, then the constraint is compatible.

8.4.1 Question & Answer

Question. What makes the constraint $\bar{D}\Phi = 0$ compatible with supersymmetry?

Answer. Suppose Φ obeys $\bar{D}\Phi = 0$. After a supersymmetry variation we must still have $\bar{D}(\Phi + \delta\Phi) = 0$. A sufficient condition is that the constraint operator commute or anticommute with the symmetry generator. Then

$$\bar{D}(Q\Phi) = \pm Q(\bar{D}\Phi) = 0,$$

and similarly for $\bar{Q}$. The transformed field stays within the constrained class. That is why the lecture first asks for operators that anticommute with the supercharges before imposing any specific constraint.

The operators used for this purpose are D and $\bar{D}$, obtained from the supercharges by flipping the sign of the space-time shift term:

$$D = \frac{\partial}{\partial\theta} - i\,\bar{\theta}\sigma^\mu\partial_\mu, \qquad \bar{D} = \frac{\partial}{\partial\bar{\theta}} - i\,\sigma^\mu\theta\,\partial_\mu. \tag{8.13}$$

These are the differential operators from which supersymmetry-compatible constraints are built.

The lecture then chooses the special condition

$$\bar{D}\Phi = 0. \tag{8.14}$$

It is important to keep the logic straight. A chiral superfield is not introduced as a standard textbook object that one is simply expected to know. It emerges as the answer to the question, "What kind of simplifying constraint can we impose without breaking supersymmetry?"

8.5 Solving $\bar{D}\Phi = 0$ and Expanding the Chiral Superfield

We now solve the constraint. The lecture first compares it with the simpler equation $\partial\Phi/\partial\bar{\theta} = 0$. That equation would merely say that Φ does not depend on $\bar{\theta}$ at all. But $\bar{D}\Phi = 0$ is not that. It says that $\bar{\theta}$ may appear, but only through a very special combination with x.

Define

$$y^\mu = x^\mu + i\,\bar{\theta}\sigma^\mu\theta. \tag{8.15}$$

The claim is that the general solution of the chiral constraint is

$$\Phi = \Phi(y, \theta). \tag{8.16}$$

In other words, there is no independent $\bar{\theta}$-dependence; the only $\bar{\theta}$-dependence is hidden inside y^μ.

The lecture checks this by the chain rule. If Φ depends on $\bar{\theta}$ only through y^μ, then

$$\frac{\partial}{\partial\bar{\theta}_\alpha}\Phi(y,\theta) = \frac{\partial\Phi}{\partial y^\mu}\,\frac{\partial y^\mu}{\partial\bar{\theta}_\alpha} = i(\sigma^\mu\theta)_\alpha\,\partial_\mu\Phi. \tag{8.17}$$

But the second term in $\bar{D}_\alpha$ contributes precisely the opposite expression. Therefore

$$\bar{D}_\alpha\Phi = \left(\frac{\partial}{\partial\bar{\theta}_\alpha} - i(\sigma^\mu\theta)_\alpha\partial_\mu\right)\Phi = 0. \tag{8.18}$$

That is the whole mechanism. The $\bar{\theta}$-variation produced by the dependence of y^μ cancels the explicit space-time derivative term in $\bar{D}$.

This is the point where the lecture pauses and gives the superfield a name. A field obeying $\bar{D}\Phi = 0$ is a *chiral superfield*. It is also, in general, complex.

Since the chiral superfield depends only on y^μ and the two components of θ, its Grassmann expansion terminates at order θ^2:

$$\Phi(y,\theta) = \phi(y) + \theta\psi(y) + \theta^2 F(y). \tag{8.19}$$

We keep the notation deliberately schematic, as the lecture does. The three component fields are a complex scalar ϕ, a two-component fermion ψ, and an auxiliary complex scalar F. Susskind already signals that F is mathematically necessary but dynamically peculiar. That warning will matter as soon as the Lagrangian is unpacked.

8.6 Superspace Actions and the Free Example $\Phi^*\Phi$

The lecture now turns from kinematics to construction. A super-Lagrangian depends on $x, \theta, \bar{\theta}$, and the action is obtained by integrating over the whole superspace:

$$S = \int d^4x\, d^2\theta\, d^2\bar{\theta}\, \mathcal{L}_{\text{sup}}. \tag{8.20}$$

The Grassmann integrals simply pick out the highest component, that is, the coefficient of $\theta^2\bar{\theta}^2$. So the whole method reduces to a peculiar but efficient piece of bookkeeping: write down a superspace expression, expand it, and keep the top Grassmann term.

The first example is

$$\mathcal{L}_{\text{sup}} = \Phi^*\Phi. \tag{8.21}$$

That looks almost embarrassingly simple. There are no derivatives of the superfield anywhere. The lecture makes a point of this, because the result will nonetheless contain ordinary kinetic terms.

For the conjugate chiral field,

$$\Phi^* = \Phi^*(x^\mu - i\,\bar{\theta}\sigma^\mu\theta, \bar{\theta}). \tag{8.22}$$

Thus the action contains

$$\Phi^*(x - i\bar{\theta}\sigma\theta, \bar{\theta})\, \Phi(x + i\bar{\theta}\sigma\theta, \theta). \tag{8.23}$$

Before expanding, Susskind uses a trick that is both simple and important. Since we are integrating over x, we may shift the integration variable without changing the integral. If we shift x so as to remove the Grassmann correction from one factor, it doubles in the other:

$$\Phi^*(x - i\bar{\theta}\sigma\theta, \bar{\theta})\, \Phi(x + i\bar{\theta}\sigma\theta, \theta) \longrightarrow \Phi^*(x, \bar{\theta})\, \Phi(x + 2i\bar{\theta}\sigma\theta, \theta). \tag{8.24}$$

The lecture lingers here because this shift is where the derivatives will come from. Without it, the super-Lagrangian $\Phi^*\Phi$ would never produce space-time propagation.

We now Taylor-expand the shifted argument. For the scalar component,

$$\phi(x + 2i\bar{\theta}\sigma\theta) = \phi(x) + 2i(\bar{\theta}\sigma^\mu\theta)\, \partial_\mu\phi(x) + (\bar{\theta}\sigma^\mu\theta)(\bar{\theta}\sigma^\nu\theta)\, \partial_\mu\partial_\nu\phi(x), \tag{8.25}$$

where, just as in the lecture, we suppress numerical coefficients beyond the clearly secure structure. The spinor term is treated similarly, but it truncates earlier because once too many θ's appear the expression vanishes. For the auxiliary term one need not expand the argument at all, since any additional shift would create too many Grassmann factors.

What survives the $d^2\theta\, d^2\bar{\theta}$ integration is therefore the $\theta^2\bar{\theta}^2$ coefficient. The board and transcript securely fix only the schematic structure, not the precise prefactors. In that schematic form the ordinary component Lagrangian is

$$\mathcal{L} \sim \phi^*\Box\phi + \bar{\psi}\,\sigma^\mu\partial_\mu\psi + F^*F. \tag{8.26}$$

Up to coefficients and integrations by parts, this is exactly what the blackboard is gathering on its lower line. The scalar term is the Klein–Gordon kinetic term, the fermionic term is the chiral Dirac kinetic term, and the last term is the algebraic F-term.

The lecture immediately draws the moral. The ordinary derivatives were not inserted by hand into the super-Lagrangian. They were generated by the shifted dependence on $x \pm i\bar{\theta}\sigma\theta$. Superspace looks abstract, but it is not inert. It turns a formally simple expression into a nontrivial space-time field theory.

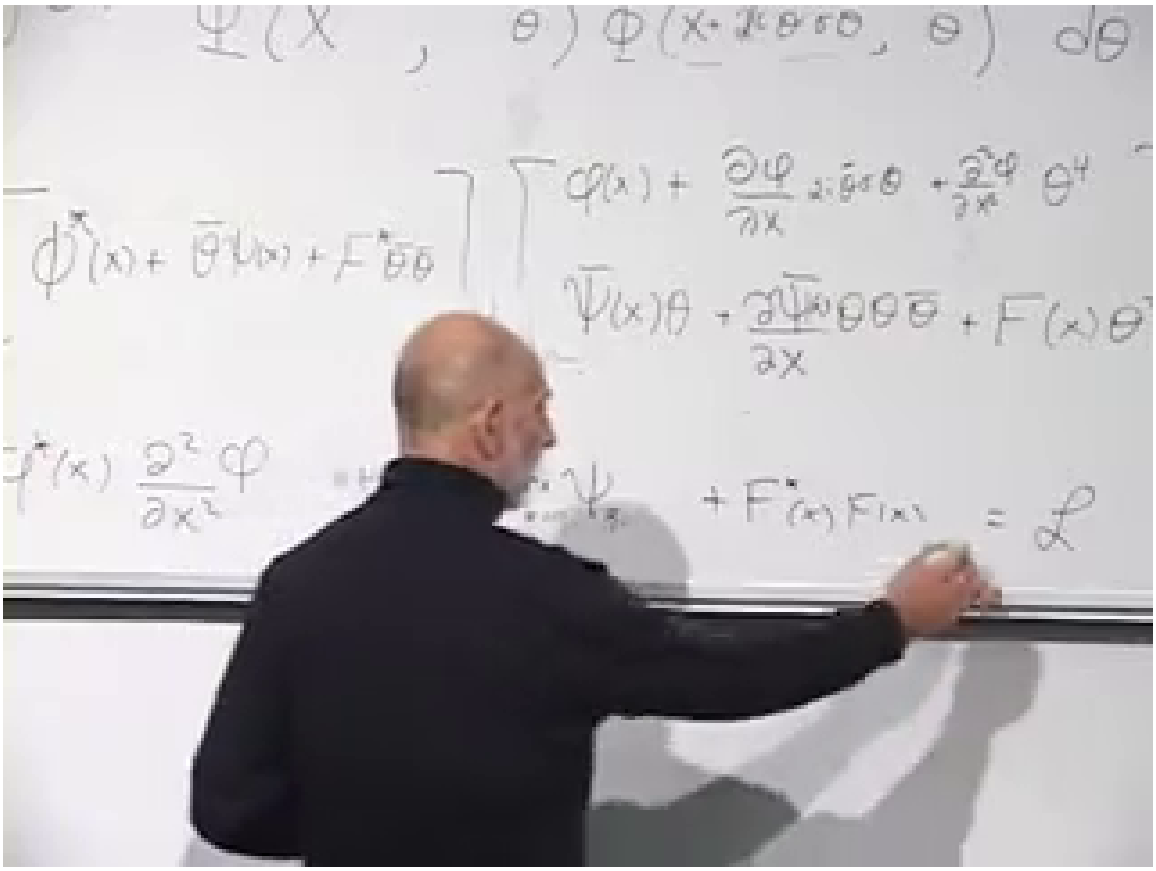

Figure 8.2: Component expansion leading to the ordinary Lagrangian.

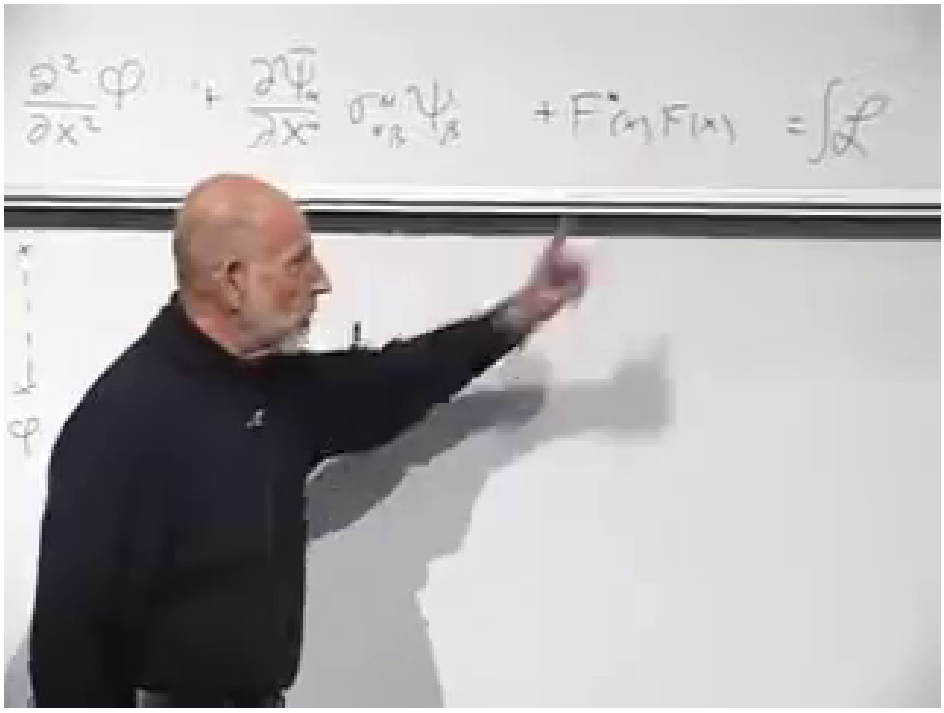

Figure 8.3: Auxiliary-field term in the simple supersymmetric Lagrangian.

8.7 Mass Terms, Interaction Terms, and the Role of the Auxiliary Field

Having obtained the free theory, the lecture asks for something more interesting. First come masses, then interactions. The guiding idea is the same throughout: write a very simple superspace expression, extract the relevant Grassmann coefficient, and let the auxiliary field translate part of the holomorphic structure into ordinary bosonic terms.

Before doing that, Susskind isolates the F-term once more. In the free theory we have

$$\mathcal{L} \supset F^* F. \tag{8.27}$$

The crop in Figure 8.3 is valuable precisely because it shows this piece at the place where the blackboard turns into the ordinary Lagrangian.

This term contains no derivatives. Therefore the Euler–Lagrange equation for F is purely algebraic. In the free case,

$$\frac{\partial \mathcal{L}}{\partial F} = 0 \quad \Rightarrow \quad F^* = 0, \qquad \frac{\partial \mathcal{L}}{\partial F^*} = 0 \quad \Rightarrow \quad F = 0. \tag{8.28}$$

This is why the lecture says that F is trivial. It is not a propagating boson in the ordinary sense.

8.7.1 Question & Answer

Question. Why is it legitimate to integrate only over $d^2\theta$ when the expression has no $\bar{\theta}$-dependence?

Answer. If one kept the full measure $d^2\theta\, d^2\bar{\theta}$, then a purely chiral expression would indeed vanish upon $\bar{\theta}$-integration. The lecture explicitly notices this and then uses the standard supersymmetric repair: when the integrand is chiral, one integrates only over the chiral measure $d^2\theta$. Susskind remarks that this deserves justification. At this point in the lecture, however, it is taken as the correct supersymmetric rule and then immediately used.

With that understood, consider the quadratic chiral term

$$S_m = \int d^4x\, d^2\theta\; m\, \Phi^2 + \text{c.c.} \tag{8.29}$$

Using the chiral expansion

$$\Phi = \phi + \theta\psi + \theta^2 F, \tag{8.30}$$

we extract the θ^2 coefficient of Φ^2. The lecture does this schematically rather than with exact combinatorial factors:

$$\Phi^2\big|_{\theta^2} \sim F\phi + \psi\psi. \tag{8.31}$$

Hence

$$\mathcal{L}_m \sim m\, F\phi + m\, \psi\psi + \text{c.c.} \tag{8.32}$$

The fermionic term is a two-component mass term; Susskind refers to it as a Majorana-type mass term.

The bosonic mass is hidden inside the auxiliary field. Combining the new term with the free F^*F piece gives

$$\mathcal{L}(F) = F^*F + m\, F\phi + m\, F^*\phi^*. \tag{8.33}$$

Its equations of motion are

$$F^* + m\phi = 0, \qquad F + m\phi^* = 0. \tag{8.34}$$

Substituting back yields

$$\mathcal{L}_{\text{boson mass}} \sim m^2\, \phi^*\phi. \tag{8.35}$$

This is the first strong supersymmetric payoff of the lecture. The same supersymmetric structure that gives a fermion mass also gives the boson a matching mass. The notation looks different — m in the fermionic equation, m^2 in the bosonic Lagrangian — but the physical mass is the same.

The next example is the cubic term

$$S_g = \int d^4x\, d^2\theta\; g\, \Phi^3 + \text{c.c.} \tag{8.36}$$

Again we keep only the θ^2 part. The lecture identifies two schematic contributions:

$$\Phi^3\big|_{\theta^2} \sim \phi\, \psi\psi + \phi^2 F. \tag{8.37}$$

So the interaction Lagrangian contains

$$\mathcal{L}_{\text{int}} \sim g\, \phi\, \psi\psi + g\, \phi^*\bar{\psi}\bar{\psi} + g\, \phi^2 F + g\, \phi^{*2} F^*. \tag{8.38}$$

The first pair of terms is the Yukawa interaction. The second pair couples scalars to the auxiliary field. Because F is not an ordinary propagating particle, these terms do not give a new long-range mode. Instead, eliminating F generates an effective quartic bosonic interaction. In schematic form,

$$\mathcal{L}_{\text{int}} \sim g\, \phi\, \psi\psi + g\, \phi^*\bar{\psi}\bar{\psi} + g^2\, \phi^2\phi^{*2}. \tag{8.39}$$

The point is not the exact coefficient. The point is the relation: the four-boson coupling is tied to the square of the Yukawa coupling.

Figure 8.4: Pedagogical redraw of the F-field propagator as a local contact. The lecture's claim is that with no derivatives the propagation collapses to a delta function at one point.

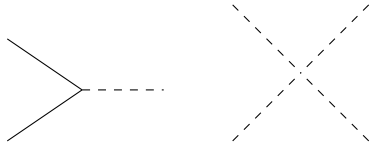

Figure 8.5: Pedagogical redraws of the interactions generated by the Φ^3 term: the Yukawa vertex and the effective four-boson contact induced by the auxiliary field.

8.8 Feynman-Diagram Interpretation and the Supersymmetric Payoff

The lecture now shifts viewpoint once more. Instead of solving equations of motion, it asks what the component terms mean diagrammatically.

First, the auxiliary field. Since F^*F contains no derivatives, the corresponding propagator does not spread from one point to another in the ordinary way. Susskind says that it is effectively a delta function: the line begins and ends at the same point. The little dashed doodle on the blackboard near Figure 8.3 supports exactly this reading, even though it is not a precise finished diagram. A clean pedagogical redraw is therefore useful.

Second, the cubic chiral term gives two visually different interactions that are numerically linked: the Yukawa vertex $\phi\psi\psi$ and the effective four-boson contact obtained after eliminating F.

This numerical linkage is the real content of supersymmetry here. One could always invent a boson and a fermion and simply assign them the same mass. That would be a free theory and nothing more. What is nontrivial is that once interactions are present, supersymmetry keeps relating apparently different couplings.

The lecture's final concrete example is the boson self-energy. There are two lowest-order contributions worth comparing. One comes from the bosonic quartic interaction. The other comes from a fermion loop built from two Yukawa vertices. The lecture does not evaluate the integrals in detail. It only asks for their structure and their coefficients. The first comes with a coefficient of order g^2. The second also comes with a coefficient of order g^2, since there is one g at each Yukawa vertex. The masses running in the loops are also matched by supersymmetry.

Now the payoff can be stated clearly. Supersymmetry does not merely place a boson and a fermion in the same multiplet. It enforces equal masses and specific relations among couplings, and those relations are precisely what allow bosonic and fermionic loop effects to cancel. The lecture stresses

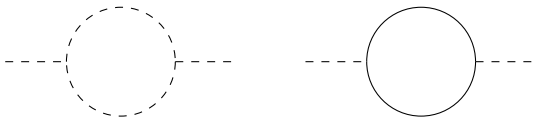

Figure 8.6: Schematic paired boson self-energy diagrams. The left loop is bosonic; the right loop is fermionic. Supersymmetry ties the masses and couplings together, while fermion and boson loops contribute with opposite signs.

that this pairing persists to all orders. That is the real force of the construction.

Only after making that point does Susskind step back and remind us that exact supersymmetry is not what nature shows us directly. So the lecture ends by pointing forward: one must next understand supersymmetry breaking, and then one may ask how the whole structure influences grand unification, where $SU(3) \times SU(2) \times U(1)$ is invited to enlarge to something like $SU(5)$.

8.9 Summary

The lecture begins from a defect in the time-only toy model and uses that defect to drive the whole discussion forward. If $\{Q^\dagger, Q\}$ closes only on the Hamiltonian, then Lorentz invariance is already in danger. That difficulty leads us first to the massless chiral fermion and its Hamiltonian $H = \sigma \cdot p$, then to the covariant chiral equation $i\sigma^\mu \partial_\mu \psi = 0$, and only then to spinorial Grassmann coordinates and the four-dimensional superalgebra.

Once superspace is in place, the lecture turns to the question of simplification. The chiral constraint $\bar{D}\Phi = 0$ is introduced not as a standard definition but as a consistency-preserving reduction of the most general superfield. Solving that constraint gives the chiral superfield $\Phi(y, \theta)$, and the superspace action principle then unfolds into an ordinary field theory. The free example $\Phi^*\Phi$ produces scalar and fermion kinetic terms together with the auxiliary-field term F^*F. The quadratic term Φ^2 gives equal boson and fermion masses. The cubic term Φ^3 gives a Yukawa coupling and an induced quartic bosonic interaction whose coefficient is fixed by supersymmetry. Finally, the lecture interprets these relations diagrammatically: equal masses and linked couplings are what make the bosonic and fermionic loop contributions cancel.

That is the proper closing note. Exact supersymmetry is extraordinarily rigid and powerful. It is not manifest as an exact symmetry of nature, but precisely for that reason the next problems are compelling: how supersymmetry breaks, and how its presence may reshape the unification of forces.

Chapter 9

Supersymmetry Breaking and the Opening of Grand Unification

These notes follow Leonard Susskind's ninth lecture in the supplementary course on supersymmetry and grand unification, with curation by LazyingArt LLC. The lecture announces its own route very clearly, and the chapter should follow that route just as clearly: first symmetry breaking in a familiar setting, then supersymmetry breaking in the simplest model from the previous lecture, then the effect of gravity, and finally the first opening move toward grand unification. The guiding idea is that symmetry breaking is first understood as the splitting of degenerate energy levels; only later do we reinterpret the same logic as the splitting of particle masses.

9.1 Symmetry Breaking as Level Splitting

Susskind begins from the most concrete possible place. Before speaking of supersymmetry at all, he wants a physical picture of symmetry breaking that one already trusts. He mentions the ferromagnet as the simplest model with the right qualitative structure: neighboring little magnetic moments prefer to align, and once they do, the system chooses a direction even though the underlying laws do not.

But before using the ferromagnet itself, he turns to an even more elementary example: an atom in a magnetic field. Suppose the atom has total angular momentum j. Then there are

$$2j + 1 \tag{9.1}$$

states, labelled by

$$m_J = -j, -j+1, \ldots, j. \tag{9.2}$$

In the absence of any preferred direction, rotational invariance forces these states to be degenerate:

$$E_{-j} = E_{-j+1} = \cdots = E_j. \tag{9.3}$$

This is the first point the lecture wants to make. Symmetry shows up in the spectrum as degeneracy.

Now let a magnetic field pick out a direction. The rotational symmetry is broken by the environment, and the degeneracy is lifted. Susskind deliberately keeps the discussion schematic, but the pattern is the familiar Zeeman pattern,

$$\Delta E_{m_J} \propto B\, m_J. \tag{9.4}$$

The exact coefficient is not the issue. What matters is that states which were identical in energy because a symmetry rotated them into one another are now split once the symmetry is broken. This is the template we will later read as boson–fermion mass splitting.

The lecture then makes the decisive move. Suppose the magnetic field is not imposed from outside, but is the field produced by a ferromagnet filling space. Then the atom again sees a distinguished direction and again sees its levels split. Yet this new direction is not built into the laws. The ferromagnet could have pointed anywhere. All uniformly aligned configurations have the same energy. The asymmetry is not in the equations; it is in the vacuum.

This is why the ferromagnet is such a powerful example. It simulates explicit symmetry breaking while actually being a case of spontaneous symmetry breaking. From the inside, the world looks asymmetric. From the outside, or more precisely from the point of view of the laws, it is not.

9.1.1 Question & Answer

Question. If one lives inside the broken phase, how can one tell the difference between spontaneous symmetry breaking and a world whose laws were asymmetric from the beginning?

Answer. The answer is the existence of Goldstone modes. If we rotate the direction of magnetization everywhere by the same amount, that costs no energy at all. Now do something only slightly less global: let the direction of magnetization vary very slowly through space. The system does not remain frozen in that distorted configuration; it develops long-wavelength spin waves, or magnons.

The important point is that the energy of these modes becomes arbitrarily small when the wavelength becomes arbitrarily large:

$$E(k) \to 0 \qquad \text{as} \qquad k \to 0. \tag{9.5}$$

Long wavelength means low momentum, and an excitation whose energy goes to zero as $k \to 0$ is massless. Thus an observer inside the ferromagnet can perform experiments, study these spin waves, and infer that the underlying world is not explicitly asymmetric. The massless modes are the telltale sign of spontaneous symmetry breaking.

The lecture wants us to carry exactly this pattern forward. A broken vacuum is not merely a background with some numbers turned on. It comes with a diagnostic structure: degenerate states split, and broken generators acting on the vacuum create new low-energy states.

9.2 Vacuum Criteria and Order Parameters

Susskind next translates the ferromagnet into the language of the vacuum. Let M denote the magnetization field. If the system is so hot, or so fluctuating, that no direction is preferred on average, then

$$M = 0, \tag{9.6}$$

and the vacuum is symmetric. If any component of M is nonzero, then the vacuum has picked a direction:

$$M \neq 0. \tag{9.7}$$

This is the order parameter. The laws remain rotationally invariant, but the vacuum does not.

He then gives the more formal statement. Let J denote the generator of rotations. Acting with J on a state gives the infinitesimal change produced by a small rotation. If the vacuum is symmetric, a small rotation changes nothing,

$$J|0\rangle = 0. \tag{9.8}$$

If the vacuum is not symmetric, then the same generator produces a genuinely new state,

$$J|0\rangle \neq 0. \tag{9.9}$$

The lecture immediately interprets that state physically: it is the infinitely long wavelength Goldstone excitation,

$$J|0\rangle \sim |\text{Goldstone}, \, k = 0\rangle. \tag{9.10}$$

This is a compact formal statement, but it carries the full physical content of the ferromagnet story. The vacuum is broken precisely when the symmetry generator does not annihilate it.

At this point the lecture has done all the preparatory work it needs. We now know what to look for when supersymmetry breaks: some component of the vacuum must play the role that M played for the ferromagnet, and the supersymmetry generator must fail to annihilate the vacuum. The only difference will be that supersymmetry generators are fermionic.

9.3 Recap of the Chiral Superfield

Only now does Susskind say, in effect: let us review supersymmetry. The point is not to introduce a new formalism from scratch. The point is to return to the one simple supersymmetric field theory already on the table and extract from it just what is needed for the vacuum analysis.

We begin with the chiral superfield Φ. In the form relevant here it depends effectively on y and θ, where

$$y^\mu = x^\mu + i\bar{\theta}\sigma^\mu\theta. \tag{9.11}$$

Its component expansion is

$$\Phi(y, \theta) = \phi(y) + \psi(y)\theta + F(y)\theta^2. \tag{9.12}$$

The field ϕ is the scalar component, ψ the fermion component, and F the auxiliary field. Susskind reminds us that the chiral superfield must be complex; trying to make it real is not stable under supersymmetry.

The basic superspace action from the previous lecture is

$$\int d^4\theta \, \Phi^\dagger \Phi. \tag{9.13}$$

To integrate over the Grassmann variables means to pick out the highest $\theta^2\bar{\theta}^2$ component of $\Phi^\dagger\Phi$. Up to the numerical factors that Susskind intentionally suppresses, this produces the ordinary component Lagrangian

$$\mathcal{L}_{\text{free}} = \partial_\mu\phi^* \, \partial^\mu\phi + \bar{\psi}\sigma^\mu\partial_\mu\psi + F^*F. \tag{9.14}$$

The first term is the scalar kinetic term, the second is the usual fermion kinetic term, and the third is algebraic. There are no derivatives on F. This is the structural fact on which the later derivation rests: F is not dynamical. It is present only to be eliminated.

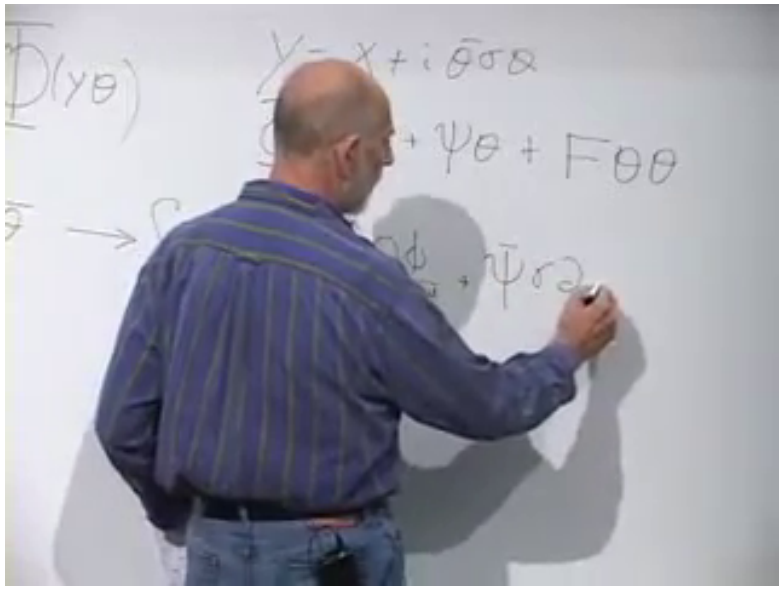

Figure 9.1: Chiral superfield recap and the fermion kinetic term.

9.4 From the Superpotential to the Scalar Potential

The lecture now adds the second ingredient. Last time the example was Φ^2, which generates a mass term. This time Susskind wants to go further, but characteristically he does so in stages. He first says: let us be a little more modest and take Φ^n, then later add different powers together into a general function.

There is a notational issue here. On the board in this stretch, the same Roman-looking letter is used too casually for two different things. To avoid confusion, we will write $W(\Phi)$ for the superpotential and $V(\phi)$ for the ordinary scalar potential that results after eliminating F.

Start from

$$\int d^2\theta\, \Phi^n + \text{h.c.} \tag{9.15}$$

with

$$\Phi = \phi + \psi\theta + F\theta^2. \tag{9.16}$$

What survives the $d^2\theta$ integral? Only terms with exactly two θ's. There are only two kinds of such terms:

1. choose one $F\theta^2$ and $n-1$ factors of ϕ, giving $nF\phi^{n-1}$;
2. choose two factors of $\psi\theta$ and $n-2$ factors of ϕ, giving $\frac{n(n-1)}{2}\phi^{n-2}\psi\psi$.

So one should read the board's arrow, quite literally, as

$$\int d^2\theta\, \Phi^n + \text{h.c.} \;\rightsquigarrow\; nF\phi^{n-1} + \frac{n(n-1)}{2}\phi^{n-2}\psi\psi + \text{h.c.} \tag{9.17}$$

The lecture explicitly pauses to say that the arrow does *not* mean equality. It means: these are the terms selected by the $d^2\theta$ integral.

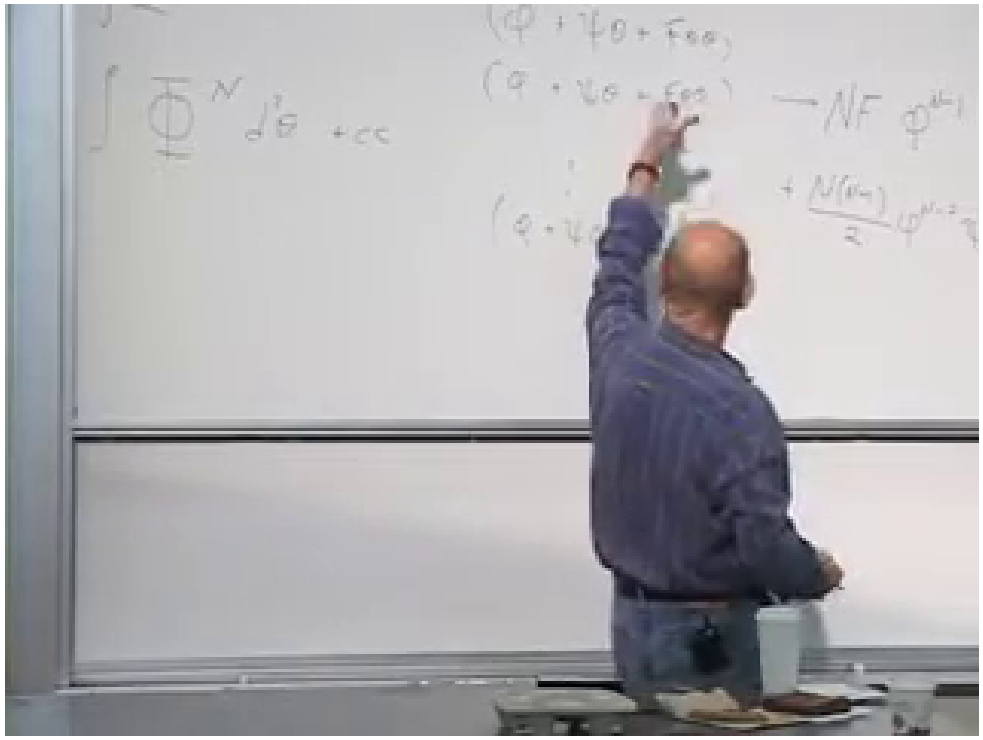

Figure 9.2: Extracting the two-θ term from Φ^n.

Now the pattern becomes transparent. If $W(\Phi) = \Phi^n$, then $W'(\phi) = n\phi^{n-1}$. More generally,

$$\int d^2\theta\, W(\Phi) + \text{h.c.} \;\rightsquigarrow\; F\, W'(\phi) - \frac{1}{2} W''(\phi)\psi\psi + \text{h.c.} \tag{9.18}$$

The lecture is not interested in the fermion bilinear at this moment, and neither are we. The vacuum question will be decided by the F-dependence.

Eliminating the auxiliary field. This is the moment at which the algebra tightens. Collect the terms containing F. In cleaned notation they are

$$\mathcal{L}(F) = F\, W'(\phi) + F^* W'(\phi)^* + F^* F. \tag{9.19}$$

The frame supports this structure, although the conjugation on the board is casual and locally unstable.

Because F has no kinetic term, the Euler–Lagrange equation reduces to a purely algebraic condition:

$$\frac{\partial \mathcal{L}}{\partial F} = 0 \quad \Rightarrow \quad W'(\phi) + F^* = 0, \tag{9.20}$$

$$\frac{\partial \mathcal{L}}{\partial F^*} = 0 \quad \Rightarrow \quad W'(\phi)^* + F = 0. \tag{9.21}$$

Thus

$$F^* = -W'(\phi), \qquad F = -W'(\phi)^*. \tag{9.22}$$

Substituting back, all F-dependence collapses into a positive definite ordinary potential,

$$V(\phi) = |W'(\phi)|^2. \tag{9.23}$$

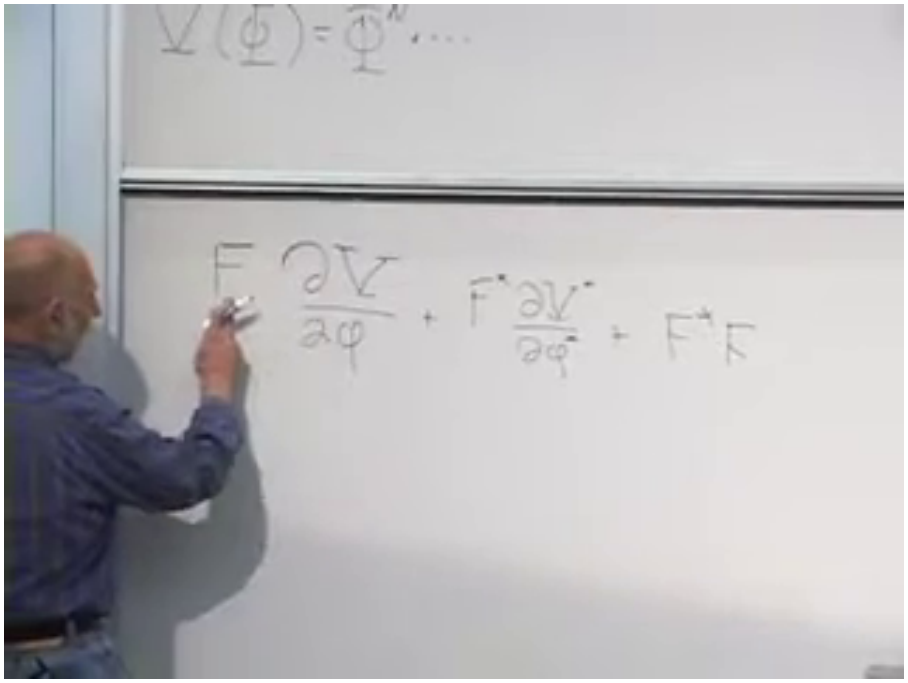

Figure 9.3: Auxiliary-field terms in the component Lagrangian.

This is the main mathematical payoff of the superspace calculation. Once the superpotential is given, the ordinary scalar potential is essentially read off by differentiation and squaring.

Worked examples. The lecture immediately uses this formula in the way it was meant to be used. If

$$W(\Phi) \sim m\Phi^2, \tag{9.24}$$

then, ignoring the numerical factor of 2 that Susskind does not track carefully on the board,

$$W'(\phi) \sim m\phi, \qquad V(\phi) \sim m^2\,\phi^*\phi. \tag{9.25}$$

This is just the ordinary scalar mass term.

If instead

$$W(\Phi) \sim g\Phi^3, \tag{9.26}$$

then

$$W'(\phi) \sim g\phi^2, \qquad V(\phi) \sim g^2\,|\phi|^4. \tag{9.27}$$

So a cubic superpotential produces a quartic scalar interaction. The lecture does not pursue the detailed Feynman rules here; it only wants the structural lesson: the ordinary potential follows directly from the superpotential.

9.4.1 Question & Answer

Question. How can a supersymmetric theory have a non-supersymmetric vacuum?

Answer. Because the equations and the vacuum are not the same thing. The theory can be supersymmetric while its minimum is not.

The crucial fact is

$$V(\phi) = |W'(\phi)|^2 \geq 0. \tag{9.28}$$

Hence there are two qualitatively different possibilities.

If the minimum occurs at zero energy,

$$V_{\text{min}} = 0, \tag{9.29}$$

then the quantity inside the absolute square must vanish:

$$W'(\phi) = 0. \tag{9.30}$$

Therefore

$$F = 0. \tag{9.31}$$

But F is the last component of the superfield. If it vanishes in the vacuum, then no superspace direction has been singled out, and the vacuum is supersymmetric.

If, on the other hand, the minimum is lifted above zero,

$$V_{\text{min}} > 0, \tag{9.32}$$

then

$$W'(\phi) \neq 0, \qquad F \neq 0. \tag{9.33}$$

This is the supersymmetric analogue of nonzero magnetization. The equations still possess the symmetry, but the vacuum has selected a direction in superspace. That is spontaneous supersymmetry breaking.

The lecture's earlier review of the ferromagnet now pays off in full. A nonzero F-component plays the role of the order parameter. It is the thing in the vacuum that breaks the symmetry, and it is the thing that splits states that supersymmetry would otherwise keep together.

9.5 Consequences of Supersymmetry Breaking

Once we allow the vacuum to break supersymmetry, the lecture unfolds the consequences one by one rather than introducing more formalism.

The first consequence is vacuum energy. If $V_{\text{min}} > 0$, the vacuum itself carries energy. In ordinary field theory one may ignore the additive normalization of the vacuum energy, but in a gravitational theory one cannot. The vacuum then gravitates. In modern language it acts as a cosmological constant. This is why Susskind suddenly asks, "Who cares about vacuum energy?" The answer is: gravity cares about it, and cares disastrously.

The second consequence is level splitting. The lecture deliberately returns to the magnetic analogy here. A nonzero F-term in the vacuum behaves, for the supermultiplet, the way the magnetic field behaved for the atom. It splits the energies, and therefore the masses, of the bosons and fermions that supersymmetry rotates into one another. The theory remains supersymmetric at the level of the equations; the solution does not.

The third consequence is the fermionic Goldstone mode. If the vacuum is not invariant, acting with the symmetry on the vacuum must produce a new zero-energy state. For ordinary symmetries

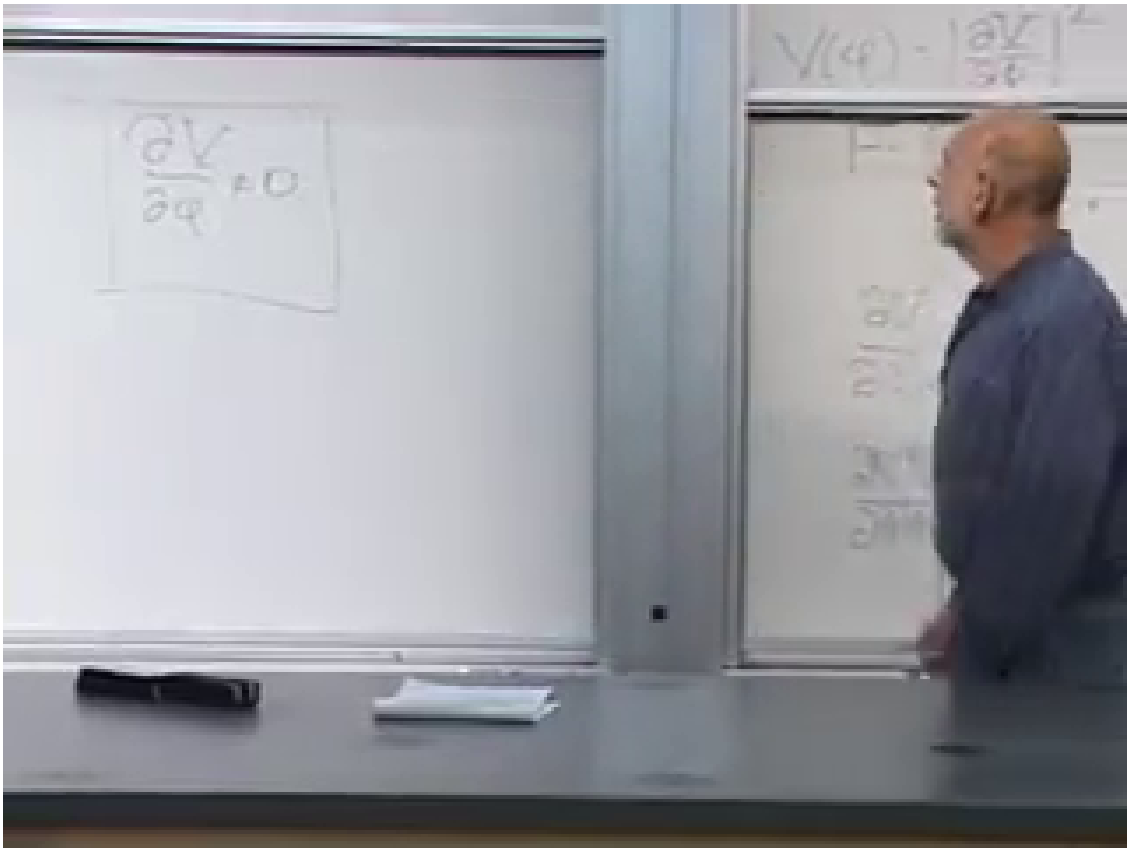

Figure 9.4: Vacuum condition and the positive supersymmetric potential.

this is a Goldstone boson. For supersymmetry it must be a fermion, because supersymmetry transformations change bosons into fermions. Thus broken supersymmetry implies a massless goldstino. Symbolically,

$$Q|0\rangle \sim |\text{goldstino}, k = 0\rangle. \tag{9.34}$$

The lecture presents this not as a separate theorem but as the direct supersymmetric analogue of the earlier ferromagnet discussion.

Hidden sectors. Susskind then introduces the jargon by returning once more to the atom and the ferromagnet. If the magnetic field of the ferromagnet were enormous, the atom would be badly distorted and no longer recognizable. If the field is weak, the atom remains recognizable and its levels are only slightly split. That is the regime of interest for phenomenology. One sector does the symmetry breaking; another sector, weakly coupled to it, merely feels the small splittings. The symmetry-breaking sector is then called hidden.

This is not presented as a beautiful mechanism. Quite the opposite: the lecture emphasizes that realistic supersymmetry breaking tends to be contrived, involving extra fields and extra sectors. But the analogy is sharp. The hidden sector is the ferromagnet; the visible sector is the atom whose levels are split only slightly.

Gravity and supergravity. At this point the lecture becomes qualitative on purpose. Gravity changes the rigid supersymmetric potential. Susskind sketches an additional contribution and then immediately hesitates over the placement of the Planck mass. That hesitation matters: the chapter should not falsely canonize a formula the lecture itself treats as uncertain. The stable lesson is simply that coupling to supergravity modifies the vacuum-energy story and can shift the effective potential downward. In favorable circumstances this may rescue the theory from an unacceptably large vacuum energy.

The super-Higgs analogy. The final supersymmetry discussion comes back to a theme the lecture announced much earlier and then held in reserve: the relation between Goldstone modes and gauge fields. A massless spin-1 particle has only two helicity states,

$$\lambda = \pm 1, \tag{9.35}$$

whereas a massive spin-1 particle has three,

$$\lambda = +1, 0, -1. \tag{9.36}$$

In the electroweak theory the missing longitudinal state is supplied by the Goldstone boson.

The lecture then applies the same logic to supergravity. A massless gravitino carries only the helicities

$$\lambda = \pm\frac{3}{2}, \tag{9.37}$$

but a massive spin-$\frac{3}{2}$ particle must carry

$$\lambda = \pm\frac{3}{2},\ \pm\frac{1}{2}. \tag{9.38}$$

The goldstino supplies the missing $\pm\frac{1}{2}$ states. Thus the goldstino combines with the gravitino in exact analogy with the way a Goldstone boson combines with a gauge boson. This is the lecture's super-Higgs picture: the broken-supersymmetry Goldstone mode completes the degrees of freedom of a massive gravitino.

By this point Susskind declares the mathematics of supersymmetry finished. The lecture will return to its physics later, but the formal story needed for symmetry breaking has now been told.

9.6 The First Step Toward $SU(5)$

The pivot to grand unification is abrupt on purpose. The lecture says, in effect, that the hard abstract mathematics is over, and now we move to something easier: a little group theory. The change of subject should feel like a genuine new beginning.

The standard model gauge group is

$$SU(3) \times SU(2) \times U(1). \tag{9.39}$$

The grand-unification question is whether this product group might be embedded in a larger, simpler group. The natural candidate is $SU(5)$, precisely because it is the smallest simple group that contains $SU(3) \times SU(2) \times U(1)$ as a subgroup.

By definition,

$$U^\dagger U = 1, \qquad \det U = 1 \tag{9.40}$$

for $U \in SU(5)$. For an infinitesimal transformation,

$$U = 1 + i\epsilon G + O(\epsilon^2). \tag{9.41}$$

Unitarity implies

$$G = G^\dagger, \tag{9.42}$$

and the special condition implies

$$\operatorname{tr} G = 0. \tag{9.43}$$

So the generators of $SU(5)$ are traceless Hermitian 5×5 matrices.

This immediately gives the counting rule. A Hermitian $n \times n$ matrix carries n^2 real parameters, and tracelessness removes one. Hence

$$\dim SU(n) = n^2 - 1, \tag{9.44}$$

so in particular

$$\dim SU(5) = 25 - 1 = 24. \tag{9.45}$$

The transcript contains a local slip before this rule is stabilized, but the correct result is the one above.

The embedding itself is done by blocks. We split the 5×5 matrix as a $2 \oplus 3$ system. The three $SU(2)$ generators τ_a live in the upper-left 2×2 block, the eight $SU(3)$ generators λ_A live in the lower-right 3×3 block, and one further diagonal generator commutes with both blocks. In the convention used in the lecture, one may write it as

$$Y \propto \operatorname{diag}\left(-1, -1, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right). \tag{9.46}$$

Its overall normalization is conventional; what matters here is its tracelessness and the different values it assigns to the 2-block and 3-block.

The lecture then makes its first particle assignment. The first two slots of the 5-component column should be a weak doublet with no color, so they are naturally ν_e and e^-. The last three slots should carry color but no weak $SU(2)$ quantum numbers. The tracelessness condition, read as a statement about charge sums inside a multiplet, then selects the anti-down quarks:

$$0 + (-1) + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 0. \tag{9.47}$$

Thus the first five-dimensional multiplet is

$$(\nu_e,\ e^-,\ \bar{d},\ \bar{d},\ \bar{d}). \tag{9.48}$$

The three copies differ only by color.

And this is exactly where the lecture stops. The remaining ten particles of one family are not forced into the same representation. Instead Susskind says that they fit into a ten-dimensional representation of $SU(5)$, and promises to return to that construction next time. The chapter should stop with the same incompleteness. Grand unification has not yet been developed; it has only been opened.

9.7 Summary

The lecture proceeds by analogy, but the analogy is never superficial. The Zeeman effect teaches us to see symmetry in the spectrum; the ferromagnet teaches us to distinguish the symmetry of the laws from the symmetry of the vacuum; the Goldstone mode teaches us how the broken vacuum announces itself. Only then do we return to the chiral superfield, derive

$$V(\phi) = |W'(\phi)|^2, \tag{9.49}$$

and ask the central vacuum question. If the minimum lies at $V = 0$, then $F = 0$ and the vacuum is supersymmetric. If the minimum lies above zero, then $F \neq 0$, the vacuum energy is nonzero, bosons and fermions split, and the goldstino appears. Gravity then turns vacuum energy from an innocuous shift into a physical problem, and supergravity offers a possible escape. Only after all of that does the lecture pivot to $SU(5)$, where the first structural embedding of $SU(3) \times SU(2) \times U(1)$ is set up and the first five-dimensional multiplet is identified.

Chapter 10

SU(5), Multiplets, and Proton Decay

These notes follow Leonard Susskind's lecture on grand unification, with curation by LazyingArt LLC, and they preserve the lecture's order of construction. The subject is not grand unified theories in general, but one concrete example, $SU(5)$, taken apart slowly enough that we can see both its beauty and its danger. The lecture begins with group theory because the particle story cannot be stated cleanly before the representation theory is in place. Only after that warmup do we fit one Standard Model family into $SU(5)$, discover the new gauge bosons, and confront the consequence that makes every grand-unified theory live dangerously: proton decay. Supersymmetry enters only at the end, not as a premise, but as a numerical improvement in the running of couplings.

10.1 SU(5) as a Concrete Grand-Unification Example

The lecture opens by narrowing the ambition. We are not going to survey all grand unified theories. We are going to take one theory, $SU(5)$, and see what makes it tick, what it predicts, and what can go wrong.

The structural claim is that the Standard Model gauge group fits naturally inside a larger simple group:

$$SU(3) \times SU(2) \times U(1) \subset SU(5). \tag{10.1}$$

So the subject is, from the start, an exercise in group theory. We are looking for a single symmetry that contains the separate color and electroweak symmetries as substructures.

The lecture is equally careful about supersymmetry. There is no deep logical statement here that supersymmetry implies grand unification, or conversely that grand unification implies supersymmetry. The point is more modest and more empirical: if one puts the two ideas together, some numerical fits get better.

That is why the lecture pauses before talking about particles and says, in effect: before we do that, we need a little group theory. Particle multiplets are group representations. If we want to classify the multiplets, we must first classify the representations.

10.2 Representations of $SU(N)$: N, $\bar{N}$, and Tensor Products

For $SU(N)$, the defining representation is simply the space of N-component complex columns,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_N \end{pmatrix}, \tag{10.2}$$

acted on by $N \times N$ special unitary matrices. In components,

$$\psi_i' = U_{ij}\psi_j. \tag{10.3}$$

This is the N-dimensional representation usually denoted just by N.

The first new representation we can make is the complex conjugate one. If we take ψ_i^*, then the transformation law is

$$\psi_i^{*\prime} = U_{ij}^*\psi_j^*. \tag{10.4}$$

This is the conjugate representation $\bar{N}$. The lecture stresses that N and $\bar{N}$ are both N-dimensional, and that there is an element of convention in deciding which one we call the particle representation and which one the antiparticle representation. Once we make that choice, however, the bookkeeping must remain fixed.

The next move is the one the lecture announces with the question, "What else can you do?" Put down two objects that each transform as N. If one is described by ψ_i and the other by ϕ_j, then the combined state is described by a two-index object

$$\psi_{ij} = \psi_i\phi_j, \tag{10.5}$$

with N^2 components. The group acts on each index separately:

$$\psi_{ij}' = U_{ik}U_{j\ell}\psi_{k\ell}. \tag{10.6}$$

This is the tensor product $N \otimes N$.

At this point the lecture introduces the key new word: irreducible. The full N^2-dimensional space is usually reducible. What that means is that there are linear combinations of the ψ_{ij} that transform among themselves and do not get mixed with the rest. Those smaller closed sectors are themselves representations.

The warmup example is ordinary spin. For $SU(2)$, the defining doublet may be thought of as the two spin states u and d. Two such doublets give four raw states, but those four states reorganize into irreducible pieces. That is the bridge from abstract tensor products to familiar physics.

10.3 Symmetric and Antisymmetric Two-Index States

The first clean split of $N \otimes N$ is into symmetric and antisymmetric tensors. The board at this stage is worth keeping, because it shows the moment when the abstract two-index object is turned into a precise condition.

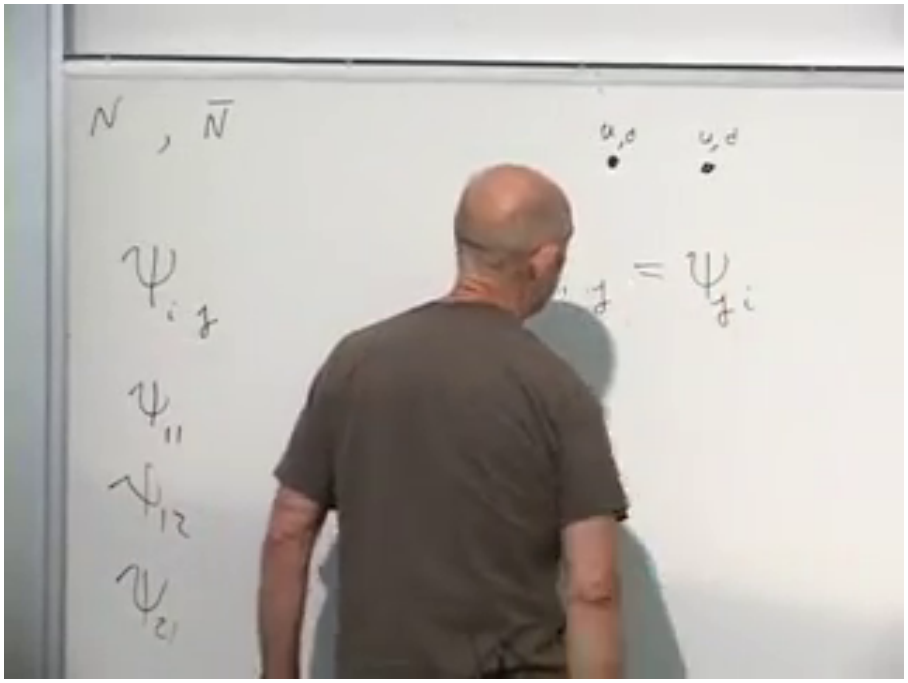

Figure 10.1: The symmetric two-index condition as it appears on the board.

The symmetric sector is defined by

$$\psi_{ij} = \psi_{ji}, \tag{10.7}$$

while the antisymmetric sector satisfies

$$\psi_{ij} = -\psi_{ji}. \tag{10.8}$$

What matters is that these two sectors do not mix under the group action. Indeed, if $\psi_{ij} = \pm\psi_{ji}$, then

$$\begin{aligned} \psi'_{ji} &= U_{jk}U_{i\ell}\psi_{k\ell} && (10.9)\\ &= U_{ik}U_{j\ell}\psi_{\ell k} && (10.10)\\ &= \pm U_{ik}U_{j\ell}\psi_{k\ell} && (10.11)\\ &= \pm\psi'_{ij}. && (10.12) \end{aligned}$$

So the symmetric and antisymmetric subspaces are each preserved by the group. That is precisely the sense in which they are separate representations.

Now we specialize to $SU(2)$, where the doublet is the spin-$\frac{1}{2}$ representation. Then

$$2 \otimes 2 = 3 \oplus 1. \tag{10.13}$$

The four raw states are

$$|uu\rangle, \qquad |ud\rangle, \qquad |du\rangle, \qquad |dd\rangle.$$

The symmetric combinations are

$$\begin{aligned} |1,1\rangle &= |uu\rangle, && (10.14)\\ |1,0\rangle &= \frac{1}{\sqrt{2}}\big(|ud\rangle + |du\rangle\big), && (10.15)\\ |1,-1\rangle &= |dd\rangle, && (10.16) \end{aligned}$$

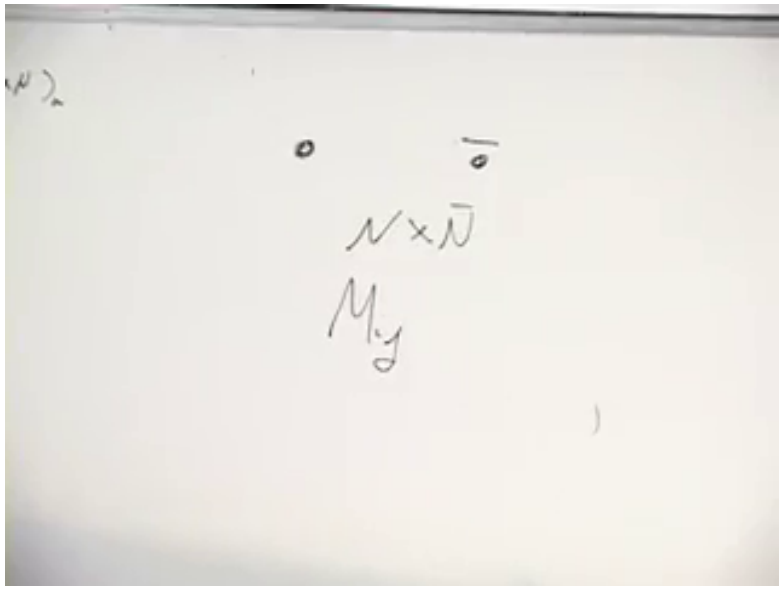

Figure 10.2: The mixed product $N \times \bar{N}$ re-expressed on the board as a matrix object.

and they form a triplet: the spin-one multiplet. The antisymmetric combination is

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|ud\rangle - |du\rangle), \tag{10.17}$$

and it is a singlet: the spin-zero state.

This is the lecture's first major payoff. Two spin-$\frac{1}{2}$ particles do indeed give four raw states, but those four states do not remain a single lump. The group itself sorts them into one triplet and one singlet. That is the pattern we will carry over to larger groups.

10.3.1 Question & Answer

Why do two N-plets give N^2 states without violating the exclusion principle?

Because the lecture is not putting two identical fermions into the same complete quantum state. It is explicitly imagining two distinguishable slots: one particle is here, the other is there. Their internal $SU(N)$ labels are being discussed while position, momentum, and the rest of the quantum numbers are being held in the background. The Pauli principle constrains the full state. Once the two particles are in different places, there is no contradiction in allowing the same internal label on both sides. So the raw counting really is N^2, and only after that counting do we ask how the group reorganizes those states into irreducible pieces.

10.4 From $N \otimes \bar{N}$ to the Adjoint, and Then to the Five-Plet

The next tensor product is conceptually different. Instead of particle with particle, we take particle with antiparticle. The board now isolates the point that this mixed product should be regarded as a matrix.

The exact board symbol below $N \times \bar{N}$ is not perfectly legible, so we use a cautious clean notation:

$$M_{ij} \in N \otimes \bar{N}. \tag{10.18}$$

The essential point is that the N^2 components can be organized as an $N \times N$ matrix. Once we do that, the natural decomposition is into trace and traceless parts:

$$M_{ij} = \frac{1}{N}(\operatorname{tr} M)\delta_{ij} + \left(M_{ij} - \frac{1}{N}(\operatorname{tr} M)\delta_{ij}\right). \tag{10.19}$$

The trace part is a singlet. The traceless part is the adjoint representation, so

$$N \otimes \bar{N} = 1 \oplus \mathrm{Adj}, \qquad \dim \mathrm{Adj} = N^2 - 1. \tag{10.20}$$

For $SU(3)$, this is the familiar statement that quark-antiquark quantum numbers split into a singlet and an octet, with the octet furnishing the gluons.

Only now does the lecture turn to $SU(5)$ itself. We do not begin by trying to unify all three generations. The lecture makes a point of refusing that larger ambition. We focus on one family. For the first family, the relevant fermions are the up and down quarks in three colors, together with the electron and neutrino.

The first five-component multiplet is then

$$\overline{\mathbf{5}} = \begin{pmatrix} \nu_L \\ e_L^- \\ \bar{d}_{rL} \\ \bar{d}_{gL} \\ \bar{d}_{bL} \end{pmatrix}. \tag{10.21}$$

This looks bizarre at first, and the lecture wants it to look bizarre. One should feel the oddity before one explains it away.

There is also a bookkeeping point here. We list only left-handed particles. The antiparticle of a right-handed electron is a left-handed positron, and similarly for the quarks. So the chiral particle content can be recorded using left-handed particles and left-handed antiparticles only. Mass terms are another story: they connect left- and right-handed fields through Higgs couplings, and the lecture explicitly treats that as a separate issue rather than as part of the grand-unified multiplet structure itself.

A note on convention. To keep the notation stable, we adopt the standard convention in which one family sits in $\overline{\mathbf{5}} \oplus \mathbf{10}$. The lecture's spoken and board language can sound like the conjugate convention $\mathbf{5} \oplus \overline{\mathbf{10}}$. We therefore keep the representation labels fixed here and translate the lecture's conjugation language when needed, instead of silently drifting with it.

10.4.1 Question & Answer

Why does the $SU(5)$ multiplet contain d-anti-quarks rather than d-quarks?

The lecture answers this by returning to the generators of the group. For $SU(N)$, the generators are Hermitian and traceless. A clean way to see the tracelessness is to look infinitesimally:

$$U(\epsilon) = e^{i\epsilon T}. \tag{10.22}$$

Because $U(\epsilon) \in SU(N)$, it has determinant one:

$$1 = \det U(\epsilon) = \det e^{i\epsilon T} = e^{i\epsilon \operatorname{tr} T}. \tag{10.23}$$

For arbitrary ϵ, this implies

$$T^\dagger = T, \qquad \operatorname{tr} T = 0. \tag{10.24}$$

Electric charge must be represented by one of the generators or, more precisely, by a linear combination of them. Therefore the charges within a single multiplet must sum to zero. To display the lecture's point cleanly, we may write the charge operator on the five-component column as

$$Q = \operatorname{diag}\left(0, -1, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \qquad \operatorname{tr} Q = 0. \tag{10.25}$$

Indeed,

$$0 + (-1) + 3\left(\frac{1}{3}\right) = 0. \tag{10.26}$$

If we had tried to place d-quarks rather than $\bar{d}$-quarks in this multiplet, the sum would have been

$$0 + (-1) + 3\left(-\frac{1}{3}\right) = -2,$$

and electric charge could not be embedded into the traceless generators in the way the lecture requires. That is why the odd assignment is not a decorative flourish. It is forced by the group theory.

10.5 The Antisymmetric Matrix and the Remaining Fermions

The five-plet is clearly not the whole family. Where are the up quarks? Where is the positron? Where do the remaining fields go? The lecture does not answer this by handing us the finished table. Instead it performs, as Susskind says, a bit of gymnastics.

The board next turns to the conjugate five-component column with quantum numbers

$$\chi = \begin{pmatrix} \bar{\nu} \\ e^+ \\ d_r \\ d_g \\ d_b \end{pmatrix}. \tag{10.27}$$

The overbar on the representation label is not the important point here; the important point is the quantum numbers. This column is an auxiliary device for doing the group theory, not a second physical five-plet that we are declaring by fiat. The move is to form the antisymmetric square of this object.

So we introduce a matrix Ψ_{ij} satisfying

$$\Psi_{ij} = -\Psi_{ji}, \qquad \Psi_{ii} = 0. \tag{10.28}$$

The diagonal entries vanish, the lower triangle is fixed by the upper triangle, and therefore the number of independent entries is

$$\frac{5 \cdot 4}{2} = 10. \tag{10.29}$$

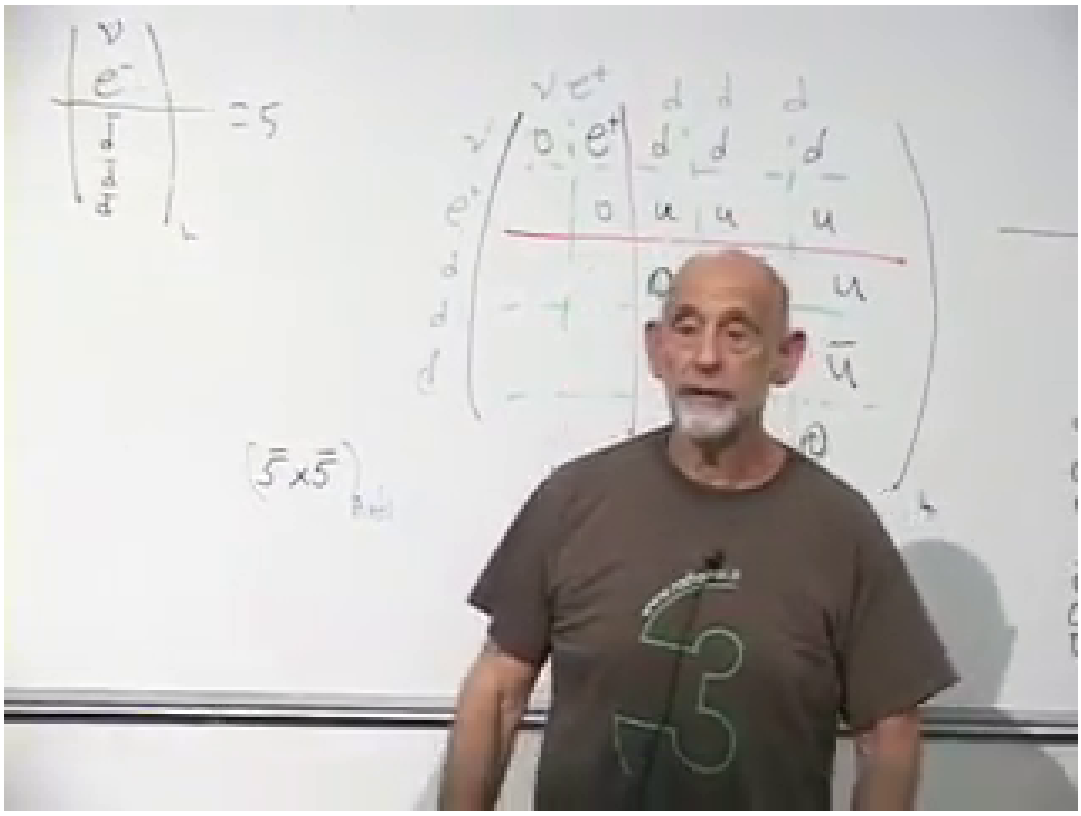

Figure 10.3: The antisymmetric 5×5 matter matrix on the board, with its visible $2 \oplus 3$ block structure.

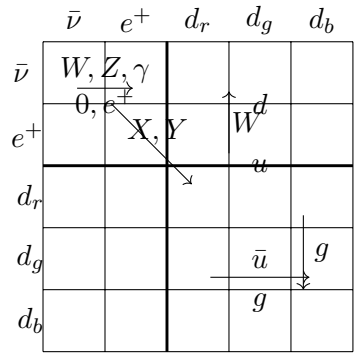

Figure 10.4: A schematic of the same $2 \oplus 3$ matrix layout: once the blocks are fixed, the gauge bosons act as horizontal or vertical shifts.

That is the ten-dimensional multiplet we were looking for.

The board evidence is particularly useful at this stage because it shows the full $2 \oplus 3$ organization.

A clean reconstruction, faithful to the lecture's content while not pretending that every obscured sign is legible on the board, is

$$\Psi \sim \left(\begin{array}{cc|ccc} 0 & e^+ & d_r & d_g & d_b \\ -e^+ & 0 & u_r & u_g & u_b \\ \hline -d_r & -u_r & 0 & \bar{u}_b & \bar{u}_g \\ -d_g & -u_g & -\bar{u}_b & 0 & \bar{u}_r \\ -d_b & -u_b & -\bar{u}_g & -\bar{u}_r & 0 \end{array}\right), \qquad \Psi_{ij} = -\Psi_{ji}. \tag{10.30}$$

The lower 3×3 block is the least secure part of the board image at the level of signs. The robust content is the antisymmetry, the $2 \oplus 3$ decomposition, and the placement of the $\bar{u}$-type entries in the color-color block.

The lecture then uses the standard color trick. An antisymmetric pair of colors behaves as an anti-color. Thus the antisymmetric color pairs carry the quantum numbers of $\bar{u}_r$, $\bar{u}_g$, and $\bar{u}_b$. With that understood, the remaining fermions fall into place. One family now fits neatly into

$$\overline{\mathbf{5}} \oplus \mathbf{10}. \tag{10.31}$$

It is also worth keeping the block structure in a cleaned schematic form, because this is exactly the layout on which the gauge-boson action will later be read. The diagram below is not a replacement for the board image. It is a clarification of the same board logic.

One should pause here in the same place the lecture pauses. This fit is mathematically striking. At the same time it is awkward in precisely the way the lecture wants us to feel: left-handed up quarks and left-handed anti-up quarks sit in one grand-unified structure, while the left-handed electron and left-handed positron live in different pieces. The point is not to smooth over the oddity too quickly.

10.6 Gauge Bosons, Leptoquark Transitions, and Proton Decay

After the break, the lecture turns from bookkeeping to dynamics. We now ask what the gauge bosons do. The answer begins from the same group-theoretic fact as before: gauge bosons live in

the adjoint representation. For $SU(5)$, that means

$$\dim \mathrm{Adj}_{SU(5)} = 5^2 - 1 = 24. \tag{10.32}$$

But the lecture does not present them as a list of twenty-four fields. It presents them as shifts.

In the five-component column, the familiar bosons occupy familiar places. The electroweak bosons act in the upper 2×2 block. The gluons act in the lower 3×3 color block. The new ingredients are the gauge bosons that connect the upper two slots with the lower three. Those are the leptoquark bosons traditionally called X and Y.

The lecture is explicit about their charges but deliberately modest about their names. The boson that connects a $\bar{d}$ to an e^- carries charge $\pm\frac{4}{3}$, while the one that connects a $\bar{d}$ to a neutrino carries charge $\pm\frac{1}{3}$. Susskind explicitly says he does not remember which label, X or Y, belongs to which charge assignment. We should preserve that caution. The stable content is the transition pattern together with the charges $\pm\frac{4}{3}$ and $\pm\frac{1}{3}$.

There is also an important physical remark in the lecture that should not be dropped: these new bosons carry color. They must, because they connect a color-carrying quark state to a colorless lepton state. In that respect they behave more like quarks than like photons. They can emit and absorb gluons, and one should not think of them as free, isolated particles at accessible scales.

How do these same gauge bosons act on the antisymmetric ten-dimensional multiplet? The lecture's picture is that they act as shifts on one index or the other. If we pretend for a moment that a matrix entry is built from a row label and a column label, then a gauge boson can be emitted from either constituent. Emission from one side gives a horizontal move; emission from the other side gives a vertical move. This is why the block picture in the previous section matters.

The weak bosons therefore do more than take $\nu \leftrightarrow e^-$ in the five-plet. Acting vertically in the antisymmetric matrix, they also take $d \to u$, which is of course the standard weak transition responsible for beta decay. Likewise the gluons mix the colors of the d- and u-type entries. None of that is new. The new physics comes from the X- and Y-type shifts between the upper and lower sectors.

Once those shifts exist, new processes exist. One can turn a quark into a lepton, or a lepton into a quark, inside the same unified gauge structure. That is the physical content of grand unification, and it is also the source of its biggest phenomenological danger.

A standard channel in the lecture is

$$p \to e^+ + \pi^0. \tag{10.33}$$

A secondary possibility, suppressed by an extra electromagnetic factor, is

$$p \to e^+ + \gamma. \tag{10.34}$$

The logic is straightforward. Start with a proton, uud. Use one of the heavy leptoquark bosons to convert a d-quark into a positron and a u-quark into an anti-u. The remaining u-quark is a spectator. The $u\bar{u}$ pair can form a neutral pion, or more rarely annihilate to a photon.

This is the lecture's physical climax. The story has turned from "this is a beautiful fit" to "this is dangerous." If protons decayed as easily as ordinary unstable particles, nuclei would not survive and neither would we.

10.6.1 Question & Answer

If the new bosons have ordinary-sized couplings, why does the proton not decay immediately?

The lecture's answer is that the only available lever is the mass of the heavy intermediate boson. The unified theory has one gauge coupling at high energies, and the X- and Y-boson couplings are not assumed to be absurdly tiny. So the suppression must come from the propagator.

At the level of a rough Feynman-diagram estimate, the amplitude behaves as

$$\mathcal{A}(p \to e^+\pi^0) \sim \frac{g^2}{M_X^2}, \tag{10.35}$$

where the numerator comes from the two gauge vertices and the denominator from the heavy propagator. Squaring the amplitude gives the rate,

$$\Gamma_p \sim \frac{g^4}{M_X^4}, \tag{10.36}$$

up to additional factors involving the proton mass and hadronic matrix elements. The lifetime is the inverse rate,

$$\tau_p \sim \Gamma_p^{-1}. \tag{10.37}$$

The lecture then compares this with the scale of microscopic physics. If everything were of order one in proton units, the natural lifetime would be the light-crossing time of a proton, roughly 10^{-23} seconds. But experimentally one finds only lower bounds, and the lecture quotes the observational scale as roughly

$$\tau_p \gtrsim 10^{33} \text{ years}. \tag{10.38}$$

So the gap to be explained is enormous. The only way to reconcile the unified interaction with proton stability is to make the exchanged boson extremely heavy:

$$M_X \sim 10^{15}\text{–}10^{16}\,\text{GeV}, \tag{10.39}$$

with the lecture settling on the memorable order of magnitude

$$M_X \gtrsim 10^{16}\,\text{GeV}. \tag{10.40}$$

That is why proton decay is not a minor side remark. It is a universal warning sign for grand unification. If one unifies quarks and leptons inside a single simple group, baryon-number-violating processes are very hard to avoid. The theory can survive only if the new bosons sit at a fantastically high scale.

10.7 Unification Scale, Running Couplings, and the Supersymmetry Epilogue

The lecture does not leave the number 10^{16} GeV hanging in isolation. It immediately asks: is that a ridiculous scale? The answer given in the lecture is no. It is still a few orders of magnitude below the Planck scale, and more importantly, there is an independent reason to find it attractive.

First, $SU(5)$ cannot be an unbroken symmetry of nature. If it were, the X and Y bosons would live in the same multiplet as the gluons, the W, the Z, and the photon, and there would be no explanation for the huge mass difference. So the theory requires a separate high-scale Higgs phenomenon that breaks the full symmetry, roughly by splitting the upper 2-block from the lower 3-block. Above that scale, however, the difference between the heavy and light gauge bosons becomes negligible, and the full $SU(5)$ symmetry should be restored as a good approximation.

That is what the lecture means by the unification scale. At energies far above M_X, even the X- and Y-boson masses become negligible, and the world should begin to look $SU(5)$-invariant. So we ask where the running couplings seem to converge.

The lecture phrases this graphically. Instead of plotting the couplings themselves, it plots their inverses against $\log \mu$:

$$\alpha_1^{-1}(\mu), \qquad \alpha_2^{-1}(\mu), \qquad \alpha_3^{-1}(\mu). \tag{10.41}$$

The three lines do not meet perfectly in the minimal nonsupersymmetric picture, but they come tantalizingly close. The near-crossing occurs at

$$\mu \sim 10^{15}\text{–}10^{16}\,\text{GeV}, \tag{10.42}$$

the same range that proton stability was already suggesting.

So the lecture closes the circle. Proton decay forces the new leptoquark bosons to be extremely heavy. The running of the Standard Model couplings suggests that the place where the separate gauge couplings begin to look unified is again around 10^{15} to 10^{16} GeV. These are not two unrelated numbers.

Supersymmetry then returns as an epilogue, exactly as promised in the opening minute of the lecture. There is no deep theoretical necessity tying it to grand unification. But if one includes the superpartner spectrum in the running, the three coupling lines meet much more cleanly, at roughly the one-percent level. That does not prove either idea. It simply makes the coincidence sharper and therefore harder to ignore.

The lecture also remarks, briefly, that a still larger unifying group can package the matter even more neatly. In the theory commonly written $SO(10)$, one family together with a right-handed neutrino fits into a single sixteen-dimensional multiplet. But that is only a glimpse of a larger story. The lecture itself remains centered on $SU(5)$.

10.8 Summary

The lecture unfolds in a deliberate order. First we build just enough representation theory to speak cleanly: the defining representation N, the conjugate representation $\bar{N}$, the symmetric and antisymmetric pieces of $N \otimes N$, and the singlet plus adjoint decomposition of $N \otimes \bar{N}$. Without that apparatus, the later particle assignments would look arbitrary.

Then the physical content appears. One Standard Model family fits into the grand-unified pattern

$$\bar{\mathbf{5}} \oplus \mathbf{10},$$

with the five-component multiplet containing ν, e^-, and the three $\bar{d}$'s, and the antisymmetric ten-dimensional matrix carrying the remaining fermions. The same group structure also forces new gauge bosons that connect quarks and leptons.

That is the beauty and the danger in one stroke. The new bosons make proton decay possible. Proton stability then drives their masses up to the neighborhood of 10^{16} GeV. The running couplings suggest that this same scale is where the broken $SU(5)$ symmetry begins to re-emerge. Supersymmetry does not create that logic, but it sharpens its numerical form. The lecture leaves us with precisely that tension: a remarkably elegant group-theoretic picture, and a universe that permits it only if the most dangerous new processes are pushed almost unimaginably far away.